Winter Term 2022/23

Numerical Methods for Partial Differential Equations I

Stephan Rave

March 9, 2023

Institute for Analysis and Numerics
Faculty of Mathematics and Computer Sciences
WWU Münster

Preface

Many physical phenomena, such as heat conduction, electrostatic fields or elastic deformations of solid bodies, can be described through elliptic partial differential equations. Only in special cases can the solutions of these equations be obtained using analytic methods. Hence, provably efficient and reliable numerical methods are required to compute approximate solutions. In this course, we will be concerned with the numerical analysis of so-called finite-element methods, which have an elegant mathematical theory and are wildly adopted in academic and industrial applications. Our focus will lie on establishing convergence rates for these methods and deriving rigorous and efficiently computable a posteriori error bounds. Further, we will consider extensions to time-dependent problems.

These lecture notes are based on notes by Mario Ohlberger, who kindly provided the LaTeX-sources to me. I thank Timo Keith for spotting numerous typos and inaccuracies.

For further reference, we refer to the monographs [Bra07b] (in German: [Bra07a]), [BS08], [Cia02], [EG04] and [EG21a; EG21b; EG21c; EG21d]. For the theory of iterative linear solvers we refer to [Saa03]. The required functional analytical background is covered in [Alt16], which is also available in German [Alt12].

Most certainly, this manuscript is not completely free of errors, and I kindly ask you to inform me of any potential typos or errors.

Contents

1 Introduction

In this chapter introduce partial differential equations and briefly show how physical principles lead to such equations. Further we sketch first ideas for the numerical treatment of partial differential equations.

1.1 Second-order partial differential equations and their classification

Partial differential equations (PDEs) are equations which relate single or multiple functions with their own partial derivatives. In this course we will only consider PDEs of second order. 'Second order' means that partial derivatives of at most second order appear in the equation. Prototypical examples of second-order PDEs are

1) the **Poisson equation** *(elliptic)*

$$-\Delta u(x) = f(x), \qquad\qquad x \in \Omega \subset \mathbb{R}^d,$$

2) the **heat equation** *(parabolic)*

$$\partial_t u(x,t) - \Delta u(x,t) = f(x,t), \qquad\qquad (x,t) \in \Omega_T \subset \mathbb{R}^d \times \mathbb{R}^+,$$

3) and the **wave equation** *(hyperbolic)*

$$\partial_{tt} u(x,t) - \Delta u(x,t) = 0, \qquad\qquad (x,t) \in \Omega_T \subset \mathbb{R}^d \times \mathbb{R}^+.$$

The classification of second-order PDE (elliptic, parabolic, hyperbolic) is based on the so-called principal symbol of the differential operator

$$Lu = \sum_{k,l=1}^{n} a_{k,l} \partial_i \partial_j u + \sum_{k=1}^{n} b_k \partial_k u + cu$$

defining the equation. In particular, for the examples above we have

$$
\begin{aligned}
&a_{k,l} = -\delta_{k,l} && b_k = 0 && c = 0 && n = d && \text{(Poisson equation)} \\
&a_{k,l} = \begin{cases} -1 & k = l \neq n \\ 0 & \text{otherwise} \end{cases} && b_k = \delta_{k,1} && c = 0 && n = d+1 && \text{(heat equation)} \\
&a_{k,l} = \begin{cases} -1 & k = l \neq n \\ 1 & k = l = n \\ 0 & \text{otherwise} \end{cases} && b_k = 0 && c = 0 && n = d+1 && \text{(wave equation)}
\end{aligned}
$$

Such a constant coefficient linear differential operator L leads via Fourier transform to the complex multivariate polynomial

$$L(\xi) := \sum_{k,l=1}^{n} a_{k,l} \cdot (i\xi_k) \cdot (i\xi_l) + \sum_{k=1}^{n} b_k \cdot (i\xi_k) + c,$$

which is called the symbol of L. The principal symbol is then the quadratic form

$$L^p(\xi) := \sum_{k,l=1}^{n} a_{k,l} \cdot (i\xi_k) \cdot (i\xi_l) = \xi^T \cdot A \cdot \xi$$

where

$$A_{i,j} := -\frac{1}{2}(a_{i,j} + a_{j,i}).$$

In the above examples we have:

$$A_{\text{poisson}} = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}, \qquad A_{\text{heat}} = \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & 0 \end{bmatrix}, \qquad A_{\text{wave}} = \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & -1 \end{bmatrix}.$$

A quadratic form is called definite if the symmetric matrix A is positive or negative definite. The form is called indefinite if A has both positive and negative eigenvalues, and the form is called degenerate if A is singular. Hence, the principal symbol of the Poisson equation is definite, the principal symbol of the heat equation is degenerate and the principal symbol of the wave equation is indefinite. In analogy to the classification of conic sections, second-order PDEs with definite, degenerate or indefinite principal symbol are called elliptic, parabolic or hyperbolic respectively. The solutions of each of these PDE types exhibit very different qualitative behaviors. In this course, we will primarily focus on elliptic PDEs, using most of times then Poisson equation as a model problem. Later, we will also discuss parabolic PDEs.

1.2 Mathematical modeling with PDEs

In this section we briefly illustrate how PDEs arise from physical principles.

1.2.1 Conservation laws

Consider the diffusion of a drop of ink in water: denoting by u the concentration of ink in water, we are interested in distribution of u in space and time, i.e., we want to determine a function $u : \Omega \times \mathbb{R}^+ \to [0, 1]$, where $\Omega \subset \mathbb{R}^d$, $d = 2, 3$ is some spatial domain of interest. To find an equation defining u, we will make use of a fundamental principle of continuum mechanics and formulate a conservation law for u:

Definition 1.1 (Conservation law)**.**

Physical principle: *Let u be an extensive state variable, e.g., mass, impulse or energy. Then the change of u over time in an arbitrary volume V can only be caused by the transport of u over the boundary of V.*

Mathematical equivalent: *For an extensive state variable u and an arbitrary volume $V \subset \Omega$ of sufficient regularity the density distribution $u(x,t)$ satisfies:*

$$\frac{d}{dt}\int_V u(x,t)\,\mathrm{d}x = -\int_{\partial V} \mathbf{q}(x,t)\cdot \mathbf{n}(x)\,\mathrm{d}\sigma(x). \tag{1.1}$$

Here, σ is the surface measure on ∂V, $\mathbf{n}$ is the unit outer normal to V and $\mathbf{q}$ the flux density of u.

To obtain an equation defining the ink distribution u from this general principle, we need to state the physical relation between the flux density $\mathbf{q}$ and u. Assuming that the water is at rest, it turns out that the flow of u is proportional to the negative gradient of its density field, i.e,

$$\mathbf{q}(x,t) = -D\nabla u(x,t).$$

Here, $\nabla := (\partial_{x_1}, \dots, \partial_{x_d})^T$ denotes the gradient w.r.t. $x \in R^d$. The proportionality constant $D > 0$ is called the diffusion coefficient.

If we now substitute this relation into the conservation law (1.1), we obtain:

$$\frac{d}{dt}\int_V u(x,t)\,\mathrm{d}x = \int_{\partial V} D\nabla u(x,t)\cdot \mathbf{n}(x)\,\mathrm{d}\sigma(x).$$

This is our mathematical model of the diffusion of a drop of ink in water.

Remember Gauß's theorem:

Theorem 1.2 (Gauß's theorem for Lipschitz domains)**.** *Let* $\Omega \subset \mathbb{R}^d$ *be a Lipschitz domain,* $\mathbf{n} \in \mathbb{R}^d$ *the unit outer normal to* Ω*, and let* $\mathbf{v} \in C^1(\Omega;\mathbb{R}^d) \cap C^0(\overline{\Omega};\mathbb{R}^d)$ *be a vector field such that* $\operatorname{div}(\mathbf{v}) \in L^1(\Omega)$*. Then:*

$$\int_\Omega \operatorname{div}(\mathbf{v})(x)\,\mathrm{d}x = \int_{\partial\Omega} \mathbf{v}(x)\cdot\mathbf{n}(x)\,\mathrm{d}\sigma(x).$$

Proof. See [Alt16, A8.8]. □

In many introductory courses a smooth boundary of $\partial\Omega$ is assumed. However, later on Ω will have a polygonal, i.e., only piece-wise smooth boundary. Hence, we have formulated the Gauß's theorem for the more general class of Lipschitz domains:

Definition 1.3 (Lipschitz domain)**.** *A bounded, connected and open subset* $\Omega \subset \mathbb{R}^d$ *is called Lipschitz domain if its boundary* $\partial\Omega$ *is locally the graph of a Lipschitz continuous function (see [Alt16, A8.2] for a formal definition).*

Using Gauß's theorem, we can reformulate our model of ink diffusion as a PDE:

$$\begin{aligned}\frac{d}{dt}\int_V u(x,t)\,\mathrm{d}x &= \int_{\partial V} D\nabla u(x,t)\cdot\mathbf{n}(x)\,\mathrm{d}\sigma(x)\\ &\overset{\text{Gauß}}{=} \int_V \operatorname{div}(D\nabla u(x,t))\,\mathrm{d}x.\end{aligned}$$

Thus, after exchanging differentiation and integration we obtain:

$$\int_V \partial_t u(x,t)\,\mathrm{d}x = \int_V \operatorname{div}(D\nabla u(x,t))\,\mathrm{d}x.$$

As this integral relation hols for arbitrary subdomains $V \subset \Omega$, we finally have:

$$\partial_t u(x,t) = \operatorname{div}(D\nabla u(x,t)) \qquad \forall\,(x,t)\in\Omega\times\mathbb{R}^+.$$

Since the diffusion coefficient D is, in our case, constant over the entire domain, we can rewrite this PDE as

$$\partial_t u(x,t) - D\Delta u(x,t) = 0.$$

We see that the mathematical model for the diffusion of our ink drop is the heat equation, which is also called *non-stationary diffusion equation* in this context.

The non-stationary diffusion equation has infinitely many solutions. To select the correct solution for our problem, we need to prescribe intial and boundary conditions for $u(x,t)$.

Definition 1.4 (Initial-boundary value problem for the non-stationary diffusion equation)**.** *Let* Ω *be domain in* $\mathbb{R}^d$*,* $T > 0$ *a final time, and let the initial values* $u_0 \in C^2(\Omega)\cap C^0(\bar{\Omega})$ *and boundary values* $g \in C^1([0,T];C^0(\partial\Omega))$ *be given. Then* $u \in C^1([0,T];C^2(\Omega)\cap C^0(\bar{\Omega}))$

is called classical solution for the Cauchy-Dirichlet problem for the non-stationary diffusion equation if

$$\begin{aligned} \partial_t u(x,t) - D\Delta u(x,t) &= 0 && \forall\,(x,t) \in \Omega \times (0,T), \\ u(x,t) &= g(x,t) && \forall\,(x,t) \in \partial\Omega \times [0,T], \\ u(x,0) &= u_0(x) && \forall x \in \Omega. \end{aligned} \tag{1.2}$$

Here, 'Cauchy' refers to the fact that we consider an initial-value problem. 'Dirichlet' refers to the fact that we prescribe the value of the solution at $\partial\Omega$.

Having obtained such a PDE from physical principles, there are many questions to be answered by mathematics:

- Is there a solution of (1.2)? If yes, is it unique?
- What regularity has the solution have?
- Is the solution bounded?
- Is there a closed-form solution?
- If not, what is a good numerical method to approximate the solution?
- Is there proof that the numerical method converges to the solution? At what rate?

We can also ask for the long-term behavior of the solution. If in (1.2) the Dirichlet-values g do not depend on time, the following theorem holds:

Theorem 1.5 (Long-term behavior)**.** *If in (1.2) we have $g(x,t) = \bar{g}(x)$ for all $t \in \mathbb{R}^+$ for a function $\bar{g} \in C^0(\partial\Omega)$, then $u(x,t)$ converges for $t \to \infty$ to a function $\bar{u} \in C^2(\Omega) \cap C^0(\bar{\Omega})$. The function $\bar{u}$ is a classical solution of the Dirichlet problem for the stationary diffusion equation, i.e,*

$$\begin{aligned} -D\Delta\bar{u}(x)) &= 0 && \forall x \in \Omega, \\ \bar{u}(x) &= \bar{g}(x) && \forall x \in \partial\Omega. \end{aligned}$$

Since we can divide by D, we see that $\bar{u}$ is a solution of the Poisson equation with zero source term f. This special case of the Poisson equation is called Laplace's equation and its solutions are called harmonic functions.

1.2.2 Energy minimization

A second fundamental concept in physics is the principle of energy minimization. As an example, consider the elastic deformation of a solid body: Let $\Omega \subset \mathbb{R}^d$ the volume in which the non-deformed body is located and let $u(x,t) \in \mathbb{R}^d$ its deformation field. With σ being the stress tensor, f an outer volume force, e.g., gravity, and $\varepsilon(u) := \frac{1}{2}(\nabla u + \nabla u^\top)$ the strain tensor, the potential energy stored in the deformed body is given by

$$E(u) = \frac{1}{2}\int_\Omega \sigma : \varepsilon(u)\,\mathrm{d}x - \int_\Omega f u\,\mathrm{d}x.$$

In this expression, the term $A : B$ for matrices $A, B \in \mathbb{R}^{n\times n}$ refers to the Frobenius inner product $A : B = \operatorname{tr}(A^\top B) = \sum_{i,j=1}^n a_{ij}b_{ij}$. For small deformations, the linear material law

$$\sigma(u) = A\,\varepsilon(u),$$

with symmetric positive A is a good approximation of σ for many materials.

We now apply the principle of energy minimization:

Definition 1.6 (Energy minimization/variational principle)**.**

Physical principle: *A physical system always strives for a state of minimal energy.*

Mathematical equivalent: *Let u be a state variable and $E(u)$ the energy of the system. Then u converges to an optimal state $\bar{u}$ minimizing E. Assuming sufficient regularity of E, this means that*

$$\frac{d}{d\xi}E(\bar{u} + \xi\varphi)\Big|_{\xi=0} = 0 \tag{1.3}$$

for arbitrary admissible variations φ.

From (1.3) and the definitions of E and σ we obtain:

$$\int_\Omega A\,\varepsilon(\bar{u}) : \varepsilon(\varphi)\,\mathrm{d}x = \int_\Omega f\varphi\,\mathrm{d}x,$$

for all admissible variations φ. Assuming that $\bar{u}$ is sufficiently smooth, we obtain with partial integration for test functions (variations) φ which vanish on $\partial\Omega$:

$$\int_\Omega -\operatorname{div}(A\varepsilon(\bar{u}))\varphi\,\mathrm{d}x = \int_\Omega f\varphi\,\mathrm{d}x.$$

We now use the fundamental lemma of calculus of variations:

Theorem 1.7 (Fundamental lemma of calculus of variations)**.** *Let $\Omega \subset \mathbb{R}^d$ be open and $u \in L^1_{loc}(\Omega)$ such that*

$$\int_\Omega u\cdot\varphi\,\mathrm{d}x = 0 \qquad \forall\varphi \in C_0^\infty(\Omega),$$

then $u \equiv 0$ almost everywhere. Here, $C_0^\infty(\Omega)$ denotes the compactly supported smooth functions of Ω.

Proof. Exercise. □

Thus, we obtain:

$$-\operatorname{div}(A\varepsilon(\bar{u})) = f.$$

In the one-dimensional case, e.g., considering the elongation of a thin wire, the PDE reduces to

$$-\partial_x(A\partial_x u) = f.$$

Another special case in two spatial dimensions is the displacement of a thin membrane, where one obtains

$$-\operatorname{div}(A\nabla u) = f.$$

1.3 Fundamental ideas of numerical methods

In this section we will briefly introduce three wildly used classes of numerical methods for the solution of PDEs, each of which is based on another fundamental idea.

1.3.1 Finite differences

Idea: Approximation of the differential operator using difference quotients.

Let u be a differentiable function. Then the partial derivative $\partial_i u(x)$ can be approximated by a difference quotient:

$$\begin{aligned}
\partial_i u(x) &\approx \partial_i^{+h} u(x) := \frac{u(x+\mathbf{e}_i h) - u(x)}{h} && \text{(forward difference)}\\
\partial_i u(x) &\approx \partial_i^{-h} u(x) := \frac{u(x) - u(x-\mathbf{e}_i h)}{h} && \text{(backward difference)}\\
\partial_i u(x) &\approx \partial_i^{c,h} u(x) := \frac{u(x+\mathbf{e}_i h) - u(x-\mathbf{e}_i h)}{2h} && \text{(backward difference).}
\end{aligned}$$

Given a regular grid where the vertices are equidistantly distributed with distance h, we then can approximate $(Lu)(x)$ (where L is the partial differential operator defining the PDE) at the vertices of the grid only using evaluations of u at these nodes. This approximates the PDE by a finite system of equations for the values of u at the vertices.

While forward, backward or central differences all converge to $\partial_i u(x)$ for $h \to 0$, not each choice necessarily leads to a converging numerical scheme. For example, for the linear transport equation

$$\partial_t u(x,t) + a\partial_x u(x,t) = 0.$$

one easily sees that the difference quotient approximating $\partial_x u$ has to be chosen depending on the sign of a. Finite difference methods for transport equations and similar conservation laws are treated in *Numerical Methods for Partial Differential Equations II.* In this course, we will use finite difference approximations only for the time discretization of parabolic problems.

1.3.2 Finite volumes

Idea: Discrete conservation principle

Partitioning $\Omega \subset \mathbb{R}^d$ into $N \in \mathbb{N}$ pairwise disjoint subdomains $T_i \subset \Omega$, such that $\overline{\Omega} = \cup_{i=1}^N \overline{T_i}$, we can employ the conservation principle 1.1 for each of these subdomains. Considering the mean value $u_i(t)$ of $u(\cdot, t)$ in T_i, we obtain using the fundamental theorem of calculus:

$$u_i(t^{n+1}) - u_i(t^n) = -\frac{1}{|T_i|} \int_{t^n}^{t^{n+1}} \int_{\partial T_i} \mathbf{q}(u(x,t), x) \cdot \mathbf{n}(x) \, \mathrm{d}\sigma(x) \, \mathrm{d}t.$$

From this identity we obtain a so-called finite volume discretization when we replace the continuous flux $\mathbf{q}$ by a numerical flux $\mathbf{q_h}$ which only depends on the averages u_i and u_j for neighboring subdomains T_j.

Finite volume methods will also be treated in *Numerical Methods for Partial Differential Equations II.*

1.3.3 Finite elements

Idea: Energy minimization in finite-dimensional subspaces.

Finite element methods are special cases of so-called Galerkin methods, which are based on the idea of energy minimization: Consider V a function space suitable for the formulation of the problem and $E : V \to \mathbb{R}$ an energy functional, assigning to each function in V its energy. In Section 1.2.2 we saw, that many physical problems can be written as an energy minimization problem:

$$u = \operatorname*{argmin}_{v \in V} E(v).$$

From this continuous minimization problem we obtain a discrete problem, by replacing the infinite-dimensional function space V by a finite-dimensional subspace $V_h \subset V$:

$$u_h = \operatorname*{argmin}_{v_h \in V_h} E(v_h).$$

As before, as a necessary condition for minimality we have:

$$\frac{d}{d\xi} E(u_h + \xi v_h)\Big|_{\xi=0} = 0 \qquad \text{for all} v_h \in V_h,$$

assuming sufficient regularity of E. As V_h is finite-dimensional, we can choose a finite basis $\langle \varphi_1, \dots, \varphi_N \rangle$ of V_h and write each $u_h \in V_h$ as $u_h = \sum_{i=1}^N u_i \varphi_i$. Then we have:

$$\frac{d}{d\xi} E\Big(\sum_{i=1}^N u_i \varphi_i + \xi \varphi_j\Big)\Big|_{\xi=0} = 0 \qquad \text{for all } j = 1, \dots, N.$$

This is a system of N linear or nonlinear equations for the unknowns u_i, $i = 1, \dots, N$, which can be solved with standard numerical methods.

Finite element approximations are obtained from this general approach by a special choice of V_h and a corresponding basis such that

$$\frac{d}{d\xi} E\Big(\varphi_i + \xi \varphi_j\Big)\Big|_{\xi=0} \neq 0$$

only for a small amount of pairs (i, j). In particular for linear problems, this minimization problem then leads to a sparse linear system of equations, which can be solved efficiently, even for many millions of unknowns.

The analysis of finite element methods for elliptic and parabolic problems will be the central topic of this course.

2 Theoretical background

In this chapter we will review some basic facts from functional analysis about Banach and Hilbert spaces. Further, we will study Sobolev spaces in more detail, as they will be central for our mathematical analysis of elliptic PDEs. Along the way we will fix some notation. We start right away by introducing multi-index notation for partial derivatives:

Definition 2.1 (Multi-indices and partial derivatives). *Let $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}_0^d$ be a multi-index. We write*

$$|\alpha| := \sum_{i=1}^{d} \alpha_i, \qquad\qquad x^\alpha := x_1^{\alpha_1} \ldots x_d^{\alpha_d}.$$

Let further $u : \mathbb{R}^d \to \mathbb{R}$ be a multivariate function, then we denote by $D^\alpha u$ the $|\alpha|$-th order partial derivative of u given by

$$D^\alpha u := \frac{\partial^\alpha u}{\partial x^\alpha} := \Big(\frac{\partial}{\partial x_1}\Big)^{\alpha_1} \ldots \Big(\frac{\partial}{\partial x_d}\Big)^{\alpha_d} u,$$

Usually, we will use the shorter notation

$$\partial_i u(x) := \partial_{x_i} u(x) := \frac{\partial}{\partial x_i} u(x).$$

for the partial derivatives.

2.1 Banach spaces

Definition 2.2 (Normed space, Banach space). *A (real) normed space is a pair $(V, \|\cdot\|)$ consisting of a vector space V and a norm $\|\cdot\| : V \to \mathbb{R}$ on V. The normed space is called a Banach space if the metric space (V, d) where $d(v, w) := \|v - w\|$ is complete, i.e., all Cauchy sequences converge.*

Examples 2.3. *1) $(\mathbb{R}, |\cdot|)$ is a Banach space.*

2) Let $X \subset \mathbb{R}^d$ and V a normed space. We denote by $C^0(X; V)$ the vector space of continuous functions on X with values in V. If X is closed and bounded, then each such function is bounded and

$$\|f\|_\infty := \sup_{x \in X} \|f(x)\|$$

defines a norm on $C^0(X;V)$. If V is a Banach space, then so is $C^0(X;V)$ with $\|\cdot\|_\infty$.

3) *Let $X \subset \mathbb{R}^d$ be open. Then we denote by $C^k(\overline{X};V)$ the space of k-times continuously differentiable V-valued functions f on X, s.t. f and all partial derivatives of f up to k-th order continuously extend to $\overline{X}$. If V is a Banach space, then so is $C^k(\overline{X};V)$ w.r.t. the norm*
$$\|f\|_{k,\infty} := \max_{|\alpha|\le k} \|D^\alpha u\|_\infty.$$

Proof. Exercise. □

Definition 2.4 (Operator)**.** *Let V, W be normed spaces, $D \subset V$. Then we call a mapping $T : D \to W$ an operator between V and W with domain D. As usual, we have:*

1) *T is continuous in $v \in D$ iff*
$$\forall \varepsilon > 0\, \exists \delta > 0\, \forall v' \in D : \quad \|v - v'\|_V < \delta \Longrightarrow \|T(v) - T(v')\|_W < \varepsilon.$$

2) *T is continuous on D iff T is continuous in v, for all $v \in D$.*

3) *T is Lipschitz continuous on D iff*
$$\exists L > 0\, \forall v, v' \in D : \quad \|T(v) - T(v')\|_W < L\|v - v'\|_V.$$

Proposition 2.5 (Continuous linear operators)**.** *Let $T : V \to W$ a linear operator. Then the following statements are equivalent:*

1) *T is continuous.*

2) *T is Lipschitz continuous.*

3) *T is bounded, i.e.*
$$\|T\| := \sup_{0\ne v\in V} \frac{\|T(v)\|}{\|v\|} < \infty.$$

Proof. 2) ⇒ 1) is trivial. For 3) ⇒ 2) note that
$$\|T(v) - T(v')\| = \|T(v - v')\| \le \|T\| \cdot \|v - v'\|,$$
for all $v, v' \in V$, so we can choose $L := \|T\|$. To show 1) ⇒ 3) we will show the contraposition and assume that T is unbounded. Then there is a sequence $0 \ne v_n \in V$ such that $\|T(v_n)\|/\|v_n\| \to \infty$. After rescaling, we can assume that $\|v_n\| = 1$; so $\|T(v_n)\| \to \infty$. Thus, for a subsequence v_{n_k} we have $\|T(v_{n_k})\| > k$. Hence,
$$\frac{v_{n_k}}{k} \to 0 \qquad \text{but} \qquad \left\|T\left(\frac{v_{n_k}}{k}\right)\right\| > 1 \quad \text{for all } k,$$

which means that T is discontinuous at 0. $\square$

Proposition 2.6. *Denote by $\mathcal{L}(V, W)$ the space of all continuous linear operators between the normed spaces V and W. Then $\|T\|$ as defined in Proposition 2.5 is a norm on $\mathcal{L}(V, W)$. If W is a Banach space, then so is $\mathcal{L}(V, W)$.*

Proof. It is trivial to check that $\|\cdot\|$ is a norm on $\mathcal{L}(V, W)$. Let T_n be a Cauchy sequence. Then for each $v \in V$, we have

$$\|T_n(v) - T_m(v)\| \leq \|T_n - T_m\| \cdot \|v\|,$$

which implies that $T_n(v)$ is a Cauchy sequence in W, converging to a $T(v)$. It is easy to see that T point-wise defined this way is indeed a linear operator.

We still need to show that T is continuous and that T_n converges to T in norm. To this end, note that with $B_1(V) := \{v \in V \mid \|v\| \leq 1\}$ we can interpret each T_n as a function in $C^{0,b}(B_1(V), W)$, the space of bounded continuous W-valued functions on $B_1(V)$, which is complete with respect to $\|\cdot\|_\infty$. Since for any $S \in \mathcal{L}(V, W)$ we have

$$\|S\| = \sup_{0 \neq v \in V} \frac{\|S(v)\|}{\|v\|} = \sup_{v \in B_1(V)} \|S(v)\| = \|S\|_{B_1(V),\infty}, \tag{2.1}$$

we obtain from the completeness of $C^{0,b}(B_1(V), W)$ that T_n converges uniformly to a bounded continuous function on $B_1(V)$, which has to agree with T as $\|\cdot\|_\infty$-convergence is stronger than pointwise convergence. By linearity of T, we have that T is continuous on V, and by (2.1) we see that T_n converges to T in norm. $\square$

Definition 2.7 (Bounded from below)**.** *We call an operator $T \in \mathcal{L}(V, W)$* bounded from below *when*

$$\inf_{0 \neq v \in V} \frac{\|T(v)\|}{\|v\|} > 0.$$

Proposition 2.8. *Let $T \in \mathcal{L}(V, W)$ be bounded from below, then T is injective. If V is a Banach space, then $\operatorname{im}(T)$ is closed in W. In particular, when $\operatorname{im}(T) = W$, then T has a continuous inverse.*

Proof. Let $T(v_n)$ be a sequence in $\operatorname{im}(T)$ which converges in W. Thus, $T(v_n)$ is a Cauchy sequence. Since

$$\inf_{0 \neq v \in V} \frac{\|T(v)\|}{\|v\|} \leq \frac{\|T(v_n) - T(v_m)\|}{\|v_n - v_m\|},$$

we get

$$\|v_n - v_m\| \leq \left(\inf_{0 \neq v \in V} \frac{\|T(v)\|}{\|v\|} \right)^{-1} \|T(v_n) - T(v_m)\|.$$

Thus, v_n is Cauchy in V. By completeness of V, v_n converges to a $v \in V$. By continuity of T, we see that $T(v_n)$ converges to $T(v) \in \operatorname{im}(T)$.

The same argument shows that $T^{-1} : \operatorname{im}(T) \to V$ is continuous. □

Definition 2.9 (Dual space)**.** *For a normed spaces V, we define the* continuous dual space *of V as*

$$V' := \mathcal{L}(V, \mathbb{R}) = \left\{ \varphi : V \to \mathbb{R} \mid \varphi \text{ linear, } \|\varphi\| = \sup_{0 \neq v \in V} \frac{|\varphi(v)|}{\|v\|} < \infty \right\}.$$

Corollary 2.10. *For each normed space V, V' is a Banach space.*

2.2 Hilbert spaces

Definition 2.11 (Hilbert space)**.** *A vector space V endowed with an inner product $(\cdot, \cdot)$ is called* pre-Hilbert space*. If V is complete w.r.t. the norm $\|\cdot\| := (\cdot, \cdot)^{1/2}$, then V is called* Hilbert space*.*

Proposition 2.12 (Parallelogram identity)**.** *Let V be a pre-Hilbert space, then for all $v, v' \in V$:*

$$2\|v\|^2 + 2\|v'\|^2 = \|v + v'\|^2 + \|v - v'\|^2. \tag{2.2}$$

Theorem 2.13 (Hilbert projection theorem)**.** *Let V be a Hilbert space and $A \subset V$ a non-empty, closed and convex subset. Then there is a unique mapping $P_A : V \to A$ such that*

$$\|v - P_A(v)\| = \operatorname{dist}(v, A) := \inf_{a \in A} \|v - a\| \qquad \text{for all } v \in V.$$

Proof. W.l.o.g. we can assume that $a = 0$ (otherwise, replace A by $A - a$). Let $a_n \in A$ be a sequence such that $\|a_n\| = \|0 - a_n\| \overset{n\to\infty}{\to} \operatorname{dist}(0, A) =: d$. Then we have

$$\left\|\frac{a_n - a_m}{2}\right\|^2 + \underbrace{\left\|\underbrace{\frac{a_n + a_m}{2}}_{\in A}\right\|^2}_{\geq d^2} \overset{(2.2)}{=} \frac{1}{2}(\|a_n\|^2 + \|a_m\|^2) \;\to\; d^2 \qquad n, m \to \infty.$$

This is only possible when a_n is a Cauchy sequence. Since V is complete and A is closed, we obtain that $a_n \to a \in A$. By continuity of the norm it follows that $\|a\| = d$. This shows existence of a $P_A(a)$. To show that $P_A(a)$ is uniquely defined, let another $a' \in A$ with $\|a'\| = d$

be given. Then with the same argument we have:

$$\left\|\frac{a-a'}{2}\right\|^2 + \underbrace{\left\|\underbrace{\frac{a+a'}{2}}_{\in A}\right\|^2}_{\geq d^2} = \frac{1}{2}(\|a\|^2 + \|a'\|^2) = d^2,$$

so $a = a'$. □

Theorem 2.14 (Orthogonal projection, orthogonal complement)**.** *Let A be a closed linear subspace of a Hilbert space V. Then the mapping $P_A : V \to A$ from Theorem 2.13 is a continuous linear operator with*

$$\|P_A\| = \begin{cases} 0 & A = \{0\} \\ 1 & \textit{otherwise} \end{cases} \qquad \textit{and} \qquad P^2 = P.$$

Further, for each $v \in V$, $P_A(v)$ is the unique element of A such that

$$(v - P_A(v),\, a) = 0 \qquad \textit{for all } a \in A. \tag{2.3}$$

Denoting by

$$A^\perp := \{a' \in X \mid (a, a') = 0 \textit{ for all } a \in A\} = \ker P_A \tag{2.4}$$

the orthogonal complement of A in V, we have that $A^\perp$ is a closed subspace of V and

$$V = A \oplus A^\perp.$$

P_A is called the orthogonal projection onto A.

Proof. Assume that there is an $a \in A$, $\|a\| = 1$, such that $(v - P_A(v),\, a) \neq 0$. Then, for $a^* := P_A(v) + (v - P_A(v),\, a) \cdot a \in A$ we have

$$\begin{aligned}
\|v - a^*\|^2 &= \|(v - P_A(v)) - (v - P_A(v),\, a) \cdot a\|^2 \\
&= \|v - P_A(v)\|^2 - 2(v - P_A(v),\, a)^2 + (v - P_A(v),\, a)^2\|a\|^2 \\
&= \|v - P_A(v)\|^2 - (v - P_A(v),\, a)^2 \\
&< \|v - P_A(v)\|^2,
\end{aligned}$$

in contradiction to the optimality of $P_A(v)$.

Conversely, assume that for $a^* \in A$ we have

$$(v - a^*, a) = 0 \qquad \text{for all } a \in A.$$

Subtracting (2.3) gives

$$(\underbrace{P_A(v) - a^*}_{\in A}, a) = 0 \qquad \text{for all } a \in A.$$

Hence, $P_A(v) - a^* = 0$, showing that $P_A(v)$ is uniquely determined by (2.3).

Now, from (2.3) it immediately follows that P_A is linear and that $A^\perp = \ker P_A$. Further,

$$\|v\|^2 = \| \underbrace{(v - P_A(v))}_{\in A^\perp} + \underbrace{P_A(v)}_{\in A} \|^2 \overset{(2.3)}{=} \|v - P_A(v)\|^2 + \|P_A(v)\|^2 \geq \|P_A(v)\|^2,$$

so, $\|P_A\| \leq 1$. Since P_A is the identity on A, $\|P_A\| = 1$ and $P_A^2 = P_A$ follow.

Finally, $A^\perp$ is closed since P_A is continuous, and we can uniquely decompose any $v \in V$ as $v = P_A(v) + (I - P_A)(v)$, so $V = A \oplus A^\perp$. □

Theorem 2.15 (Riesz representation theorem)**.** *Let V be a Hilbert space and let $f \in V'$. Then there is a unique $v_f \in V$ such that*

$$f(v) = (v, v_f) \qquad \textit{for all } v \in V. \tag{2.5}$$

The mapping $\Phi : V \to V'$, $\Phi(v) := (\cdot, v)$ is an isometric isomorphism.

Proof. Let $0 \neq f \in V'$ be given. Then $A := \ker f$ is a closed linear subspace of V, and by Theorem 2.14 we have $V = A \oplus A^\perp$. Since $\dim(\operatorname{im} f) = 1$, we also have $\dim A^\perp = 1$, so there is a $\bar{v}_f$ s.t. $A^\perp = \operatorname{span}\{\bar{v}_f\}$, and we let

$$v_f := \frac{f(\bar{v}_f)}{\|\bar{v}_f\|^2} \bar{v}_f.$$

Then for arbitrary $v = v_A + \lambda \bar{v}_f \in V$ we have

$$(v, v_f) = (v_A, v_f) + \lambda(\bar{v}_f, v_f) = \lambda f(\bar{v}_f) = f(v).$$

This shows existence of a v_f as in (2.5). Further, v_f is unique as for a second v_f^* also satisfying (2.5) we have

$$\|v_f - v_f^*\|^2 = (v_f - v_f^*, v_f) - (v_f - v_f^*, v_f^*) = f(v_f - v_f^*) - f(v_f - v_f^*) = 0.$$

Note that the existence of v_f implies the surjectivity of Φ. It is clear that Φ is linear. Finally, for any $v \in V$ we have

$$\|\Phi(v)\| = \sup_{0 \neq v' \in V} \frac{(v', v)}{\|v'\|} \overset{\text{C.S.}}{\leq} \sup_{0 \neq v' \in V} \frac{\|v'\| \cdot \|v\|}{\|v'\|} = \|v\|.$$

On the other hand, $\Phi(v)[v] = (v, v) = \|v\|^2$, so $\|\Phi(v)\| \geq \|v\|$. This shows that Φ is an isometry. □

Second proof. We give a second proof for the existence of v_f using the principle of energy minimization. For given f define the functional $E : V \to \mathbb{R}$ via

$$E(v) := \frac{1}{2}\|v\|^2 - f(v).$$

Let v_f be a minimizer of $E(v)$,

$$v_f := \arg\min_{v \in V} E(v).$$

Then for arbitrary $v \in V$ consider the function

$$\begin{aligned} g_v(\varepsilon) &:= E(v_f + \varepsilon v) \\ &= \tfrac{1}{2}\|v_f + \varepsilon v\|^2 - f(v_f + \varepsilon v) \\ &= \tfrac{1}{2}\|v_f\| + \varepsilon(v_f, v) + \tfrac{1}{2}\varepsilon^2\|v_f\|^2 - f(v_f) - \varepsilon f(v). \end{aligned}$$

Since v_f is a minimizer of E we obtain:

$$0 = g_v'(\varepsilon)\Big|_{\varepsilon=0} = \left[(v_f, v) + \varepsilon\|v_f\|^2 - f(v)\right]\Big|_{\varepsilon=0} = (v_f, v) - f(v).$$

Hence, v_f satisfies (2.5).

It remains to show that E indeed has a minimizer. To this end, first note that E is bounded from below due to

$$E(v) \geq \tfrac{1}{2}\|v\|^2 - \underbrace{\|f\| \cdot \|v\|}_{\leq \frac{1}{2}\|f\|^2 + \frac{1}{2}\|v\|^2} \geq -\tfrac{1}{2}\|f\|^2 > -\infty$$

Here we have used Young's inequality (Theorem 2.16).

Hence, there exists a minimizing sequence $v_n \in V$, such that

$$\lim_{n\to\infty} E(v_n) = \inf_{v \in V} E(v) > -\infty.$$

Using the parallelogram identity (2.2), we see that v_n is a Cauchy sequence:

$$\begin{aligned} \left\|\frac{v_m - v_n}{2}\right\|^2 &= \tfrac{1}{2}\|v_m\|^2 + \tfrac{1}{2}\|v_n\|^2 - \left\|\frac{v_m + v_n}{2}\right\|^2 \\ &= E(v_m) + E(v_n) + \underbrace{f(v_m) + f(v_n) - \left\|\frac{v_m + v_n}{2}\right\|^2}_{= -2\underbrace{E(\frac{v_m+v_n}{2})}_{\geq \inf_{v\in V} E(x)}} \\ &\leq E(v_m) + E(v_n) - 2\inf_{v \in V} E(v) \\ &\xrightarrow{m,n\to\infty} 0. \end{aligned}$$

By completeness of V we obtain that v_n converges to a $v \in V$, which, by continuity of E, is a minimizer of E. □

Theorem 2.16 (Young's inequality)**.** *Let $a, b \geq 0$ and $p, q > 1$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then:*

$$ab \leq \tfrac{1}{p} a^p + \tfrac{1}{q} b^q. \tag{2.6}$$

In particular, for $p = q = 2$ and $\varepsilon > 0$:

$$ab \leq \tfrac{\varepsilon}{2} a^2 + \tfrac{1}{2\varepsilon} b^2. \tag{2.7}$$

Proof. Let $f : \mathbb{R}^+ \to \mathbb{R}^+$ be a continuous, strictly monotone function, then

$$ab \leq \int_0^a f(x)\,\mathrm{d}x + \int_0^b f^{-1}(y)\,\mathrm{d}y. \tag{2.8}$$

Choosing $f(x) = x^{p-1}$ in (2.8) yields (2.6) since $(p-1)(q-1) = 1$, so $f^{-1}(y) = y^{q-1}$. (2.8) can be interpreted geometrically as the statement that each point in $[0, a] \times [0, b]$ lies either below the graph of $f(x)$ in the xy-plane or above. In the latter case, the point lies below the graph of f^{-1} in the yx-plane.

(2.7) follows from (2.8) by substituting $\sqrt{\varepsilon} a$, $1/\sqrt{\varepsilon} b$ for a and b. □

2.3 Bilinear forms and operators

Definition 2.17. *Let V, W be normed spaces. Then we denote by* $\operatorname{Bil}(V, W)$ *the space of continuous bilinear forms $V \times W \to \mathbb{R}$.*

Proposition 2.18. *A bilinear form b on normed spaces V, W is continuous iff*

$$\|b\| := \sup_{0 \neq v \in V} \sup_{0 \neq w \in W} \frac{b(v, w)}{\|v\| \cdot \|w\|} < \infty. \tag{2.9}$$

(2.9) defines a norm on $\operatorname{Bil}(V, W)$*.*

Proof. Exercise. □

Proposition 2.19. *Let V, W be normed spaces. Then*

$$\begin{aligned} \Phi : \operatorname{Bil}(V, W) &\to \mathcal{L}(V, W') \\ b &\mapsto [v \mapsto b(v, \cdot)] \end{aligned} \tag{2.10}$$

is an isometric isomorphism between $\operatorname{Bil}(V, W)$ *and* $\mathcal{L}(V, W')$*. In particular,* $\operatorname{Bil}(V, W)$ *is a Banach space.*

Proof. Exercise. □

Corollary 2.20. *Let V be a normed space and let W be a Hilbert space. Then*

$$\begin{aligned}\Psi\,\mathrm{Bil}(V,W) &\to \mathcal{L}(V,W)\\ b &\mapsto [v \mapsto w_{b(v,\cdot)}]\end{aligned}$$

is an isometric isomorphism. Here, $w_{b(v,\cdot)}$ denotes the Riesz representative of the linear functional $b(v,\cdot)$ in W. The inverse of Ψ is given by:

$$\begin{aligned}\Psi^{-1}: \mathcal{L}(V,W) &\to \mathrm{Bil}(V,W)\\ T &\mapsto [(v,w) \mapsto (T(v),w)].\end{aligned}$$

Definition 2.21. *A bilinear form $b : V \times W \to \mathbb{R}$ on normed spaces V, W is called* inf-sup *stable iff*

$$\inf_{0 \neq v \in V} \sup_{0 \neq w \in W} \frac{b(v,w)}{\|v\| \cdot \|w\|} > 0.$$

Proposition 2.22. *A bilinear form $b \in \mathrm{Bil}(V,W)$ is inf-sup stable iff the corresponding operator in $\mathcal{L}(V,W')$ (or $\mathcal{L}(V,W)$ when W is a Hilbert space) is bounded from below.*

Proof. Exercise. □

2.4 L^p-spaces

Definition and Proposition 2.23 (L^p-spaces)**.** *Let $X \subset \mathbb{R}^d$ be measurable. For a measurable function $u : X \to \mathbb{R}$ and $p \in [1,\infty)$, let*

$$\|u\|_p := \|u\|_{L^p(X)} := \Big(\int_X |u|^p \,\mathrm{d}x\Big)^{1/p},$$

where $\mathrm{d}x$ denotes integration w.r.t. the Lebesgue measure restricted to X. Further, let

$$\|u\|_\infty := \|u\|_{L^\infty(X)} := \operatorname*{ess\,sup}_{x \in X} |u(x)| := \inf_{\substack{N \subset X\\ \lambda(N)=0}} \sup_{x \in X \setminus N} |u(x)|,$$

with λ denoting the Lebesgue measure. Then for $p \in [1,\infty]$, $\|\cdot\|_p$ is a norm on the quotient space

$$L^p(X) := \{u : X \to \mathbb{R} \text{ measurable} \,|\, \|u\|_p < \infty\} \,/\, \{u \,|\, u(x) = 0 \;\; \lambda\text{-a.e.}\},$$

making it a Banach space. As usual, we identify individual functions on X with their equivalence classes in $L^p(X)$.

The norm on $L^2(X)$ is induced by an inner product

$$(u,v)_2 := (u,v)_{L^2(X)} := \int_X u \cdot v \,\mathrm{d}x,$$

making it a Hilbert space.

Finally, let

$$L^1_{\mathrm{loc}}(X) := \{u : X \to \mathbb{R} \text{ measurable} \mid u \in L^1(K)\ \forall K \subset X \text{ compact}\}$$

be the space of locally integrable functions on X.

Theorem 2.24 (Hölder inequality)**.** *Let $u, v : X \to \mathbb{R}$ be measurable and $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then*

$$\|uv\|_1 \le \|u\|_p \|v\|_q. \tag{2.11}$$

In particular, if $u \in L^p(X)$ and $v \in L^q(X)$, then $uv \in L^1(X)$. In case $p = q = 2$, the Hölder inequality becomes the Cauchy-Schwarz inequality for the $(\cdot,\cdot)_2$-inner product.

Theorem 2.25. *Let $p \in [1, \infty)$ and let $q \in [1, \infty]$ s.t. $\frac{1}{p} + \frac{1}{q} = 1$. Then $L^p(X)' = L^q(X)$ via the isomorphism*

$$\begin{aligned} L^q(X) &\to (L^p(X))' \\ u &\mapsto \int_X (\cdot) \cdot u \, \mathrm{d}x. \end{aligned}$$

2.5 Sobolev spaces

Definition 2.26 (Space of test functions)**.** *Let $\Omega \subset \mathbb{R}^d$ be an open non-empty set. Then we denote by $C_0^\infty(\Omega) \subset C^\infty(\Omega)$ the space of smooth functions on Ω with compact support.*

$C_0^\infty(\Omega)$ is neither a normed space nor a Fréchet space, but it can be given an LF-space topology as the inductive limit of the spaces $C^\infty(K)$, $K \subset \Omega$ of smooth functions on Ω with support in K. With respect to this topology, a sequence of functions $u_n \in C_0^\infty(\Omega)$ converges iff all but finitely many u_n are supported on the same compact set K and all partial derivatives of the u_n converge uniformly.[1] A distribution Λ is then nothing but a continuous linear functional on $C_0^\infty(\Omega)$. In particular, every function in $u \in L^1_{\mathrm{loc}}(\Omega)$ can be identified as a distribution via

$$u[\varphi] := \int_\Omega u\varphi \, \mathrm{d}x \qquad \forall \varphi \in C_0^\infty(\Omega).$$

Assuming that u is sufficiently smooth we then have for any multi-index $\alpha \in \mathbb{N}_0^d$:

$$\begin{aligned} (D^\alpha u)[\varphi] &= \int_\Omega D^\alpha u\varphi \, \mathrm{d}x \\ &= (-1)^{|\alpha|} \int_\Omega u D^\alpha \varphi \, \mathrm{d}x = (-1)^{|\alpha|} u \left[D^\alpha \varphi\right]. \end{aligned}$$

[1] However, $C_0^\infty(\Omega)$ is not a sequential space, i.e. its topology cannot be fully described by convergence of sequences.

The latter expression makes sense for arbitrary distributions $\Lambda \in C_0^\infty(\Omega)'$, so we define:

$$(D^\alpha \Lambda)[\varphi] := (-1)^{|\alpha|}\Lambda[D^\alpha \varphi]. \tag{2.12}$$

Note that not every distribution is a function. The most prominent example is the family of Dirac distributions $\delta_x \in C_0^\infty(\Omega)'$, $x \in \Omega$ given by

$$\delta_x[\varphi] := \varphi(x).$$

Note that $D^\alpha \delta_x$ is a distribution which cannot be written as integration w.r.t. a measure.

In the following we will not need the theory of distributions. However, we will work a lot with weakly differentiable functions, i.e., functions whose distributional derivatives (2.12) are again represented by functions:

Definition 2.27 (Weak derivative)**.** *Let $\alpha \in \mathbb{N}_0^d$ be a multi-index and let $\Omega \subset \mathbb{R}^d$ an open, non-empty set. A function $u \in L^1_{\mathrm{loc}}(\Omega)$ is said to have an α-th* weak derivative *if there is a function $v_\alpha \in L^1_{\mathrm{loc}}(\Omega)$ s.t.*

$$\int_\Omega u D^\alpha \varphi = (-1)^{|\alpha|} \int_\Omega v_\alpha \varphi \qquad \forall \varphi \in C_0^\infty(\Omega). \tag{2.13}$$

We write $v_\alpha = D^\alpha u$.

Example 2.28. *Let $\Omega = (-1,1)$ and $u(x) = |x|$. Then $u'(x) := \operatorname{sign}(x)$ is the first weak derivative of u. u does not have a second weak derivative.*

Proof. Exercise. □

Definition and Proposition 2.29 (Sobloev spaces)**.** *Let $m \in \mathbb{N}_0$, $p \in [1,\infty]$, then we define the Sobolev spaces*

$$H^{m,p}(\Omega) := \{u \in L^1_{\mathrm{loc}}(\Omega) \mid D^\alpha u \in L^p(\Omega)\ \forall |\alpha| \le m\}.$$

Each Sobolev space $H^{m,p}(\Omega)$ is a Banach space with the norm

$$\|u\|_{H^{m,p}(\Omega)} := \Big(\sum_{|\alpha| \le m} \|D^\alpha u\|^p_{L^p(\Omega)}\Big)^{1/p},$$

if $1 \le p < \infty$ and

$$\|u\|_{H^{m,\infty}(\Omega)} := \max_{|\alpha| \le m} \|D^\alpha u\|_{L^\infty(\Omega)}$$

if $p = \infty$, respectively.

For $p = 2$ we have that $H^{m,2}(\Omega)$ is a Hilbert space with inner product

$$(u, v)_{H^m(\Omega)} := \sum_{|\alpha| \leq m} (D^\alpha u, D^\alpha v)_{L^2(\Omega)}.$$

We abbreviate $H^m(\Omega) := H^{m,2}(\Omega)$

Finally, we introduce the Sobolev semi-norms

$$|u|_{H^{m,p}(\Omega)} := \Big(\sum_{|\alpha| = m} \|D^\alpha u\|^p_{L^p(\Omega)} \Big)^{1/p}$$

for $1 \leq p < \infty$ and

$$|u|_{H^{m,\infty}(\Omega)} := \max_{|\alpha| = m} \|D^\alpha u\|_{L^\infty(\Omega)}.$$

Remark 2.30. *In the literature, one often finds the notation $W^{m,p}(\Omega)$ instead of $H^{m,p}(\Omega)$.*

Examples 2.31. *1) Let $\Omega := B_{1/2}(\mathbb{R}^2)$ and $u(x) := \log|\log|x||$. We have $u \in H^1(\Omega)$, but $u \notin C^0(\Omega)$. I.e., Sobolev functions in higher spatial dimensions are not necessarily continuous.*

2) Let $\Omega := B_1(\mathbb{R}^d)$ and $u(x) = |x|^s$, $s \in \mathbb{R}$. Then we have $u \in H^{1,p}(\Omega)$ if $s > 1 - \frac{d}{p}$.

Proof. Exercise. □

Sobolev spaces can be seen as complete L^p-analogs of the spaces $C^m(\Omega)$. Indeed, we will show that $H^{m,p}(\Omega) \cap C^m(\Omega)$ is dense in $H^{m,p}(\Omega)$. For this, we need an important technical tool:

Definition 2.32 (Dirac sequence)**.** *We call a sequence of functions $\varphi_k \in C_0^\infty(\mathbb{R}^d)$ a* Dirac sequence *if*

$$\varphi_k \geq 0, \qquad \int_{\mathbb{R}^d} \varphi_k \,\mathrm{d}x = 1 \qquad \forall k$$

and

$$\lim_{k \to \infty} \int_{\mathbb{R}^n \setminus B_\delta(0)} \varphi_k \,\mathrm{d}x = 0 \qquad \forall \delta > 0.$$

If $\varphi \in C_0^\infty(B_1(0))$ with $\varphi \geq 0$ and $\int_{\mathbb{R}^d} \varphi \,\mathrm{d}x = 1$, then a Dirac sequence is obtained by setting

$$\varphi_k(x) := (1/k)^{-d} \varphi\left(\frac{x}{1/k}\right).$$

Such a sequence is called standard Dirac sequence*. The functions φ_k are also called* mollifiers.

Theorem 2.33. *Let φ_k be a Dirac sequence, $p \in [1, \infty]$ and $u \in L^p(\mathbb{R}^d)$. We define the convolution of u with φ_k as the function*

$$(u * \varphi_k)(x) := \int_{\mathbb{R}^d} u(y)\varphi_k(x-y)\,\mathrm{d}y.$$

Then the following statements hold:

1) *For all k, $(u * \varphi_k) \in L^p(\mathbb{R}^d)$ and $\|u * \varphi_k\|_p \leq \|\varphi_k\|_1 \cdot \|u\|_p$.*

2) *For all k, $(u * \varphi_k) \in C^\infty(\mathbb{R}^d)$ and $D^\alpha(u * \varphi_k) = (u * D^\alpha \varphi_k)$.*

3) *If $p < \infty$, then $(u * \varphi_k) \overset{k\to\infty}{\longrightarrow} u \quad$ in $L^p(\mathbb{R}^d)$.*

Proof of 1). Let $p < \infty$, then

$$\begin{aligned}
\|u * \varphi_k\|_p^p &= \int_{\mathbb{R}^d} \left| \int_{\mathbb{R}^d} u(y)\varphi_k(x-y)\,\mathrm{d}y \right|^p \mathrm{d}x \\
&= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} \left| u(y)\varphi_k(x-y)^{1/p}\varphi_k(x-y)^{1/q} \right| \mathrm{d}y \right)^p \mathrm{d}x \\
&\overset{(2.11)}{\leq} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |u(y)|^p |\varphi_k(x-y)| \,\mathrm{d}y \cdot \left(\int_{\mathbb{R}^d} |\varphi_k(x-y)| \,\mathrm{d}y \right)^{p/q} \mathrm{d}x \\
&= \|\varphi_k\|_1^{p/q} \cdot \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |u(y)|^p |\varphi_k(x-y)| \,\mathrm{d}y\,\mathrm{d}x \\
&\overset{\text{Fubini}}{=} \|\varphi_k\|_1^{p/q} \cdot \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |u(y)|^p |\varphi_k(x-y)| \,\mathrm{d}x\,\mathrm{d}y \\
&= \|\varphi_k\|_1^{p/q+1} \cdot \|u\|_p^p \\
&= \|\varphi_k\|_1^p \cdot \|u\|_p^p,
\end{aligned}$$

where the last equality holds due to $p/q + 1 = p(1/q + 1/p) = p$. The case $p = \infty$ is easier. $\square$

Proof of 2). To show that $(u * \varphi_k)$ is continuous note that

$$\begin{aligned}
|(u * \varphi_k)(x) - (u * \varphi_k)(x')| &\leq \int_{\mathbb{R}^d} |u(y)||(\varphi_k(x-y) - \varphi_k(x'-y)|\,\mathrm{d}y \\
&\overset{(2.11)}{\leq} \|u\|_p \left(\int_{\mathbb{R}^d} |\varphi_k(x-y) - \varphi_k(x'-y)|^q \,\mathrm{d}y \right)^{1/q}
\end{aligned}$$

Since $\varphi_k \in C_0^\infty(X)$, for $x' \to x$ the functions $\varphi_k(x - \cdot) - \varphi_k(x' - \cdot)$ are uniformly bounded by an integrable function and pointwise convergent to zero. By the dominated convergence theorem we get that integral on the right-hand side to zero as well, showing continuity of $(u * \varphi_k)$ in x.

Similarly, we have

$$\left|\frac{1}{h}\Big((u*\varphi_k)(x+he_i)-(u*\varphi_k)(x)\Big)-(u*\partial_i\varphi_k)(x)\right|$$
$$\leq \|u\|_p\left(\int_{R^d}\left|\frac{1}{h}\Big(\varphi_k(x+he_i-y)-\varphi_k(x-y)\Big)-\partial_i\varphi_k(x-y)\right|^q\mathrm{d}y\right)^{1/q}\xrightarrow{h\to 0}0,$$

showing that $\partial_i(u*\varphi_k)$ exists and equals $(u*\partial_i\varphi_k)$. The claim now follows by induction. □

Proof of 3). Since $\int_{\mathbb{R}^d}\varphi_k\,\mathrm{d}x=1$ we can write

$$(u*\varphi_k-u)(x)=\int_{\mathbb{R}^d}(u(y)-u(x))\varphi_k(x-y)\,\mathrm{d}y$$

Further, choose an arbitrary $\delta>0$ and decompose φ_k as $\varphi_k\cdot\mathbb{1}_{B_\delta(\mathbb{R}^d)}+\varphi_k\cdot\mathbb{1}_{\mathbb{R}^d\setminus B_\delta(\mathbb{R}^d)}=:\varphi_k^\delta+\check{\varphi}_k^\delta$. Then

$$\|u*\varphi_k-u\|_p\leq\left\|\int_{\mathbb{R}^d}(u(y)-u(x))\varphi_k^\delta(x-y)\,\mathrm{d}y\right\|_p+\left\|\int_{\mathbb{R}^d}(u(y)-u(x))\check{\varphi}_k^\delta(x-y)\,\mathrm{d}y\right\|_p.\tag{2.14}$$

Using the same Hölder-inequality argument as in the proof of 1) we get for the first norm:

$$\begin{aligned}\left\|\int_{\mathbb{R}^d}(u(y)-u(x))\varphi_k^d(x-y)\,\mathrm{d}y\right\|_p^p&\leq\left\|\varphi_k^\delta\right\|_1^{p/q}\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}|u(y)-u(x)|^p\varphi_k(x-y)\,\mathrm{d}y\,\mathrm{d}x\\&\leq\left\|\varphi_k^\delta\right\|_1^{p/q}\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}|u(x+h)-u(x)|^p\varphi_k(-h)\,\mathrm{d}h\,\mathrm{d}x\\&\leq\left\|\varphi_k^\delta\right\|_1^{p/q}\int_{\mathbb{R}^d}\|u(x+h)-u(x)\|_p^p\varphi_k(-h)\,\mathrm{d}h\\&\leq\left\|\varphi_k^\delta\right\|_1^p\cdot\sup_{|h|<\delta}\|u(x+h)-u(x)\|_p^p.\end{aligned}$$

Here, the supremum only goes goes over $|h|<\delta$ since φ_k^δ is supported on $B_\delta(\mathbb{R}^d)$. Now for $p<\infty$ one can show that for all $u\in L^p(\mathbb{R}^d)$ one has

$$\lim_{h\to 0}\|u(\cdot+h)-u(\cdot)\|_p=0$$

(exercise). Hence, the supremum over $|h|<\delta$ gets arbitrarily small for δ small enough. For the second norm in (2.14) we get with the same argument

$$\left\|\int_{\mathbb{R}^d}(u(y)-u(x))\check{\varphi}_k^d(x-y)\,\mathrm{d}y\right\|_p^p\leq\left\|\check{\varphi}_k^\delta\right\|_1^p\cdot\sup_{h\in\mathbb{R}^d}\|u(x+h)-u(x)\|_p^p.$$

Here we can use that by definition of a Dirac sequence, $\left\|\check{\varphi}_k^\delta\right\|_1$ tends to zero for $k\to\infty$. □

Theorem 2.34. *Let $\Omega\subset\mathbb{R}^d$ be an open non-empty set and $p\in[1,\infty)$. Then $C^\infty(\Omega)\cap H^{m,p}(\Omega)$ is dense $H^{m,p}(\Omega)$.*

Sketch of proof. For $\Omega = \mathbb{R}^d$ and $u \in H^{m,p}(\mathbb{R}^d)$ we have from Theorem 2.33 that

$$C^\infty(\mathbb{R}^d) \cap L^p(\mathbb{R}^d) \ni (u * \varphi_k) \xrightarrow{\|\cdot\|_p} u$$

for some Dirac sequence φ_k. Further we have

$$\begin{aligned} D^\alpha(u * \varphi_k)(x) &\overset{2.33}{=} (u * D^\alpha \varphi_k)(x) \\ &= \int_{R^d} u(y)(D^\alpha \varphi_k)(x-y)\,\mathrm{d}y \\ &= (-1)^{|\alpha|} \int_{\mathbb{R}^d} u(y) D^\alpha(\varphi_k(x-\cdot))(y)\,\mathrm{d}y \\ &\overset{(2.13)}{=} \int_{\mathbb{R}^d} D^\alpha u(y)\varphi_k(x-y)\,\mathrm{d}y \\ &= ((D^\alpha u) * \varphi_k)(x). \end{aligned}$$

Thus, again by Theorem 2.33, $D^\alpha(u * \varphi_k)(x) \in L^p(\mathbb{R}^d)$ and $D^\alpha(u * \varphi_k) \longrightarrow D^\alpha u$ in $L^p(\mathbb{R}^d)$, showing that $(u * \varphi_k)$ converges to u in $H^{m,p}(\mathbb{R}^d)$.

For $\Omega \neq \mathbb{R}^d$ assume that $u \in H^{m,p}(\Omega)$ has compact support $K \subset \Omega$. Since, in this case, $\operatorname{dist}(K, \partial\Omega) > 0$, we can choose φ_k as a standard Dirac sequence such that for k large enough we have that the support of $(u * \varphi_k)$ is contained in Ω. Then the same arguments as above can be applied to show that $(u * \varphi_k)$ converges to u in $H^{m,p}(\Omega)$.

For a function in $H^{m,p}(\Omega)$ with non-compact support we can employ a partition-of-unity argument to decompose u as an infinite sum of compactly supported functions to which the previous arguments can be applied individually ([Alt16, Theorem 4.24]). □

The denseness of $C^\infty(\Omega)$ in $H^{m,p}(\Omega)$ can be used in many cases to extend statements about smooth functions to Sobolev functions. For example:

Proposition 2.35 (Product rule for Sobolev functions)**.** *Let $\Omega \subset \mathbb{R}^d$ be open and non-empty, $p, q \in [1, \infty]$ s.t. $1/p + 1/q = 1$ and $u \in H^{m,p}(\Omega)$, $v \in H^{m,q}(\Omega)$. Then $u \cdot v \in H^{m,1}(\Omega)$ and for a multi-index α with $|\alpha| \leq m$ we have*

$$D^\alpha(u \cdot v) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta u \cdot D^{\alpha-\beta} v.$$

Proof. Exercise. □

2.5.1 Boundary values of Sobloev functions

When discussing boundary value problems, we need to consider the restricution of Sobolev functions to $\partial\Omega$. However, as L^p-functions, Sobolev functions are a priori only defined up to sets of measure zero, such as $\partial\Omega$. Ideed, we have seen in Examples 2.31 that Sobolev functions can have poles, so point-wise evaluation of Sobolev functions does not make any sense in general. The trace theorem shows, that it is still meaningful to speak of boundary values of a Sobloev function in an L^p-sense:

Theorem 2.36 (Trace theorem)**.** *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain and $1 \leq p \leq \infty$. Then there is a unique continuous linear Operator $S : H^{1,p}(\Omega) \to L^p(\partial\Omega)$, such that*

$$S(u) = u|_{\partial\Omega} \qquad \forall u \in H^{1,p}(\Omega) \cap C^0(\bar{\Omega}).$$

We call $S(u)$ the trace *or* weak boundary values *of u. When the meaning is clear, we will usually write u instead of $S(u)$.*

Proof. See [Alt16, A8.6]. □

In particular, we will often work with functions with zero boundary values:

Definition and Proposition 2.37 (Sobolev functions with zero boundary values)**.** *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain. For $1 \leq p < \infty$ and $m \in \mathbb{N}$ we define*

$$H_0^{m,p}(\Omega) := \overline{C_0^\infty(\Omega)}^{||\cdot||_{H^{m,p}(\Omega)}} = \{u \in H^{m,p}(\Omega) \mid Su = 0\}.$$

$H_0^{m,p}(\Omega)$ is a Banach space. $H_0^{m,2}(\Omega)$ is a Hilbert space.

Proof. By continuity of S it is clear that $\overline{C_0^\infty(\Omega)}^{||\cdot||_{H^{m,p}(\Omega)}} \subseteq \{u \in H^{m,p}(\Omega) \mid Su = 0\}$. For the converse direction see [Alt16, A8.10]. By construction, $H_0^{m,p}(\Omega)$ is a closed subspace of $H^{m,p}(\Omega)$, so it inherits its completeness. □

We will often work with the dual of $H_0^{m,p}(\Omega)$ for which we introduce the following commonly used notation:

Definition 2.38 (Dual space of $H_0^{m,p}(\Omega)$)**.** *Let $p, q \in [1, \infty]$ s.t. $1/p + 1/q = 1$. Then we denote by*

$$H^{-m,q}(\Omega) := \left(H_0^{m,p}(\Omega)\right)'$$

the continous dual space of $H_0^{m,p}(\Omega)$. We write $H^{-m}(\Omega)$ for $H^{-m,2}(\Omega)$.

Note that with this notation we have that the distributional derivatives ∂_i uniquely extend to continuous mappings $\partial_i : L^q(\Omega) = H^{0,q}(\Omega) \to H^{-1,q}(\Omega)$, since the test functions $C_0^\infty(\Omega)$ are dense in $H_0^{1,p}(\Omega)$. In general, there is not a unique extension to $H^{1,p}(\Omega)$.

Using the trace theorem, we can also extend Gauß's theorem to Sobolev functions:

Theorem 2.39 (Gauß's theorem for Sobolev functions). *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain, $n \in \mathbb{R}^d$ the unit outer normal to Ω.*

1) Let $v \in H^{1,1}(\Omega)$, then:

$$\int_\Omega \partial_i v \,\mathrm{d}x = \int_{\partial\Omega} v n_i \,\mathrm{d}\sigma \qquad i = 1, \ldots, d.$$

2) Let $v \in (H^{1,1}(\Omega))^d$, then:

$$\int_\Omega \mathrm{div}(v) \,\mathrm{d}x = \int_{\partial\Omega} v \cdot n \,\mathrm{d}\sigma.$$

3) Let $p, q \in [1, \infty]$ s.t. $1/p + 1/q = 1$, and let $u \in H^{1,p}(\Omega)$, $v \in H^{1,q}(\Omega)$. Then:

$$\int_\Omega \partial_i u \cdot v + u \cdot \partial_i v \,\mathrm{d}x = \int_{\partial\Omega} u v n_i \,\mathrm{d}\sigma \qquad i = 1, \ldots, d.$$

Proof. Exercise. □

2.5.2 Embedding theorems

Having L^p-integrable weak derivatives imposes additional local regularity constraints on functions in $H^{m,p}(\Omega)$ compared to functions in $L^p(\Omega)$. Some consequences are expressed in the Sobolev embedding theorems:

Theorem 2.40 (1. Sobolev embedding theorem). *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain, $m_1, m_2 \in \mathbb{N}_0$ and $p_1, p_2 \in [1, \infty)$.*

1) If

$$m_1 - \frac{d}{p_1} \geq m_2 - \frac{d}{p_2} \qquad \text{and} \qquad m_1 \geq m_2$$

then the identity defines a continuous embedding

$$H^{m_1,p_1}(\Omega) \hookrightarrow H^{m_2,p_2}(\Omega).$$

2) If

$$m_1 - \frac{d}{p_1} > m_2 - \frac{d}{p_2} \qquad \text{and} \qquad m_1 > m_2,$$

then the embedding

$$H^{m_1,p_1}(\Omega) \hookrightarrow H^{m_2,p_2}(\Omega).$$

is compact, i.e., bounded sequences in H^{m_1,p_1} have converging subsequences in $H^{m_2,p_2}(\Omega)$.

Proof. See [Alt16, p. 10.9]. □

Theorem 2.41 (2. Sobolev embedding theorem)**.** *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain, $m_1, m_2 \in \mathbb{N}_0$, $p \in [1, \infty)$ and $\alpha \in [0, 1]$.*

1) If

$$m_1 - \frac{d}{p} = m_2 + \alpha \qquad \text{and} \qquad 0 < \alpha < 1,$$

then the identity defines a continuous embedding

$$H^{m_1,p}(\Omega) \hookrightarrow C^{m_2,\alpha}(\bar{\Omega}).$$

2) If

$$m_1 - \frac{d}{p} > m_2 + \alpha,$$

then the embedding

$$H^{m_1,p}(\Omega) \hookrightarrow C^{m_2,\alpha}(\bar{\Omega}).$$

is compact.

Proof. See [Alt16, p. 10.13]. □

Example 2.42. *In one spatial dimension ($d = 1$) we have $1 - \frac{1}{2} > 0$, so functions in $H^1(\Omega)$ are continuous by Theorem 2.41 . For $d > 1$ we have $1 - \frac{d}{2} \leq 0$, so Theorem 2.41 cannot be applied.*

2.5.3 Poincaré inequality

Finally, we will show the Poincaré inequality, which is a quantitative version of the statement a functions with vanishing derivatives and zero boundary values must be constant zero.

Theorem 2.43 (Poincaré inequality)**.** *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain with diameter $D := \operatorname{diam}(\Omega)$ and let $p \in [1, \infty)$. Then there is a constant $c_p \leq D$, such that for all $v \in H_0^{1,p}(\Omega)$ we have*

$$\|v\|_{L^p(\Omega)} \leq c_p \|\nabla v\|_{L^p(\Omega)} = c_p |v|_{1,p} \tag{2.15}$$

In particular, $|\cdot|_{1,p}$ is in fact a norm on $H_0^{1,p}(\Omega)$, which is equivalent to $\|\cdot\|_{1,p}$.

Proof. First we note that it is sufficient to show (2.15) for $v \in C_0^\infty(\Omega)$. Indeed, by density of $C_0^\infty(\Omega)$ in $H_0^{1,p}(\Omega)$, we have for each $v \in H_0^{1,p}(\Omega)$ a sequence $v_n \in C_0^\infty(\Omega)$ converging to v in $\|\cdot\|_{1,p}$. Then:

$$\begin{aligned}
\|v\|_p &\le \|v_n\|_p + \|v_n - v\|_p \\
&\le c_p|v_n|_{1,p} + \|v_n - v\|_p \\
&\le c_p|v|_{1,p} + c_p|v_n - v|_{1,p} + \|v_m - v\|_p \\
&\to c_p|v|_{1,p} \qquad \text{for } m \to \infty.
\end{aligned}$$

So let us now assume $v \in C_0^\infty(\Omega)$ and extend v to $\mathbb{R}^d$ by the function

$$\tilde{v}(x) := \begin{cases} v(x) & x \in \Omega \\ 0 & \text{otherwise.} \end{cases}$$

Further, w.l.o.g. we can assume that $\Omega \subset [0, D]^d$. Using the fundamental theorem of calculus we then have:

$$\tilde{v}(x) = \int_0^{x_1} \partial_{x_1}\tilde{v}(s, x_2, \dots, x_d)\,\mathrm{d}s \qquad \forall x \in \mathbb{R}^d.$$

Thus, with q s.t. $1/p + 1/q = 1$:

$$\begin{aligned}
|\tilde{v}(x)|^p &= \Big|\int_0^{x_1} \partial_{x_1}\tilde{v}(s, x_2, \dots, x_d)\,\mathrm{d}s\Big|^p \\
&\le \Big(\int_0^D |\partial_{x_1}\tilde{v}(s, x_2, \dots, x_d)| \cdot 1\,\mathrm{d}s\Big)^p \\
&\overset{(2.11)}{\le} \Big(\int_0^D |\partial_{x_1}\tilde{v}(s, x_2, \dots, x_d)|^p\,\mathrm{d}s\Big)^{p/p} \cdot \Big(\int_0^D 1^q\,\mathrm{d}s\Big)^{p/q} \\
&= D^{p/q}\int_0^D |\partial_{x_1}\tilde{v}(s, x_2, \dots, x_d)|^p\,\mathrm{d}s.
\end{aligned}$$

Integration w.r.t. x_1 then gives:

$$\int_0^D |\tilde{v}(x)|^p\,\mathrm{d}x_1 \le D^{p/q+1}\int_0^D |\partial_{x_1}\tilde{v}(s, x_2, \dots, x_d)|^p\,\mathrm{d}s.$$

Finally, integration over $x_2, \dots, x_d$ yields:

$$\int_{\mathbb{R}^d} |\tilde{v}(x)|^p\,\mathrm{d}x \le D^{p/q+1}\int_{\mathbb{R}^d} |\partial_{x_1}\tilde{v}(x)|^p\,\mathrm{d}x$$

Noting that $\tilde{v}$ is supported on $\bar{\Omega}$ and that $p/q + 1 = p$ we obtain:

$$\Big(\int_\Omega |v(x)|^p\,\mathrm{d}x\Big)^{1/p} \le D\Big(\int_\Omega |\partial_{x_1}v(x)|^p\,\mathrm{d}x\Big)^{1/p} \le D\Big(\int_\Omega |\nabla v(x)|^p\,\mathrm{d}x\Big)^{1/p}.$$

□

3 Weak solutions of elliptic boundary value problems

3.1 Strong formulation of the Poisson problem

We formally define the Poisson boundary value problem we will use in the following as a model problem for an elliptic PDE:

Definition 3.1 (Classical solution/strong form of the Poisson Problem)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with smooth boundary and let $f \in C^0(\Omega)$. $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$ is then called a* classical solution *of the homogeneous Dirichlet problem for the Poisson equation iff*

$$-\Delta u(x) = f(x) \qquad \forall x \in \Omega, \tag{3.1}$$

$$u(x) = 0 \qquad \forall x \in \partial\Omega. \tag{3.2}$$

(3.1) is also called the strong form of the Poisson equation.

Remark 3.2 (Other boundary conditions)**.** *Instead of homogeneous Dirichlet boundary values, we can also consider other boundary conditions. The most important alternatives are the following:*

- ***Dirichlet condition:***

  $$u(x) = g_D(x) \qquad \forall x \in \partial\Omega,$$

  for a given function $g_D \in C^0(\partial\Omega)$.

- ***Neumann condition:***

  $$\nabla u(x) \cdot n(x) = g_N(x) \qquad \forall x \in \partial\Omega,$$

  for a given function $g_N \in C^0(\partial\Omega)$.

- ***Robin condition:***

  $$\nabla u(x) \cdot n(x) + \alpha(x) u(x) = g_R(x) \qquad \forall x \in \partial\Omega,$$

  for given functions $\alpha, g_R \in C^0(\partial\Omega)$.

Further, we can consider mixed boundary conditions, where on different parts of $\partial\Omega$ different boundary conditions are emposed.

Remark 3.3. *For simplicity, we will consider the Poisson problem in the following. However, what we say can be easily generalized to diffusion-advection-reaction equations of the form*

$$\begin{aligned} -\nabla \cdot (A(x)\nabla u(x) + b(x)u(x)) + c(x)u(x) &= f(x) && \forall x \in \Omega, \\ u(x) &= 0 && \forall x \in \partial\Omega. \end{aligned}$$

where

$$A \in C^1(\Omega; \mathbb{R}^{d\times d}), \qquad b \in C^1(\Omega; \mathbb{R}^d) \qquad c, f \in C^0(\Omega).$$

Here, A corresponds to a non-isotropic symmetric positive definite diffusion tensor, b to a divergence-free velocity field and $c > 0$ to a reaction rate.

3.2 Weak formulation of the Poisson problem

Definition 3.4 (Weak form of the Poisson problem)**.** *Let Ω be a Lipschitz domain and $f \in L^2(\Omega)$. A function $u \in H_0^1(\Omega)$ is called* weak solution *of the homogeneous Dirichlet boundary values problem for the Poisson equation iff*

$$\int_\Omega \nabla u(x) \cdot \nabla \varphi(x) \,\mathrm{d}x = \int_\Omega f(x)\varphi(x)\,\mathrm{d}x \qquad \forall \varphi \in H_0^1(\Omega). \tag{3.3}$$

Proposition 3.5. *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with smooth boundary, $f \in C^0(\Omega)$. A function $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$ is a classical solution of the Poisson problem in the sense of Definition 3.1 iff it is a weak solution in the sense of Definition 3.4.*

Proof. Let u be a classical solution. Then for each test function $\varphi \in H_0^1(\Omega)$ we have via partial integration

$$\int_\Omega \nabla u(x) \cdot \nabla \varphi(x)\,\mathrm{d}x = \int_\Omega -\Delta u(x)\varphi(x)\,\mathrm{d}x = \int_\Omega f(x)\varphi(x)\,\mathrm{d}x.$$

So u is also a weak solution. Conversely, if u is a weak solution in $C^2(\Omega) \cap C^0(\bar{\Omega})$, we can again perform partial integration to obtain:

$$\int_\Omega -\Delta u(x)\varphi(x)\,\mathrm{d}x = \int_\Omega f(x)\varphi(x)\,\mathrm{d}x \qquad \forall \varphi \in H_0^1(\Omega).$$

Hence, by Theorem 1.7 we have

$$-\Delta u(x) = f(x) \qquad \forall x \in \Omega.$$

□

Definition 3.6 (Abstract elliptic problem in weak formulation)**.** *Let V be a Hilbert space, $\ell \in V'$ and $b \in Bil(V, V)$ a coercive continuous bilinear form on V. Here, we call b coercive iff there is a $c_b > 0$ such that*

$$b(v, v) \geq c_b \|v\|^2 \qquad \forall v \in V.$$

Then we call $u \in V$ a weak solution of the abstract elliptic problem in weak formulation iff

$$b(u, v) = \ell(v) \qquad \forall v \in V. \tag{3.4}$$

Proposition 3.7. *The weak Poisson problem (3.3) is an abstract elliptic problem in weak formulation where $V = H_0^1(\Omega)$ and*

$$b(u, v) := \int_\Omega \nabla u(x) \cdot \nabla v(x) \, \mathrm{d}x, \qquad \ell(v) := \int_\Omega f(x) v(x) \, \mathrm{d}x \qquad \forall u, v \in H_0^1(\Omega).$$

Proof. We know that V is a Hilbert space, and it is clear that b is a bilinear form and ℓ a linear form. Further, b is continuous since

$$\begin{aligned} b(u, v) &= \int_\Omega \nabla u(x) \cdot \nabla v(x) \, \mathrm{d}x \\ &\leq \|\nabla u\|_2 \cdot \|\nabla v\|_2 \\ &= |u|_{1,2} \cdot |v|_{1,2} \leq \|u\|_{1,2} \cdot \|v\|_{1,2}, \end{aligned}$$

thanks to the L^2-Cauchy-Schwarz inequality, so $\|b\| \leq 1$. Similarly one has

$$\ell(v) = \int_\Omega f(x) v(x) \, \mathrm{d}x \leq \|f\|_2 \cdot \|v\|_2 \leq \|f\|_2 \cdot \|v\|_{1,2},$$

so $\|\ell\| \leq \|f\|_2$. Further, b is coercive since

$$b(u, u) = \int_\Omega |\nabla u(x)|^2 \, \mathrm{d}x = |u|_{1,2}^2 \geq \frac{1}{1 + c_p^2} \|u\|_{1,2}^2,$$

where we have used the Poincaré inequality (2.15). $\square$

We want to show that an abstract elliptic problem in weak formulation has a unique solution. We will show a more general result:

Theorem 3.8 (Banach-Nečas-Babuška)**.** *Let V be a Banach space, W a Hilbert space, $b \in \mathrm{Bil}(V, W)$ and $\ell \in W'$. Assume that b is inf-sup-stable with constant c_b', i.e.,*

$$\inf_{0 \neq v \in V} \sup_{0 \neq w \in W} \frac{b(v, w)}{\|v\| \cdot \|w\|} \geq c_b' > 0,$$

and non-degenerate in the second variable, i.e.,

$$(b(v,w) = 0 \quad \forall v \in V) \;\Rightarrow\; w = 0 \tag{3.5}$$

for all $w \in W$. Then the abstract weak/variational problem

$$b(u,w) = \ell(w) \qquad \forall w \in W \tag{3.6}$$

has a unique solution $u \in V$ satisfying

$$\|u\| \le \frac{1}{c_b'}\|\ell\|.$$

Proof. Let $B : V \to W'$ be the continuous linear operator associated with b. Then (3.6) is equivalent to

$$B(u) = \ell.$$

Thus, the theorem is proved when we can show that B is invertible. To this end, note that since b is inf-sup stable we have that B is bounded from below with the same constant c_b'. By Proposition 2.8, $\operatorname{im}(B)$ is closed and B is an isomorphism onto its image. It remains to show that $\operatorname{im}(B) = W'$. By the Riesz representation theorem, $\operatorname{im}(B) = W'$ iff

$$\mathcal{A} := \{w_{B(v)} \,|\, v \in V\} = W,$$

where $w_{B(v)}$ again denotes the Riesz representative of $B(v)$ in W. Since $\mathcal{A}$ is closed in W, it suffices to show that $\mathcal{A}^\perp = \{0\}$ by Theorem 2.14. For that note that

$$\begin{aligned} w \in \mathcal{A}^\perp \Leftrightarrow 0 = (w, w_{B(v)}) = B(v)[w] = a(v,w) \quad \forall v \in V \\ \overset{(3.5)}{\Rightarrow} w = 0. \end{aligned}$$

Finally note that

$$\|u\| = \left\|B^{-1}\ell\right\| \le \left\|B^{-1}\right\| \cdot \|\ell\| \le \frac{1}{c_b'}\|\ell\|.$$

$\square$

Corollary 3.9 (Lax-Milgram)**.** *Every abstract elliptic problem in weak formulation has a unique solution u with*

$$\|u\| \le \frac{1}{c_b}\|\ell\|.$$

In particular, the Poisson problem (3.3) has a unique weak solution u satisfying

$$\|u\|_{1,2} \le (1 + c_p^2) \cdot \|f\|_2$$

Proof. With $W := V$ we have

$$\inf_{0 \ne v \in V} \sup_{0 \ne w \in V} \frac{b(v,w)}{\|v\|\|w\|} \ge \inf_{0 \ne v \in V} \frac{b(v,v)}{\|v\|^2} \ge c_b,$$

so $c_b' \geq c_b > 0$. Further

$$(b(v,w) = 0 \ \ \forall v \in V) \implies \underbrace{b(w,w)}_{\geq c_b \|w\|^2} = 0,$$

so b is also non-degenerate in the second variable.

The statements about the Poisson problem directly follow from Proposition 3.7 and its proof. □

Remark 3.10. *1) If, in addition, b is symmetric, then the statement of Theorem 3.8 directly follows from the Riesz representation theorem by showing that b defines an inner product on V w.r.t. which V is a Hilbert space and ℓ a continuous linear functional. As in the proof of the representation theorem we obtain that u is a minimizer of the energy functional*

$$I(v) := \frac{1}{2} b(v,v) - \ell(v) \qquad \text{over } V.$$

For the Poisson problem, we obtain that u minimizes

$$I(v) := \frac{1}{2} |v|_{1,2}^2 - (f,v)_2 \qquad \text{over } H_0^1(\Omega).$$

2) Theorem 3.8 remains true if W is a reflexive Banach space, e.g., $H^{m,p}(\Omega)$, $1 < p < \infty$.

Remark 3.11 (Non-homogeneous boundary values)**.** *For the Poisson problem*

$$-\Delta u = f \ \text{ on } \Omega, \qquad u = g_D \ \text{ on } \partial\Omega,$$

with non-homogeneous Dirichlet boundary values g_D, consider

$$H^1_{g_D}(\Omega) := \{u \in H^1(\Omega) \,|\, u|_{\partial\Omega} = g_D\}.$$

Then we call $u \in H^1_{g_D}(\Omega)$ a weak solution of the problem iff

$$(\nabla u, \nabla \varphi)_2 = (f, \varphi)_2 \qquad \forall \varphi \in H_0^1(\Omega).$$

Again one easily checks that a smooth function u is a weak solution iff it is a classical solution. However, note that $H^1_{g_D}(\Omega)$ is only an affine space. To show existence and uniqueness of solutions we 'deaffinize' the problem: assuming that g_D extends to $\tilde{g}_D \in H^1(\Omega)$, let $\tilde{u} := u - \tilde{g}_D \in H_0^1(\Omega)$. Then u is a weak solution iff $\tilde{u}$ is the unique solution of the weak problem

$$(\nabla \tilde{u}, \nabla \varphi)_2 = (f, \varphi)_2 - (\nabla \tilde{g}_D, \nabla \varphi)_2 \qquad \forall \varphi \in H_0^1(\Omega).$$

3.3 Regularity theory for elliptic boundary value problems

In the proof of Theorem 3.8 we have seen that the solution operator B^{-1} associated with the weak formulation of the Poisson problem continuously maps $H^{-1}(\Omega)$ onto $H_0^1(\Omega)$. As one immediately sees in one spatial dimension, if the right-hand side f has higher regularity then H^{-1} one can expect correspondingly higher regularity of the solution u. E.g., if $f \in C^0(\bar{\Omega})$, then by the fundamental theorem of calculus, we immediately see that the equation

$$u''(x) = f(x)$$

has a solution in $C^2(\Omega)$. In multiple spatial dimensions, the situation is more complicated [Gri11, Sec. 1.1]. However, assuming $f \in C^{0,\alpha}(\bar{\Omega})$, $0 < \alpha < 1$ and a sufficiently regular $\partial\Omega$, one can indeed show that $u \in C^{2,\alpha}(\bar{\Omega})$ [Gri11, Sec. 6.3.2]. This is the regularizing property of elliptic PDEs. In the following, we will be interested in the following Sobolev-space version of this statement:

Theorem 3.12. *Let Ω be a Lipschitz domain, $f \in L^2(\Omega)$ and u the unique solution of the corresponding weak Poisson problem from Definition 3.4. If Ω is convex or $\partial\Omega$ has $C^{1,1}$ regularity, then $u \in H^2(\Omega)$ and there is a constant C independent of f such that*

$$\|u\|_{H^2(\Omega)} \leq C\|f\|. \tag{3.7}$$

Proof. The statement for domains with smooth boundary can be found in [Alt16, A12.3]. The statement for convex domains can be found in [Gri11, Thm 3.2.1.2]. □

Remark 3.13. *For domains with insufficient regularity, Theorem 3.12 no longer holds (see exercises).*

4 Finite Element methods for elliptic PDEs

This chapter will be concerned with the numerical approximation of elliptic PDEs of second order using finite element methods (FEMs). As discussed in Chapter 1, finite element methods are a special (Ritz-)Galerkin methods, which we will discuss first in an abstract setting.

4.1 Ritz-Galerkin method and abstract error bound

Definition 4.1 (Ritz-Galerkin method)**.** *Let an abstract elliptic problem in weak formulation of the form*

$$b(u, v) = \ell(v) \qquad \forall v \in V$$

as in Definition 3.6 be given. I.e., V is a Hilbert space, $b \in \mathrm{Bil}(V, V)$ coercive and $\ell \in V'$. Further, let $V_h \subset V$ be a closed subspace of V. Then the Ritz-Galerkin approximation *$u_h \in V_h$ of u is given by*

$$b(u_h, v_h) = \ell(v_h) \qquad \forall v_h \in V_h. \tag{4.1}$$

(We use the subscript h to indicate the dependence of V_h on some discretization parameter h.)

Proposition 4.2. *The Ritz-Galerkin approximation u_h is uniquely defined.*

Proof. (4.1) is an abstract elliptic problem in weak formulation since continuity and coercivity is inherited by subspaces. (In general, this is not true for inf-sup stability!) Hence, (4.1) has a unique solution by Corollary 3.9. □

Proposition 4.3 (Galerkin orthogonality)**.** *For the approximation error $u - u_h$ the following orthogonality property holds:*

$$b(u - u_h, v_h) = 0 \qquad \forall v_h \in V_h. \tag{4.2}$$

If b is symmetric, then (V, b) is a Hilbert space and $u_h = P_{V_h,b}u$ is the b-orthogonal projection of u onto V_h.

Proof. We have

$$b(u - u_h, v_h) = b(u, v_h) - b(u_h, v_h) = \ell(v_h) - \ell(v_h) = 0 \qquad \forall v_h \in V_h,$$

since u_h is a solution of (4.1), $V_h \subset V$ and u is a solution of (3.4).

(V, b) is a Hilbert space since the norm induced by b is equivalent to the norm on V (see Proposition 4.6). In particular, V_h is a closed subspace of (V, b). $u_h = P_{V_h,b}(u)$ follows since (4.2) characterizes the orthogonal projection by Theorem 2.14. □

Theorem 4.4 (Céa's lemma)**.** *Let V, V_h, b, u and u_h be given as in Definition 4.1. Then the following abstract error bound holds:*

$$\|u - u_h\| \leq \frac{\|b\|}{c_b} \inf_{v_h \in V_h} \|u - v_h\|. \tag{4.3}$$

Thus, u_h is a quasi-best approximation of u in V_h.

Proof. For arbitrary $v_h \in V_h$ we have

$$\begin{aligned} c_b\|u - u_h\|^2 &\leq b(u - u_h, u - u_h) \\ &\overset{(4.2)}{=} b(u - u_h, u - v_h) \\ &\leq \|b\|\|u - u_h\|\|u - v_h\|. \end{aligned}$$

Dividing by $c_b\|u - u_h\|$ and taking the infimum over all $v_h \in V_h$ yields the claim. □

Remark 4.5. *(4.3) shows that if we can construct a good approximation space V_h such that the best-approximation error $\inf_{v_h \in V_h} \|u - v_h\|$ is small, the corresponding Galerkin method will yield a good approximation u_h (up to an additional factor of $\|b\|/c_b$). Further, we automatically get a rigorous bound for $\|u - u_h\|$ from a rigorous bound for the best-approximation error. Hence, our only remaining job is to find a good V_h.*

Proposition 4.6. *Let b be symmetric. Then the* energy norm *$\|\cdot\|_b$ induced by b on V given by*

$$\|u\|_b^2 := b(u, u) \qquad \forall u \in V$$

is equivalent to the original norm on V. More precisely, we have

$$c_b^{1/2}\|u\| \leq \|u\|_b \leq \|b\|^{1/2}\|u\|. \tag{4.4}$$

Proposition 4.7. *Let V, V_h, b, u and u_h be given as in Definition 4.1 and additionally assume that b is symmetric. Then we have:*

$$\|u - u_h\|_b = \inf_{v_h \in V_h} \|u - v\|_b. \tag{4.5}$$

The bound (4.3) improves to:

$$\|u - u_h\| \le \left(\frac{\|b\|}{c_b}\right)^{1/2} \inf_{v_h \in V_h} \|u - v_h\|. \tag{4.6}$$

Proof. (4.5) directly follows from Proposition 4.3 as the orthogonal projection gives a best approximation in the projection space. (4.6) follows from (4.5) using (4.4). ☐

Example 4.8 (Possible choices of V_h)**.** *If we choose the approximation space $V_h \subset V$ to be a finite-dimensional (making it automatically closed), then (4.1) becomes a finite-dimensional system of linear equations that can be solved on a computer. Apart from the finite-element spaces V_h we will consider in the remainder of this course, other choices are considered in the literature. For instance:*

- *Polynomial spaces $V_h := \mathbb{P}^{k(h)}(\Omega) \cap \{v_h \in C^0(\bar{\Omega}) |\ v_h = 0 \text{ auf } \partial\Omega\}$, where $\mathbb{P}^{k(h)}(\Omega)$ is the space of polynomials over Ω with degree $\le k(h)$. The corresponding Galerkin methods are called* spectral methods. *Spectral methods are well-suited for simple domains Ω and smooth f, in which case exponential convergence of u_h to u w.r.t. k can be shown.*

- *$V_h := \operatorname{span}\{u_i \in X \,|\, \Delta u_i = \lambda_i u_i, i = 1, \dots, k(h)\}$, where (λ_i, u_i) is the i-th eigenpair of the Laplacian.*

Proposition 4.9 (Matrix-vector form of a Ritz-Galerkin method)**.** *Let V, V_h, b and u_h be given as in Definition 4.1 and let $N(h) := \dim(V_h) < \infty$. If $\{\varphi_1, \dots, \varphi_{N(h)}\}$ is a basis of V_h, then u_h has a unique coordinate representation $u_h = \sum_{i=1}^{N(h)} \underline{u}_{h,i}\varphi_i$. Inserting this identity into (4.1), we obtain*

$$\sum_{i=1}^{N(h)} b(\varphi_i, \varphi_j)\underline{u}_{h,i} = b\left(\sum_{i=1}^{N(h)} \underline{u}_{h,i}\varphi_i, \varphi_j\right) = \ell(\varphi_j) \quad \forall j = 1, \dots, N(h).$$

Thus defining the matrix and vector

$$\underline{b}_{ji} := b(\varphi_i, \varphi_j), \qquad \underline{\ell}_i := \ell(\varphi_i) \qquad \forall i, j = 1, \dots, N(h)$$

We obtain that $u_h \in V_h$ is the solution of (4.1) iff the coefficient vector $\underline{u} \in \mathbb{R}^{N(h)}$ satisfies

$$\underline{b} \cdot \underline{u}_h = \underline{\ell}.$$

$\underline{b}$ is also called the stiffness matrix.

4.2 Finite element methods

Finite element methods use approximation spaces V_h which are based on a partitioning of the computational domain Ω into non-overlapping subdomains T_i of simple geometric structure (triangles, tetrahedrons, etc.). On each T_i local approximation spaces with an appropriate basis are constructed. These *finite elements* are then glued together to build a global approximation space V_h. Due to this construction, the global approximation properties of V_h can be deduced from the local approximation properties of the individual finite elements.

We formalize this notion of a finite element as follows:

Definition 4.10 (Finite element)**.** *Let*

1) $T \subset \mathbb{R}^d$ be the closure of a Lipschitz domain (element domain)*,*

2) $\mathcal{P}$ a k-dimensional function space over T (space of shape functions)*,*

3) $\mathcal{N} := \{N_1, \ldots, N_k\}$ a basis of $\mathcal{P}'$ (local degrees of freedom)*.*

Then the triple $(T, \mathcal{P}, \mathcal{N})$ is called a finite element*.*

The dual basis $\Phi := \{\varphi_1, \ldots, \varphi_k\}$ of $\mathcal{N}$, i.e., $N_i(\varphi_j) = \delta_{i,j}$, $1 \le i, j \le k$, is called nodal basis *of $\mathcal{P}$.*

The most basic class of finite elements are so-called Lagrange elements on simplicial meshes, which we will study first.

4.2.1 Simplicial Lagrange elements

Definition 4.11 (Simplex)**.** *Let $s \in \{0, \ldots, d\}$, and let $a_0, \ldots a_s \in \mathbb{R}^d$ be points such that the vectors $(a_j - a_0)_{j=1,\ldots,s}$ are linear independent. Then*

$$T := \{x \in \mathbb{R}^d \,|\, x = \sum_{i=0}^{s} \lambda_i a_i, 0 \le \lambda_i,\ \sum_{i=0}^{s} \lambda_i = 1\}$$

is called a non-degenerate s-simplex in $\mathbb{R}^d$. The points $a_0, \ldots a_s$ are called the vertices of the simplex.

For $r \in \{0, \ldots, s\}$ and $\{\tilde{a}_0, \ldots, \tilde{a}_r\} \subseteq \{a_0, \ldots a_s\}$, we call the simplex

$$\tilde{T} := \{x \in \mathbb{R}^d \,|\, x = \sum_{i=0}^{r} \lambda_i \tilde{a}_i, 0 \le \lambda_i, \sum_{i=0}^{r} \lambda_i = 1\}$$

an r-face of T. The 1-faces of T are called edges of T. The $s-1$-faces of T are called facets.

For fixed d, we denote by $\hat{T}$ the simplex with vertices $a_0 = 0$, $a_i = e_i$, $i = 1, \ldots, d$. $\hat{T}$ is called the standard or unit d-simplex.

For each simplex T, the diameter of T is given by

$$h(T) := \operatorname{diam}(T) = \max_{i,j=0}^{s} |a_i - a_j|.$$

By

$$\rho(T) := 2 \sup \left\{ R \,\middle|\, \exists x_0 \in \mathbb{R}^d : \, B_R(x_0) \cap \left\{ \sum_{i=0}^{s} \lambda_i a_i \mid \sum_{i=0}^{s} \lambda_i = 1 \right\} \subset T \right\}$$

we denote the inner sphere diameter T. Further we let

$$\sigma(T) := \frac{h(T)}{\rho(T)}.$$

Remark 4.12. *0-faces are points, 1-faces are line segments, 2-faces are triangles, 3-faces are tetrahedra. Note that our definition of the standard d-simplex uses a different convention as in algebraic topology, where the standard d-simplex is usually defined to be the convex hull of the unit vectors $e_1, \dots, e_{d+1} \in \mathbb{R}^{d+1}$.*

Definition 4.13 (Barycentric coordinates)**.** *Let $T \subset \mathbb{R}^d$ be a non-degenerate s-simplex. The barycentric cooredinates $\lambda_0, \dots, \lambda_s \in [0,1]$ of a point $x \in T$ are given by the unique solution of the linear system*

$$x = \sum_{i=0}^{s} \lambda_i a_i, \quad \sum_{i=0}^{s} \lambda_i = 1.$$

The barycenter (center of mass) x_s of T is given by

$$x_s := \frac{1}{s+1} \sum_{i=0}^{s} a_i$$

with barycentric coordinates $\lambda_i = \frac{1}{s+1}$. The vertex a_k has the barycentric coordinates $\lambda_i = \delta_{k,i}$.

Remark 4.14. *A point $x \in T$ with barycentric coordinates λ_i is the barycenter of the masses λ_i attached to the points a_i.*

Definition and Proposition 4.15 (Reference map)**.** *Let $\hat{T}$ be the standard d-simplex. Then for each d-simplex $T \subset \mathbb{R}^d$, there is a unique affine map $F : \hat{T} \to T$ such that $F(e_j) = a_j$, $j = 1, \dots, d$ and $F(0) = a_0$ called the* reference map *for T. Writing F as $F(x) = Ax + b$, $A \in \mathbb{R}^{d \times d}, b \in \mathbb{R}^d$, we have*

$$\|\nabla F\| = \|A\| \leq \frac{h(T)}{\rho(\hat{T})},$$

$$\left\|(\nabla F^{-1})\right\| = \left\|A^{-1}\right\| \leq \frac{h(\hat{T})}{\rho(T)},$$

$$c\rho(T)^d \leq |\det(\nabla F)| = |\det(A)| \leq C h(T)^d.$$

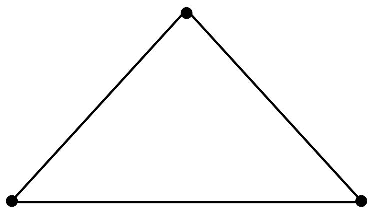


Figure 4.1: Linear Lagrange element.

Proof. Exercise. □

Definition and Proposition 4.16 (Linear simplicial Lagrange element)**.** *Let $T \subset \mathbb{R}^d$ be a d-simplex, and let $\mathbb{P}^1(T)$ be the space of affine linear functions over T. Prescribing values at all vertices a_i, $i = 0, \ldots, d$ of T uniquely defines functions in $\mathbb{P}^1(T)$. Thus, $\mathcal{N} := \{\delta_{a_i} \mid i = 0, \ldots d\}$, where $\delta_{a_i}(v) := v(a_i)$, defines a basis of $\mathbb{P}^1(T)'$. We call the finite element $(T, \mathbb{P}^1(T), \mathcal{N})$* linear simplicial Lagrange element*. The nodal shape function basis is given by the functions λ_i.*

Proof. To show that each $v \in \mathbb{P}^1(T)$ is uniquely determined by its values at a_i note that for each $v(x) = \alpha_0 + \sum_{j=1}^d \alpha_j x_j$ and $T \ni x = \sum_{i=0}^d \lambda_i a_i$ with $\sum_{i=0}^d \lambda_i = 1$ we have

$$
\begin{aligned}
v(x) = v(\sum_{i=0}^d \lambda_i a_i) &= \alpha_0 + \sum_{j=1}^d \alpha_j (\sum_{i=0}^d \lambda_i a_i)_j \\
&= \sum_{i=0}^d \lambda_i (\alpha_0 + \sum_{j=1}^d \alpha_j (a_i)_j) = \sum_{i=0}^d \lambda_i v(a_i).
\end{aligned}
$$

Thus, in particular, $\delta(v) = 0 \ \forall \delta \in \mathcal{N}$ implies $v = 0$. The claim now follows using elementary linear algebra arguments noting that $|\mathcal{N}| = \dim \mathbb{P}^1(T)$ (exercise). □

Example 4.17 (Lineare Lagrange element for $d = 2$)**.** *Consider the standard 2-simplex $\hat{T}$ with vertices $\hat{a}_0 = (0, 0)$, $\hat{a}_1 = (1, 0)$ and $\hat{a}_2 = (0, 1)$. Then the nodal shape function basis is given by*

$$
\begin{aligned}
\hat{\varphi}_0(x, y) &= 1 - x - y, \\
\hat{\varphi}_1(x, y) &= x, \\
\hat{\varphi}_2(x, y) &= y.
\end{aligned}
$$

In particular, an arbitrary $\hat{v} \in \mathbb{P}^1(\hat{T})$ is given by

$$
\hat{v}(x, y) = \sum_{i=0}^2 \hat{v}(\hat{a}_i) \hat{\varphi}_i(x, y).
$$

For an arbitrary triangle $T \subset \mathbb{R}^2$ we obtain the corresponding linear Lagrange element via the reference map $F : \hat{T} \to T$ (Definition and Proposition 4.15), i.e., the nodal basis of $\mathbb{P}^1(T)$ is *given by*

$$\varphi_i(x,y) := \hat{\varphi}_i(F^{-1}(x,y)).$$

To see this note that $\varphi_i \in \mathbb{P}^1(T)$ *since* F *is affine. Further,* F^{-1} *maps the vertices* a_0, a_1, a_2 *of* T *onto* $\hat{a}_0, \hat{a}_1, \hat{a}_2$*. In particular, for arbitrary* $v \in \mathbb{P}^1(T)$ *we have*

$$v(x,y) = \sum_{i=0}^{2} v(a_i)\varphi_i(x,y) = \sum_{i=0}^{2} v(a_i)\hat{\varphi}_i(F^{-1}(x,y)).$$

Remark 4.18. *Example 4.17 illustrates that it is sufficient to introduce finite elements on a reference geometry and then obtain finite elements on other geometries of the same type via the reference map.*

In order to build a global approximation space V_h from finite elements we need to consider a suitable triangulation of Ω with d-simplices:

Definition 4.19 (Admissible triangulation)**.** *Let* $\Omega \subset \mathbb{R}^d$ *be a domain, and let*

$$\mathcal{T}_h := \{T_j \,|\, j = 1, \dots m\}$$

be a set of d*-simplices in* $\mathbb{R}^d$*.* $\mathcal{T}_h$ *is called* admissible triangulation *of* Ω *if*

- $\bar{\Omega} = \cup_{j=1}^m T_j$.
- *For* $T_1, T_2 \in \mathcal{T}_h$ *with* $T_1 \neq T_2$ *we either have* $T_1 \cap T_2 = \emptyset$*, or* $T_1 \cap T_2$ *is an* s*-face of both* T_1 *and* T_2*.*

Further we define $h := \max_{T\in\mathcal{T}_h} h(T)$ *and* $\rho := \min_{T\in\mathcal{T}_h} \rho(T)$*.*

We can now define the global finite element space V_h:

Definition and Proposition 4.20 (Linear finite-element space S_h^1)**.** *1) Let* $\mathcal{T}_h$ *an admissible triangulation of* $\Omega \subset \mathbb{R}^d$*. Then the linear finite-element space* $S_h^1 := S^1(\mathcal{T}_h)$ *is given by*

$$S_h^1 := \{v_h \in C^0(\Omega) \,|\, v_h|_T \in \mathbb{P}^1(T) \;\forall T \in \mathcal{T}_h\}.$$

We have $S_h^1 \subset H^1(\Omega)$*.*

2) Let $\{\bar{a}_j \,|\, j = 1, \dots, N\}$ *be the set of all vertices of all* $T \in \mathcal{T}_h$*. Then a function* $v_h \in S_h^1$ *is uniquely defined by prescribing the values* $v_h(\bar{a}_j)$ *of* v_h *in all* $\bar{a}_j$*. In particular,* $\dim(S_h^1) = N$*. A basis of* S_h^1 *is given by the functions* $\bar{\varphi}_i \in S_h^1$ *such that*

$$\bar{\varphi}_i(\bar{a}_j) = \delta_{ij}, \quad i,j = 1, \dots, N.$$

This basis is called the nodal basis *of* S_h^1*.*

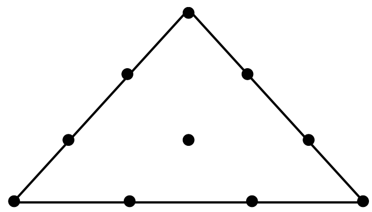


Figure 4.2: Cubic Lagrange element.

3) Let $(\hat{T}, \mathbb{P}^1(\hat{T}), \hat{\mathcal{N}})$ be the linear Lagrange element on the standard simplex $\hat{T}$, and let $v_h \in S_h^1$ be given by

$$v_h(x) = \sum_{i=1}^{N} v_h(\bar{a}_i)\bar{\varphi}_i(x).$$

Then for an arbitrary $T \in \mathcal{T}_h$ with vertices $a_0, \ldots, a_d$ we have

$$v_h|_T(x) = \sum_{i=0}^{d} v_h(a_i)\hat{\varphi}_i(F^{-1}(x)),$$

where $F : \hat{T} \to T$ is the reference map and $\hat{\varphi}_i$ the nodal shape function basis for $\hat{T}$.

Proof. To see that v_h is uniquely defined by its values at $\bar{a}_i$, note that prescribing these values uniquely defines $v_h|_{T_j}$ for each $T_j \in \mathcal{T}_h$. It remains to show that the resulting piecewise defined function is continuous across intersections $S = T_1 \cap T_2$. Since $\mathcal{T}_h$ is admissible, we know that S is an s-face of T_1 and T_2 with vertices contained in $\{\bar{a}_i\}$. Thus, both $v_h|_{T_1}$ and $v_h|_{T_2}$ are affine linear functions on S, which agree at the vertices of S. Hence, both functions must agree on S. $\square$

Definition 4.21 (Linear finite-element method)**.** *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$ and let*

$$b(u, v) = \ell(v), \quad \forall v \in V$$

be an abstract elliptic problem with $V = H_0^1(\Omega)$. Let

$$V_h := S_{h,0}^1 := S_h^1 \cap \{v \in C^0(\bar{\Omega}) \mid v|_{\partial\Omega} = 0\}$$

Then the corresponding Ritz-Galerkin method

$$b(u_h, v_h) = \ell(v_h), \quad \forall v_h \in V_h$$

is called linear finite element method.

An extension to higher-degree polynomials is straightforward. However, we need to be careful how to choose the local degrees of freedom in order to obtain a basis for the resulting conforming finite-element space. I.e., we need to make sure that shape functions with the same local degrees of freedom on a shared face S restrict to the same function on S.

Definition and Proposition 4.22 (Higher-order Lagrange element). *Let $T \subset \mathbb{R}^d$ be a d-simplex, and let $\mathbb{P}^k(T)$ denote the space of polynomial functions over T of maximum total degree k with dimension $g_k := \binom{d+k}{k}$. Further, let the k-th order Lagrange lattice $G_k(T)$ be given by*

$$G_k(T) := \left\{ x = \sum_{j=0}^{d} \lambda_j a_j \,\middle|\, \lambda_j \in \left\{ \frac{m}{k} \,\middle|\, m = 0, \ldots, k \right\},\ \sum_{j=0}^{d} \lambda_j = 1 \right\},$$

and let $\mathcal{N} := \{\delta_g \,|\, g \in G_k(T)\}$. Then $(T, \mathbb{P}^k(T), \mathcal{N})$ defines a finite element called simplicial Lagrange element of order k.

Proof. [EG21a, p. 7.4]. □

Definition 4.23 (Lagrange finite-element space S_h^k). *1) Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$. Then the Lagrange finite-element space $S_h^k := S_h^k(\mathcal{T}_h)$ is given by*

$$S_h^k := \{v_h \in C^0(\Omega) |\ v_h|_T \in \mathbb{P}^k(T)\ \forall T \in \mathcal{T}_h\}.$$

We have $S_h^k \subset H^1(\Omega)$.

2) Let the k-the order Lagrange lattice $G_k(\mathcal{T}_h)$ on $\mathcal{T}_h$ be given by

$$G_k(\mathcal{T}_h) := \{\bar{g}_j \,|\, j = 1, \ldots, N\} := \bigcup_{T \in \mathcal{T}_h} G_k(T).$$

Then a function $v_h \in S_h^k$ can be uniquely defined by prescribing the values $v_h(\bar{g}_j)$ of v_h in all $\bar{g}_j$. The nodal basis of S_h^k is given by the functions $\bar{\varphi}_i \in S_h^k$ such that

$$\bar{\varphi}_i(\bar{g}_j) = \delta_{ij}, \quad i, j = 1, \ldots, N$$

3) Let $(\hat{T}, \mathbb{P}^k(\hat{T}), \hat{\mathcal{N}})$ be the k-th order Lagrange element on the standard simplex $\hat{T}$ and let $v_h \in S_h^k$ be given by

$$v_h(x) = \sum_{i=1}^{N} v_h(\bar{g}_i) \bar{\varphi}_i(x).$$

Then for an arbitrary $T \in \mathcal{T}_h$ with Lagrange points $g_i \in G_k(T)$ we have

$$v_h|_T(x) = \sum_{i=1}^{g_k} v_h(g_i) \hat{\varphi}_i(F^{-1}(x)),$$

where $F : \hat{T} \to T$ is the reference map and $\hat{\varphi}_i$ denotes the nodal shape function basis of $\mathbb{P}^k(\hat{T})$.

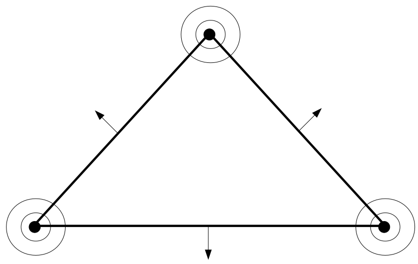


Figure 4.3: Argyris element.

Proof. Exercise. □

4.2.2 Some further finite elements

Remark 4.24 (Other element domains)**.** *Rectangles or cuboids are also frequently used as element domains. As shape function spaces (on the reference domain) one uses tensor-product polynomial spaces*

$$Q^k([0,1]^d) := \bigotimes_{i=1}^{d} \mathbb{P}^k([0,1]) = \left\{ \sum_{\alpha \in \mathbb{N}_0^d,\, |\alpha_i| \le k} c_\alpha x^\alpha \;\middle|\; c_\alpha \in \mathbb{R} \right\}.$$

E.g., for $d = 2$ each $p \in Q^1([0,1]^2)$ has the form

$$p(x, y) = a + bx + cy + dxy.$$

The corresponding Lagrange finite-element spaces are denoted by Q_h^k.

For geometries of the form $\Omega = \Omega' \times (0, 1)$ prismatic element domains are often used.

While the spaces S_h^k, Q_h^k are contained in $H^1(\Omega)$, they do not have H^2-regularity. As an example of a finite element which leads to H^2-conforming finite-element spaces, we mention the Argyris element:

Definition 4.25 (Argyris element on triangles)**.** *Let $T \subset \mathbb{R}^2$ be a 2-simplex with vertices $\{a_i \,|\, i = 0, \ldots 2\}$. Denote by $\{b_i \,|\, i = 0, \ldots 2\}$ the midpoints of the edges of T. Then the functionals $p(a_i)$, $\nabla p(a_i)_k$, $\nabla^2 p(a_i)_{k,l}$ and $\nabla p(b_i) \cdot n_i$, $i = 0, \ldots, d$, $k, l = 1, \ldots, 2$ uniquely determine polynomials $p \in \mathbb{P}^5(T)$. The corresponding finite element with domain T and shape function space $\mathbb{P}^5(T)$ is called the Argyris element.*

4.3 Local interpolation error bounds

In the following, our goal will be to show a priori convergence rates for Lagrange finite-element methods using the finite-element spaces S_h^k. Thanks to Céas lemma (Theorem 4.4), it suffices to bound the best-approximation error for u in S_h^k. We will do so by showing that the nodal interpolation operator I_h, which assigns to u the finite-element function which agrees with u on all Lagrange nodes in $G_k(\mathcal{T}_h)$, gives indeed a good approximation of u (cf. Theorem 4.34). In more detail, we want to show estimates of the form

$$\|u - I_h(u)\|_X \leq Ch^\alpha \tag{4.7}$$

with a rate α, which is as large as possible, while the constant C is controlled.

In this section we will start by showing local interpolation error estimates of the form (4.7) for a single $T \in \mathcal{T}_h$. Since all $T \in \mathcal{T}_h$ are affine-equivalent to the standard simplex $\hat{T}$, which has a fixed diameter h, we expect to obtain the factor h^α from the rescaling of the local functions w.r.t. $\hat{T}$. As we will see, the order α appearing due to rescaling will correspond to the order of derivatives appearing in the Sobolev semi-norms (Corollary 4.32). Thus, we first want to derive an interpolation error bound on $\hat{T}$ which only involves derivatives with highest order. To this end, we will use the Poincaré inequality for functions with zero mean:

Theorem 4.26 (Poincaré inequality for functions with zero mean)**.** *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain and $p \in [1, \infty]$. Then there is a constant C only depending on Ω and p s.t. for all functions $u \in H^{1,p}(\Omega)$ with*

$$\int_\Omega u \,\mathrm{d}x = 0,$$

we have

$$\|u\|_p \leq C|u|_{1,p}.$$

Proof. By Theorem 2.40, $H^{1,p}(\Omega)$ compactly embeds into $L^p(\Omega)$. (This is also true for $p = \infty$, since $H^{1,\infty}(\Omega) \hookrightarrow H^{1,p'}(\Omega) \hookrightarrow C^{0,\alpha}(\Omega) \hookrightarrow L^\infty(\Omega)$ for $d < p' < \infty$.) The claim follows via proof by contradiction, assuming that there is a sequence u_n with $\int_\Omega u_n \,\mathrm{d}x = 0$, $\|u_n\|_p = 1$ but $|u_n|_{1,p} \to 0$. □

Corollary 4.27. *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain, $p \in [1, \infty]$ and $m \in \mathbb{N}$. Then there is a constant C only depending on Ω, p and m s.t. for all functions $u \in H^{m,p}(\Omega)$ with*

$$\int_\Omega D^\alpha u = 0 \qquad \forall |\alpha| \leq m - 1,$$

we have

$$\|u\|_{m,p} \leq C|u|_{m,p}.$$

Proof. By Theorem 4.26 we have $|u|_{k,p} \leq C_m |u|_{k+1,p}$ for $k = 0, \dots, m-1$. Thus, the claim follows via induction. $\square$

We now use the Poincaré inequality to show a version of the so-called Bramble-Hilbert Lemma, which bounds the error for the best-approximation with polynomials by the highest-order Sobolev semi-norm of the interpolated function.

Lemma 4.28. *For each $u \in H^{k,p}(\Omega)$ there is a unique $q \in \mathbb{P}^k(\Omega)$, such that*

$$\int_\Omega D^\alpha (u - q)\,\mathrm{d}x = 0 \qquad \forall |\alpha| \leq k. \tag{4.8}$$

Proof. We note that for given u, (4.8) is a system of $\binom{d+k}{k}$ linear equations over the vector space $\mathbb{P}^k(\Omega)$ of the same dimensions. Thus, (4.8) has a unique solution iff the homogeneous system

$$\int_\Omega D^\alpha q\,\mathrm{d}x = 0 \qquad \forall |\alpha| \leq k \tag{4.9}$$

has the unique solution 0. To see this note that $D^\alpha q$ is constant for $|\alpha| = k$. Thus, (4.9) implies $D^\alpha q = 0$ for $|\alpha| = k$. We conclude that $D^\alpha q$ must be constant for all $|\alpha| = k-1$. Via induction it follows that $q = 0$. $\square$

Theorem 4.29 (Bramble-Hilbert)**.** *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain, $p \in [1, \infty]$ and $k \in \mathbb{N}_0$. Then there is a constant C only depending on Ω, p and k such that for all $u \in H^{k+1,p}(\Omega)$ we have*

$$\inf_{q \in \mathbb{P}^k(\Omega)} \|u - q\|_{k+1,p} \leq C |u|_{k+1,p}(\Omega).$$

Proof. For given u, choose q according to Lemma 4.28. Then by Corollary 4.27 we have

$$\|u - q\|_{k+1,p} \leq C |u - q|_{k+1,p} \overset{q \in \mathbb{P}^k}{=} |u|_{k+1,p}.$$

$\square$

In particular, we get the following abstract error bound for interpolation with polynomials:

Corollary 4.30. *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain and let $k, m \in \mathbb{N}_0$, $p, p' \in [1, \infty]$ be chosen such that*

$$H^{k+1,p}(\Omega) \subseteq H^{m,p'}(\Omega).$$

Further assume that

$$I \in \mathcal{L}(H^{k+1,p}(\Omega), H^{m,p'}(\Omega))$$

is a continuous linear (interpolation) operator, which is the identity on $\mathbb{P}^k(\Omega)$. Then there is a constant C only depending on $\Omega, k, p, \|I\|$, such that for all $u \in H^{k+1,p}(\Omega)$ we have

$$\|u - I(u)\|_{m,p'} \leq C|u|_{k+1,p}.$$

Proof. For arbitrary $q \in \mathbb{P}^k(\Omega)$ we have

$$\begin{aligned}\|u - I(u)\|_{m,p'} &= \|u - I(q) + I(q) - I(u)\|_{m,p'} \\ &= \|(u-q) - I(u-q)\|_{m,p'} \\ &\leq (\|\mathrm{id}_{H^{k+1,p} \to H^{m,p'}}\| + \|I\|) \cdot \|u - q\|_{k+1,p}.\end{aligned}$$

Taking the infimum over q, the claim follows from Theorem 4.29. □

To obtain an interpolation error bound for $T \in \mathcal{T}_h$, we now study how the Sobolev semi-norms transform under affine mappings:

Proposition 4.31 (Transformation of Sobolev norms)**.** *Let $\hat{\Omega}, \Omega \subset \mathbb{R}^d$ be open, non-empty sets, and let $F(\hat{x}) = A\hat{x} + b$, $A \in \mathbb{R}^{d\times d}$, $b \in \mathbb{R}^d$ be an affine map such that $F(\hat{\Omega}) = \Omega$. Then for each $m \in \mathbb{N}_0$, $p \in [1, \infty]$ and $u \in H^{m,p}(\Omega)$ we have $\hat{u} := u \circ F \in H^{m,p}(\hat{\Omega})$ and the following norm estimates hold true:*

$$\begin{aligned}|\hat{u}|_{H^{m,p}(\hat{\Omega})} &\leq C_1 \|A\|^m |\det A|^{-1/p} |u|_{H^{m,p}(\Omega)}, \\ |u|_{H^{m,p}(\Omega)} &\leq C_2 \left\|A^{-1}\right\|^m |\det A|^{1/p} |\hat{u}|_{H^{m,p}(\hat{\Omega})}.\end{aligned}$$

The constants C_1, C_2 only depend on d, m and p.

Proof. First assume that $u \in C^\infty(\Omega)$. Using the chain rule we can show via induction that we have

$$\partial_{\hat{x}_{i_1}} \cdots \partial_{\hat{x}_{i_k}} (u \circ F)(\hat{x}) = \sum_{j_1,\ldots,j_k=1}^{d} (\partial_{x_{j_1}} \cdots \partial_{x_{j_k}} u)(F(\hat{x})) A_{j_1,i_1} \cdots A_{j_k,i_k}. \tag{4.10}$$

To show that (4.10) also holds for $u \in H^{m,p}(\Omega)$ in the weak sense, first note that for arbitrary $u \in H^{m,p}(\Omega)$ the right-hand side is in $L^p(\Omega)$ and, in particular, in $L^1_{loc}(\Omega)$. Further for an arbitrary test function $\hat{\varphi} \in C_0^\infty(\hat{\Omega})$, with $\varphi := \hat{\varphi} \circ F^{-1}$ we have

$$\begin{aligned}
&\int_{\hat{\Omega}} \sum_{j_1,\dots,j_k=1}^{d} (\partial_{x_{j_1}} \cdots \partial_{x_{j_k}} u)(F(\hat{x})) A_{j_1,i_1} \cdots A_{j_k,i_k} \cdot \hat{\varphi}(\hat{x}) \,\mathrm{d}\hat{x} \\
&\quad = \int_{\Omega} \sum_{j_1,\dots,j_k=1}^{d} (\partial_{x_{j_1}} \cdots \partial_{x_{j_k}} u)(x) A_{j_1,i_1} \cdots A_{j_k,i_k} \cdot \varphi(x) \cdot |\det A| \,\mathrm{d}x \\
&\quad = (-1)^k \int_{\Omega} \sum_{j_1,\dots,j_k=1}^{d} u(x) A_{j_1,i_1} \cdots A_{j_k,i_k} \cdot (\partial_{x_{j_1}} \cdots \partial_{x_{j_k}} \varphi)(x) \cdot |\det A| \,\mathrm{d}x \\
&\quad = (-1)^k \int_{\hat{\Omega}} \hat{u}(\hat{x}) \cdot \sum_{j_1,\dots,j_k=1}^{d} A_{j_1,i_1} \cdots A_{j_k,i_k} (\partial_{x_{j_1}} \cdots \partial_{x_{j_k}} \varphi)(F(\hat{x})) \,\mathrm{d}\hat{x} \\
&\quad \overset{(4.10)}{=} (-1)^k \int_{\hat{\Omega}} \hat{u}(\hat{x}) \cdot (\partial_{\hat{x}_{j_1}} \cdots \partial_{\hat{x}_{j_k}} \hat{\varphi})(\hat{x}) \,\mathrm{d}\hat{x}.
\end{aligned}$$

To show the norm estimate, note that with $A_{i,j} \leq \|A\|$ for all i, j, we obtain:

$$|\partial_{\hat{x}_{i_1}} \cdots \partial_{\hat{x}_{i_k}} (u \circ F)(\hat{x})| \leq \sum_{j_1,\dots,j_k=1}^{d} |\partial_{x_{j_1}} \cdots \partial_{x_{j_k}} u|(F(\hat{x})) \cdot \|A\|^k.$$

This leads us to

$$\sum_{|\alpha|=m} |D_{\hat{x}}^\alpha \hat{u}|^p(\hat{x}) \leq C_1^p \|A\|^{pm} \sum_{|\alpha|=m} |D_x^\alpha u|^p(F(\hat{x})).$$

Using the integration by substitution, we obtain:

$$\begin{aligned}
|\hat{u}|^p_{H^{m,p}(\hat{\Omega})} &= \sum_{|\alpha|=m} \int_{\hat{\Omega}} |D_{\hat{x}}^\alpha \hat{u}|^p(\hat{x}) \,\mathrm{d}\hat{x} \\
&\leq C_1^p \|A\|^{pm} \sum_{|\alpha|=m} \int_{\hat{\Omega}} |D_x^\alpha u|^p(F(\hat{x})) \,\mathrm{d}\hat{x} \\
&= C_1^p \|A\|^{pm} \sum_{|\alpha|=m} \int_{\Omega} |D_x^\alpha u|^p(x) |\det A|^{-1} \,\mathrm{d}x \\
&= C_1^p \|A\|^{pm} |\det A|^{-1} |u|^p_{H^{m,p}(\Omega)}.
\end{aligned}$$

The second inequality follows by reversing the roles of Ω and $\hat{\Omega}$. □

Corollary 4.32 (Transformation of Sobolev functions on simplices)**.** *Let $T \subset \mathbb{R}^d$ be a d-simplex and let $\hat{T}$ be the standard d-simplex and $F : \hat{T} \to T$, $F(\hat{x}) = A\hat{x} + b$ be the reference map. Then for each $m \in \mathbb{N}_0$, $p \in [1, \infty]$ and $u \in H^{m,p}(T)$ we have $\hat{u} := u \circ F \in H^{m,p}(\hat{T})$ and the following* *norm estimates hold true:*

$$|\hat{u}|_{H^{m,p}(\hat{T})} \leq C_1 \Big(\frac{h(T)}{\rho(\hat{T})}\Big)^m \rho(T)^{-d/p} |u|_{H^{m,p}(T)},$$
$$|u|_{H^{m,p}(T)} \leq C_2 \Big(\frac{h(\hat{T})}{\rho(T)}\Big)^m h(T)^{d/p} |\hat{u}|_{H^{m,p}(\hat{T})}.$$

The constants C_1, C_2 only depend on d, m and p.

Proof. Proposition 4.31 and Definition and Proposition 4.15. □

Combining Corollary 4.32 with the interpolation error bound on the unit simplex we finally obtain the desired result:

Theorem 4.33 (Local interpolation error estimate)**.** *Let $T \subset \mathbb{R}^d$ be a d-simplex, $\hat{T}$ the standard d-simplex and $F : \hat{T} \to T$, $F(y) = Ay + b$ the reference map. Further, let $k, m \in \mathbb{N}_0$, $p, p' \in [1, \infty]$ be chosen such that*

$$H^{k+1,p}(\hat{T}) \subseteq H^{m,p'}(\hat{T}).$$

Further assume that

$$I_0 \in \mathcal{L}(H^{k+1,p}(\hat{T}), H^{m,p'}(\hat{T}))$$

is a continuous linear (interpolation) operator, which is the identity on $\mathbb{P}^k(\hat{T})$. Then

$$I(u) := I_0(u \circ F) \circ F^{-1}$$

is a continuous linear (interpolation) operator $I \in \mathcal{L}(H^{k+1,p}(T), H^{m,p'}(T))$ with the interpolation error bound

$$|u - Iu|_{H^{m,p'}(T)} \leq C\sigma(T)^{m+\frac{d}{p}}\ h(T)^{k+1-m+d(\frac{1}{p'}-\frac{1}{p})} |u|_{H^{k+1,p}(T)}$$

for all $u \in H^{k+1,p}(T)$. The constant C only depends on $\hat{T}, k, m, p$ and $\|I_0\|$.

Proof. Using Corollary 4.32 and Corollary 4.30 we have:

$$\begin{aligned}
|u - Iu|_{H^{m,p'}(T)} &\le C\Big(\frac{h(\hat{T})}{\rho(T)}\Big)^m h(T)^{d/p'} |(u - Iu) \circ F|_{H^{m,p'}(\hat{T})} \\
&= C\Big(\frac{h(\hat{T})}{\rho(T)}\Big)^m h(T)^{d/p'} |(\hat{u} - I_0\hat{u})|_{H^{m,p'}(\hat{T})} \\
&\le C'\Big(\frac{h(\hat{T})}{\rho(T)}\Big)^m h(T)^{d/p'} |\hat{u}|_{H^{k+1,p}(\hat{T})} \\
&\le C''\Big(\frac{h(\hat{T})}{\rho(T)}\Big)^m h(T)^{d/p'} \Big(\frac{h(T)}{\rho(\hat{T})}\Big)^{k+1} \rho(T)^{-d/p} |u|_{H^{k+1,p}(T)} \\
&= C''' h(T)^{k+1+d/p'} \rho(T)^{-d/p-m} |u|_{H^{k+1,p}(T)} \\
&= C''' h(T)^{k+1-m+d(1/p'-1/p)} \sigma(T)^{d/p+m} |u|_{H^{k+1,p}(T)}.
\end{aligned}$$

□

4.4 Global interpolation error bounds and a priori estimate

Since the Sobolev semi-norms $|\cdot|_{m,p}$ are integrals over Ω, which can be written as a sum of integrals over all $T \in \mathcal{T}_h$, all that is left to do is to find a global S_h^k-valued interpolation operator which is the identity on each $\mathbb{P}^k(T)$ and then apply Theorem 4.33 for each T. By construction of the Lagrange lattice $G_k(\mathcal{T}_h)$ (cf. Definition 4.23), we can simply interpolate at the Lagrange nodes:

Theorem 4.34 (Global interpolation error estimate)**.** *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$, $k \in \mathbb{N}$ and S_h^k the corresponding Lagrange finite-element space of k-th order. If $\frac{d}{2} < p \le \infty$, then the interpolation operator*

$$I_h(u) \in S_h^k, \quad I_h(u)(x) = u(x) \qquad \forall x \in G_k(\mathcal{T}_h)$$

is well defined for all $u \in H^{2,p}(\Omega)$. For all $u \in H^{k+1,p}(\Omega)$, and $m \in \{0,1\}$ the following interpolation error estimate holds:

$$|u - I_h u|_{H^{m,p}(\Omega)} \le C\Big(\sum_{T\in\mathcal{T}_h} \sigma(T)^{(m+\frac{d}{p})p} h(T)^{(k+1-m)p} |u|^p_{H^{k+1,p}(T)}\Big)^{1/p}$$

The constant C only depends on d, k, m and p. Assuming $\sigma(T) \le \sigma$ for all $T \in \mathcal{T}_h$, we obtain:

$$|u - I_h u|_{H^{m,p}(\Omega)} \le C\sigma^{m+\frac{d}{p}} h^{k+1-m} |u|_{H^{k+1,p}(\Omega)}. \tag{4.11}$$

Proof. Due to the assumptions on p we have that $H^{2,p}(\Omega)$ continuously embeds into $C^0(\bar{\Omega})$

(Theorem 2.41). Thus, $I_h \in \mathcal{L}(H^{k+1,p}, H^{m,p})$. Using Theorem 4.33, we obtain for $u \in H^{k+1,p}(\Omega)$:

$$\begin{aligned} |u - I_h u|^p_{H^{m,p}(\Omega)} &= \sum_{T \in \mathcal{T}_h} |u - I_h u|^p_{H^{m,p}(T)} \\ &\leq C \sum_{T \in \mathcal{T}_h} \sigma(T)^{mp+d} h(T)^{(k+1-m)p} |u|^p_{H^{k+1,p}(T)}. \end{aligned}$$

□

Remark 4.35. *Using $|\det(A)| = |\det(A^{-1})|^{-1}$ and $p = p'$, the norm transformation estimates can be improved to obtain σ^m instead of $\sigma^{m+\frac{d}{p}}$ as a factor in the interpolation error bound.*

Combining Theorem 4.34 with Céa's Lemma, we obtain convergence of finite-element methods:

Theorem 4.36 (A priori error bound)**.** *Let $d \leq 3$ and let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$ such that $\sigma(T) \leq \sigma$ for all $T \in \mathcal{T}_h$. Let $f \in L^2(\Omega)$ and let $u \in H^1_0(\Omega)$ be the weak solution of the homogeneous Dirichlet problem for the Poisson equation with right-hand side f, i.e.,*

$$\int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x = \int_\Omega f v \, \mathrm{d}x \qquad \forall v \in H^1_0(\Omega).$$

Further, for $k \in \mathbb{N}$ let $u_h \in S^k_{h,0}$ be the k-th order finite-element approximation of u, i.e.,

$$\int_\Omega \nabla u_h \cdot \nabla v_h \, \mathrm{d}x = \int_\Omega f v_h \, \mathrm{d}x \qquad \forall v_h \in S^k_{h,0}.$$

Assuming $u \in H^{k+1}(\Omega)$ we have:

$$\|u - u_h\|_{1,2} \leq C h^k |u|_{k+1,2}. \tag{4.12}$$

The constant C only depends on d, k, σ and Ω.

Proof. By construction of I_h we have $I_h(u) \in S^k_{h,0}$ as u vanishes on $\partial\Omega$. Using Theorem 4.4 and Theorem 4.34 ($d/2 < 2 = p$), we obtain:

$$\begin{aligned} \|u - u_h\|_{1,2} &\leq \frac{\|b\|}{c_b} \inf_{v_h \in S^k_{h,0}} \|u - v_h\|_{1,2} \\ &\leq \frac{\|b\|}{c_b} \|u - I_h(u)\|_{1,2} \\ &\leq \frac{\|b\|}{c_b} c \sigma^{1+\frac{d}{2}} h^k |u|_{k+1,2} \end{aligned}$$

□

Remarks 4.37. *1) To obtain an error estimate, we had to assume that $u \in H^2(\Omega)$. By Theorem 3.12, this is true for convex domains or domains or domains with $C^{1,1}$ boundary. The latter obviously does not hold for domains with an admissible triangulation $\mathcal{T}_h$, which have polygonal boundaries. However, if the domains $\bar{\Omega}_h := \bigcup_{T \in \mathcal{T}_h} T$ approximate a domain Ω with $C^{1,1}$ boundary, the theory can still be applied.*

2) Theorem 4.36 holds for all PDEs which lead to abstract elliptic problems in weak formulation (cf. Remark 3.3).

4.4.1 Aubin-Nitsche Lemma and L^2-error bound

Since $\|u - u_h\|_2 \leq \|u - u_h\|_{1,2}$, the a priori bound (4.12) also gives us a bound on the L^2-error:

$$\|u - u_h\|_2 \leq Ch^k |u|_{k+1,2}.$$

However, from Theorem 4.34 we know that

$$\|u - I_h u\|_2 \leq Ch^{k+1} |u|_{k+1,2}.$$

The question remains whether the Galerkin approximation gives us suboptimal convergence rates in the L^2-norm or whether our result can be improved by one factor of h. Indeed, this is possible using the Aubin-Nitsche Lemma:

Theorem 4.38 (Aubin-Nitsche Lemma)**.** *Let V be a Hilbert space and $u \in V$ the solution of an abstract elliptic problem in weak formulation of the form*

$$b(u, v) = \ell(v) \qquad \forall v \in V.$$

Let V_h be a closed subspace of V and $u_h \in V_h$ the Galerkin approximation

$$b(u_h, v_h) = \ell(v_h) \qquad \forall v_h \in V_h.$$

Further, let W be a Hilbert space and $\Phi : V \to W$ a continuous linear operator. Then we have:

$$\|\Phi(u - u_h)\|_W \leq \|b\| \|u - u_h\|_V \sup_{0 \neq w \in W} \frac{\inf_{v_h \in V_h} \|v_w - v_h\|_V}{\|w\|_W},$$

where for given $w \in W$, v_w is the solution of the dual (adjoint) problem

$$b^*(v_w, u) = (w, \Phi(u))_W \qquad \forall u \in V,$$

and $b^*(v, u) := b(u, v)$ for all $u, v \in V$.

Proof. With $e_h := u - u_h$ we have for arbitrary $v_h \in V_h$

$$\begin{aligned}
\|\Phi(e_h)\|_W^2 &= (\Phi(e_h), \Phi(e_h))_W \\
&\overset{\text{def } v_w}{=} b^*(v_{\Phi(e_h)}, e_h) \\
&\overset{\text{def } b^*}{=} b(e_h, v_{\Phi(e_h)}) \\
&\overset{\text{Gal. orth.}}{=} b(e_h, v_{\Phi(e_h)} - v_h) \\
&\leq \|b\| \|e_h\|_V \left\| v_{\Phi(e_h)} - v_h \right\|_V .
\end{aligned}$$

Taking the infimum over $v_h \in V_h$ and dividing by $\|\Phi(e_h)\|_H$ yields the claim. □

Corollary 4.39 (L^2-error bound)**.** *Let the assumptions of Theorem 4.36 be satisfied, and further assume that the H^2-regularity bound (3.7) holds for the given domain Ω and arbitrary right-hand sides $f \in L^2(\Omega)$. Then there is a constant C only depending on d, k, σ, Ω such that*

$$\|u - u_h\|_2 \leq C h^{k+1} |u|_{k+1,2},$$

for all $u \in H^{k+1}(\Omega)$.

Proof. Using Theorem 4.38 with $V = H_0^1(\Omega)$, $W = L^2(\Omega)$ and Φ the inclusion, we have

$$\begin{aligned}
\|u - u_h\|_2 &\leq \|b\| \cdot \|u - u_h\|_{1,2} \cdot \sup_{0 \neq w \in L^2(\Omega)} \frac{\inf_{v_h \in V_h} \|v_w - v_h\|_{1,2}}{\|w\|_2} \\
&\overset{(4.12)}{\leq} \|b\| \cdot C h^k |u|_{k+1,2} \cdot \sup_{0 \neq w \in L^2(\Omega)} \frac{\|v_w - I_h(v_w)\|_{1,2}}{\|w\|_2} \\
&\overset{(4.11),(3.7)}{\leq} \|b\| \cdot C h^k |u|_{k+1,2} \cdot \sup_{0 \neq w \in L^2(\Omega)} \frac{C' h |v_w|_{2,2}}{C'' |v_w|_{2,2}} \\
&= C''' h^{k+1} |u|_{k+1,2}.
\end{aligned}$$

□

4.5 A posteriori estimates and adaptivity

The a priori estimates we proved in the last section show that finite-element approximations indeed converge to the solution of a given elliptic problem when the maximum mesh diameter h tends to zero. Moreover we gained information on the asymptotic rate with which convergence takes place. However, while the obtained rates are optimal, the involved constants, e.g. the Poincaré constant for functions of mean zero or the constant in the regularity estimate (3.7), are hard to determine in practice, and we gain no insight into the spatial distribution of the error.

In the following we will derive so-called a posteriori error bounds of the form

$$\|u - u_h\| \leq \eta_h(u_h), \tag{4.13}$$

with a computable error estimator $\eta_h(u_h)$. Moreover, $\eta_h(u_h)$ will be a sum of local contributions over the elements and facets of $\mathcal{T}_h$. The will allow us to devise adaptive algorithms, which only refine $\mathcal{T}_h$ in regions where the estimated error is too large.

However, only proving a bound of the form (4.13) is insufficient, as $\eta_h(u_h)$ could over-estimate the real error by orders of magnitude. This observation leads us to the following definition:

Definition 4.40 (Reliable and efficient error estimator)**.** *An error estimator $\eta_h(u_h)$ for the error $\|u - u_h\|$ is called reliable iff*

$$\|u - u_h\| \leq \eta_h(u_h).$$

η_h is called efficient with constant $C > 0$ iff

$$C \cdot \eta_h(u_h) \leq \|u - u_h\|.$$

4.5.1 Residual-based error estimators

In the following we will focus on residual-based error estimators. For a given $n \times n$-linear system $Ax = f$, the residual $R(x')$ of a potential solution x' is given by

$$R(x') := f - Ax \in \mathbb{R}^n$$

Hence, the residual encodes the local violation of equation system by our approximation x', In particular, $R(x')$ can be computed without knowledge of the true solution x.

We can see an abstract elliptic problem of the form

$$b(u, v) = \ell(v) \qquad \forall v \in V,$$

as an infinite system of equations over the infinite-dimensional space V. The residual then becomes an element of the dual space of V:

Proposition 4.41 (Residual-based error bound)**.** *Let V be a Hilbert space and $u \in V$ the solution of an abstract elliptic problem of the form*

$$b(u, v) = \ell(v) \qquad \forall v \in V.$$

Then for arbitrary $v \in V$ we define the residual of $R(v) \in V'$ of v as the functional given by

$$R(v)[w] := \ell(w) - b(v, w).$$

We define the error estimator

$$\eta(v) := \frac{1}{c_b}\|R(v)\|_{V'}$$

where c_b denotes the coercivity (or inf-sup) constant of b. Then we have:

$$\|u - v\| \leq \eta(v) \leq \frac{\|b\|}{c_b}\|u - v\|,$$

i.e., $\eta(v)$ is a reliable and efficient error estimator for $\|u - v\|$ with efficiency constant $c_b/\|b\|$.

Proof. Let $B : V \to V'$ be the operator associated with b (Proposition 2.19). In particular, $\|B\| = \|b\|$ and B is bounded from below with constant c_b (Proposition 2.22), i.e., $\|B^{-1}\| \leq 1/c_b$. Noting that

$$B(u - v) = \ell - b(v, \cdot) = R(v),$$

we have

$$\|u - v\| = \left\|B^{-1}(R(v))\right\| \leq \left\|B^{-1}\right\|\|R(v)\| \leq \frac{1}{c_b}\|R(v)\|$$

and

$$\|R(v)\| \leq \|B\|\|u - v\| = \|b\|\|u - v\|.$$

□

Corollary 4.42 (Error bound for the Poisson problem)**.** *For the Poisson problem (3.3) in weak formulation let the residual Operator $R : H_0^1(\Omega) \to H^{-1}(\Omega)$ be given by*

$$R(v) := (f, \cdot)_2 - b(v, \cdot) \qquad \forall v \in H_0^1(\Omega).$$

For the error estimator

$$\eta(v) := (1 + c_p^2) \cdot \|R(v)\|_{-1,2}$$

where c_p is the Poincaré constant from Theorem 2.43 we have for arbitrary $v \in H_0^1(\Omega)$:

$$\|u - v\|_{1,2} \leq \eta(v) \leq (1 + c_p^2)\|u - v\|_{1,2}.$$

Proof. In the proof of Proposition 3.7 we have seen that $\|b\| = 1$ and $c_b = 1/(1 + c_p^2)$. □

We have seen that the norm of $R(v)$ can give us an upper bound on the global approximation error $\|u - v\|$. However, the computation of this norm would, by definition of the norm on V', entail the solution of an infinite-dimensional minimization problem. Further, the norm of $R(v)$ does not give us any information about the error distribution. Therefore we want to localize the residual:

Lemma 4.43 (Localization of the residual)**.** *For the weak Poisson problem (3.3) let $u_h \in S_{h,0}^k$ the finite-element approximation for an admissible triangulation $\mathcal{T}_h$. Then for all $\varphi \in H_0^1(\Omega)$*

and all $\varphi_h \in S^k_{h,0}$ we have

$$R(u_h)[\varphi] = \sum_{T\in\mathcal{T}_h} (f + \Delta u_h, \varphi - \varphi_h)_{L^2(T)} + \sum_{S\in\mathcal{F}^i_h} (\llbracket \nabla u_h \cdot n \rrbracket, \varphi - \varphi_h)_{L^2(S)}. \tag{4.14}$$

Here,

$$\mathcal{F}_h := \{S \mid S \text{ facet of } T,\ T \in \mathcal{T}_h\}, \qquad \mathcal{F}^i_h := \{S \in \mathcal{F}_h \mid S \not\subset \partial\Omega\}$$

denote the sets of facets and of inner facets of $\mathcal{T}_h$. Further, Δu_h denotes the classical Laplacian of u_h (as a polynomial on T), and $\llbracket \nabla u_h \cdot n \rrbracket := (\nabla u_h|_{T_2} - \nabla u_h|_{T_1}) \cdot n$ denotes the jump of the normal derivative of u_h across the facet S shared by T_1 and T_2 with n pointing from T_1 to T_2.

Proof. By definition of the residual we have

$$R(u_h)[\varphi] = \int_\Omega f\varphi \,\mathrm{d}x - \int_\Omega \nabla u_h \cdot \nabla\varphi \,\mathrm{d}x = \sum_{T\in\mathcal{T}_h} \int_T f\varphi - \nabla u_h \cdot \nabla\varphi \,\mathrm{d}x.$$

Since $u_h|_T \in \mathbb{P}^k(T)$ we can perform partial integration on T to obtain:

$$\int_T \nabla u_h \cdot \nabla\varphi = \int_{\partial T} \nabla u_h|_T \cdot n\varphi \,\mathrm{d}x + \int_T -\Delta u_h \varphi \,\mathrm{d}\sigma.$$

As φ vanishes on $\partial\Omega$ we obtain:

$$R(u_h)[\varphi] = \sum_{T\in\mathcal{T}_h} \Big((f + \Delta u_h, \varphi)_{L^2(T)} - \sum_{S\subset\partial T\setminus\partial\Omega} (\nabla u_h|_T \cdot n, \varphi)_{L^2(S)} \Big).$$

Noting that each $S \in \mathcal{F}^i_h$ appears twice in the double sum and collecting the corresponding terms we obtain

$$R(u_h)[\varphi] = \sum_{T\in\mathcal{T}_h} (f + \Delta u_h, \varphi)_{L^2(T)} + \sum_{S\subset\mathcal{F}^i_h} (\llbracket \nabla u_h \cdot n \rrbracket, \varphi)_{L^2(S)}.$$

Finally note that $R(u_h)[\varphi_h] = 0$ for all $\varphi_h \in S^k_h$ since u_h is the Galerkin approximation. Hence $R(u_h)[\varphi] = R(u_h)[\varphi - \varphi_h]$, from which the claim follows. ☐

Remark 4.44. *To obtain a localized bound for $\|R(u_h)\|_{H^{-1}(\Omega)}$, we could proceed with*

$$|R(u_h)(\varphi)| \le \sum_{T\in\mathcal{T}_h} \|f + \Delta u_h\|_{L^2(T)} \|\varphi - \varphi_h\|_{L^2(T)} + \sum_{S\in\mathcal{F}^i_h} \|\llbracket \nabla u_h \cdot n \rrbracket\|_{L^2(S)} \|\varphi - \varphi_h\|_{L^2(S)}.$$

Finally we can take the infimum of this estimate over all $\varphi_h \in S^k_{h,0}$, which gives us the terms $\inf_{\varphi_h\in S^k_{h,0}} \|\varphi - \varphi_h\|_{L^2(T)}$, that we would like to bound using an interpolation error estimate. However, φ only has H^1-regularity, so we cannot use Lagrange interpolation as we did in for the a priori estimate to bound this infimum. Hence, we need to develop an interpolation operator, *which is well-defined for arbitrary H^1-functions.*

4.5.2 H^1-Interpolation

As point-evaluations are not well-defined for functions in $H^1(\Omega)$ with $\Omega \subset \mathbb{R}^d$, $d > 1$, we want to construct an interpolation operator $I : H^1(\Omega) \to S_h^k$ which is based on local averages.

A classical approach to define an H^1-interpolation operator goes back to Clément [Clé75], who defined $I(v) \in S_h^k$ to be the finite-element function such that for all Lagrange nodes $x \in G_k(\mathcal{T}_h)$:

$$I(v)[x] = P_{\omega_0(x)}(v),$$

where

$$\omega_0(x) := \{T \in \mathcal{T}_h \mid x \in T\} \tag{4.15}$$

and $P_{\omega_0(x)} : L^2(\Omega) \to \mathbb{P}^k(\omega_0(x))$ is the L^2-orthogonal projection. We will pursue a slightly different approach as presented in [EG17] by considering element-wise L^2-projections onto broken finite-element spaces which are then averaged in the Lagrange nodes. An advantage of this approach is that the resulting interpolation operator can be implemented more easily in a computer program by avoiding projections onto local patch spaces $\omega_0(x)$. Further, the approach easily generalizes to vector-valued finite-element spaces.

As a new tool, we will required a quantified version of the trace theorem (Theorem 2.36):

Proposition 4.45 (Scaled trace inequality). *Let $T \subset \mathbb{R}^d$ be a d-simplex and let S be a facet of T. Then there is a constant C only depending on d, such that for $v \in H^1(T)$ we have:*

$$\|v\|_{L^2(S)} \leq C\sigma(T)^{d/2}\Big(h(T)^{-1/2}\|v\|_{L^2(T)} + h(T)^{1/2}|v|_{H^1(T)}\Big). \tag{4.16}$$

Proof. Exercise. □

Proposition 4.46 (L^2-projection onto broken finite element space). *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$ with $\sigma(T) \leq \sigma$ for all $T \in \mathcal{T}_h$ and let*

$$S_h^{k,b} := \{v \in L^2(\Omega) \mid v|_T \in \mathbb{P}^k(T)\ \forall T \in \mathcal{T}_h\}$$

be the corresponding broken finite-element space of piecewise polynomial functions of order k. Denoting by $P_{S_h^{k,b}}$ the L^2-orthogonal projection onto $S_h^{k,b}$, for each $v \in H^1(\Omega)$, $T \in \mathcal{T}_h$ and $m \in \{0,1\}$ we have

$$|v - P_{S_h^{k,b}}(v)|_{H^m(T)} \leq Ch(T)^{1-m}|v|_{H^1(T)}. \tag{4.17}$$

The constant C only depends on d, k, m and σ. Further, for any inner facet S of T we have:

$$\left\| [\![P_{S_h^{k,b}}(v)]\!] \right\|_{L^2(S)} \leq C(h(T)^{1/2}|v|_{H^1(T)} + h(T')^{1/2}|v|_{H^1(T')}), \tag{4.18}$$

where $T' \in \mathcal{T}_h$ denotes the element sharing S with T. Similarly, for $v \in H_0^1(\Omega)$ and a boundary facet S of T we have:

$$\left\| P_{S_h^{k,b}}(v) \right\|_{L^2(S)} \leq Ch(T)^{1/2}|v|_{H^1(T)}. \tag{4.19}$$

In both cases, the constant C only depends on d, k and σ.

Proof. Denote by $P_T^k : H^1(T) \to \mathbb{P}^k(T)$ the L^2-orthogonal projection onto the local polynomial space on T. By definition of $S_h^{k,b}$ we have $P_{S_h^{k,b}}(v)|_T = P_T^k(v|_T)$. Since the reference map $F : \hat{T} \to T$ is affine, we observe that

$$\begin{aligned}
0 &= (v|_T - P_T^k(v|_T), q)_{L^2(T)} \\
&= \int_T (v|_T - P_T^k(v|_T)) \cdot q \,\mathrm{d}x \\
&= \int_{\hat{T}} (v|_T \circ F - P_T^k(v|_T) \circ F) \cdot q \circ F \,\mathrm{d}x \cdot |\det A|
\end{aligned}$$

for all $q \in \mathbb{P}^k(T)$. As $\circ F$ bijectively maps $\mathbb{P}^k(T)$ onto $\mathbb{P}^k(\hat{T})$ it follows that

$$P_T^k(v|_T) \circ F = P_{\hat{T}}^k(v|_T \circ F).$$

Since $P_{\hat{T}}^k$ is the identity on $\mathbb{P}^k(\hat{T})$, (4.17) follows directly from Theorem 4.33.

To show (4.18) note that since v has a unique trace on S we have

$$\begin{aligned}
&\left\| [\![P_{S_h^{k,b}}(v)]\!] \right\|_{L^2(S)} \\
&\quad = \left\| [\![v - P_{S_h^{k,b}}(v)]\!] \right\|_{L^2(S)} \\
&\quad \leq \left\| v - P_T^k(v) \right\|_{L^2(S)} + \left\| v - P_{T'}^k(v) \right\|_{L^2(S)} \\
&\quad \leq C\Big(h(T)^{-1/2}\left\| v - P_T^k(v) \right\|_{L^2(T)} + h(T)^{1/2}|v - P_T^k(v)|_{H^1(T)}\Big) \\
&\qquad\quad + C\Big(h(T')^{-1/2}\left\| v - P_{T'}^k(v) \right\|_{L^2(T')} + h(T')^{1/2}|v - P_{T'}^k(v)|_{H^1(T')}\Big) \\
&\quad \leq C'\Big(h(T)^{1/2}|v|_{H^1(T)} + h(T)^{1/2}|v|_{H^1(T)}\Big) \\
&\qquad\quad + C'\Big(h(T')^{1/2}|v|_{H^1(T')} + h(T')^{1/2}|v|_{H^1(T')}\Big).
\end{aligned}$$

The proof for (4.19) is the same. □

Proposition 4.47 (Averaging Operator)**.** *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$ with $\sigma(T) \leq T$ for all $T \in \mathcal{T}_h$. Let S_h^k be the associated Lagrange finite-element space with Lagrange lattice $G_k(\mathcal{T}_h)$, and let $S_h^{k,b}$ be the corresponding broken space. Further, for $x \in \overline{\Omega}$ let $\omega_0(x) := \{T \in \mathcal{T}_h \mid x \in T\}$. Then we define the averaging operator $I_{av} : S_h^{k,b} \to S_{h,0}^k$ s.t. for $v_h \in S_h^{k,b}$ and $x \in G_k(\mathcal{T}_h)$ we have*

$$I_{av}(v_h)[x] = \begin{cases} \frac{1}{|\omega_0(x)|} \sum_{T \in \omega_0(x)} v_h|_T(x) & x \in \Omega \\ 0 & x \in \partial\Omega. \end{cases}$$

Then, with $\omega_1(T) := \{S \text{ facet of } T' \in \mathcal{T}_h \mid S \cap T \neq \emptyset\}$ we have for all $v_h \in S_h^{k,b}$, $T \in \mathcal{T}_h$ and $m \in \{0,1\}$:

$$|v_h - I_{av}(v_h)|_{H^m(T)} \leq Ch(T)^{-m+1/2} \sum_{S \in \omega_1(T)} \|[\![v_h]\!]\|_{L^2(S)}.$$

Here we use the convention $\|[\![v_h]\!]\|_{L^2(S)} = \|v_h\|_{L^2(S)}$ for $S \subset \partial\Omega$. The constant C only depends on d, k, m and σ.

Proof. First assume that $T \cap \partial\Omega = \emptyset$. Noting that all (semi-)norms on the finite-dimensional space $\mathbb{P}^k(\hat{T})$ can be bounded by an arbitrary norm on $\mathbb{P}^k(\hat{T})$, we have for $v_h \in S_h^{k,b}$:

$$\begin{aligned}
&|v_h - I_{av}(v_h)|_{H^m(T)} \\
&\quad \leq Ch(T)^{-m+d/2} |(v_h - I_{av}(v_h)) \circ F|_{H^m(\hat{T})} \\
&\quad \leq C'h(T)^{-m+d/2} \max_{\hat{x} \in G_k(\hat{T})} |((v_h|_T - I_{av}(v_h)) \circ F)(\hat{x})| \\
&\quad = C'h(T)^{-m+d/2} \max_{x \in G_k(T)} \Big| v_h|_T(x) - 1/|\omega_0(x)| \sum_{T' \in \omega_0(x)} v_h|_{T'}(x) \Big| \\
&\quad \leq C'h(T)^{-m+d/2} \max_{x \in G_k(T)} \sum_{T' \in \omega_0(x)} 1/|\omega_0(x)| \cdot \underbrace{\Big| v_h|_T(x) - v_h|_{T'}(x) \Big|}_{A_x^{T'}}.
\end{aligned}$$

$A_x^{T'}$ vanishes for $T = T'$. If T and T' share a common facet S (on which x then has to lie), we have

$$A_x^{T'} \leq \|[\![v_h]\!]\|_{L^\infty(S)}.$$

Otherwise there is a sequence of elementes $T_k \in \omega_0(x)$, $k = 1, \dots K$ such that T_k and T_{k+1} share a common facet S, $x \in S$ and $T_1 = T$, $T_K = T'$. Note that $K \leq |\omega_0(x)| \leq C$, where C only depends on d and σ. We then have

$$\begin{aligned}
A_x^{T'} &\leq \sum_{k=1}^{K-1} \Big| v_h|_{T_k}(x) - v_h|_{T_{k+1}}(x) \Big| \\
&\leq \sum_{k=1}^{K-1} \|[\![v_h]\!]\|_{L^\infty(T_k \cap T_{k+1})} \\
&\leq \sum_{S \in \omega_1(T)} \|[\![v_h]\!]\|_{L^\infty(S)}.
\end{aligned}$$

Further note that for a $d-1$-simplex S with reference simplex $\hat{S}$ and any $q \in \mathbb{P}_k(S)$ the following inverse inequality holds:

$$\begin{aligned} \|q\|_{L^\infty(S)} &= \|\hat{q}\|_{L^\infty(\hat{S})} \\ &\le C\|\hat{q}\|_{L^2(\hat{S})} \\ &\le C' h(S)^{-(d-1)/2}\|q\|_{L^2(S)}. \end{aligned}$$

Here, we again have exploited that all norms on $\mathbb{P}_k(\hat{S})$ are equivalent. Further note that $h(T) \le Ch(S)$ for all $S \in \omega_1(T)$, where C only depends on σ. Thus we obtain:

$$\begin{aligned} |v_h - I_{av}(v_h)|_{H^m(T)} &\le C' h(T)^{-m+d/2} \max_{x \in G_k(T)} \sum_{S \in \omega_1(T)} \|[\![v_h]\!]\|_{L^\infty(S)} \\ &\le C'' h(T)^{-m+d/2} \sum_{S \in \omega_1(T)} h(S)^{-(d-1)/2} \|[\![v_h]\!]\|_{L^2(S)} \\ &\le C''' h(T)^{-m+1/2} \sum_{S \in \omega_1(T)} \|[\![v_h]\!]\|_{L^2(S)} \end{aligned}$$

Finally, if $T \cap \partial\Omega \neq \emptyset$, then for any $x \in G_k(T) \cap \partial\Omega$ we have

$$|v_h(x) - I_{av}(v_h)(x)| = |v_h(x)| \le \|v_h\|_{L^\infty(S)} \le \|[\![v_h]\!]\|_{L^\infty(S)}$$

where $x \in S \in \omega_1(T) \cap \partial\Omega$. $\square$

Corollary 4.48 (H^1_0-Interpolation)**.** *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$ with $\sigma(T) \le T$ for all $T \in \mathcal{T}_h$. Then $I_h^{1,0} := I_{av} \circ P_{S_h^{k,b}} : H^1_0(\Omega) \to S^k_{h,0}$ is the identity on $S^k_{h,0}$. Further, for $v \in H^1_0(\Omega)$, $m \in \{0,1\}$ we have:*

$$|v - I_h^{1,0}(v)|_{H^m(T)} \le Ch(T)^{1-m}|v|_{H^1(\omega_0(T))},$$

with the local element patch $\omega_0(T)$ defined as in (4.15). The constant C only depends on d, k, m and σ.

Proof. For $v \in S^k_{h,0}$ we have $v|_T \in \mathbb{P}^k(T)$. Hence, $P_{S_h^{k,b}}(v) = v$. In particular, $P_{S_h^{k,b}}(v)$ is continuous in all Lagrange nodes and vanishes on $\partial\Omega$. Thus, also $I_{av}(P_{S_h^{k,b}}(v)) = I_{av}(v) = v$.

Further, we have

$$
\begin{aligned}
|v - I_h^{1,0}(v)|_{H^m(T)} &\le |v - P_{S_h^{k,b}}(v)|_{H^m(T)} + |P_{S_h^{k,b}}(v) - I_{av}(P_{S_h^{k,b}}(v))|_{H^m(T)} \\
&\le Ch(T)^{1-m}|v|_{H^1(T)} + C'h(T)^{-m+1/2} \sum_{S\in\omega_1(T)} \left\| [\![P_{S_h^{k,b}}(v)]\!] \right\|_{L^2(S)} \\
&\le Ch(T)^{1-m}|v|_{H^1(T)} + C'h(T)^{-m+1/2} \sum_{T'\in\omega_0(T)} C''h(T')^{1/2}|v|_{H^1(T')} \\
&\le C'''h(T)^{1-m} \sum_{T'\in\omega_0(T)} |v|_{H^1(T')} \\
&\le C''''h(T)^{1-m}|v|_{H^1(\omega_0(T))}.
\end{aligned}
$$

Here we exploited $h(T') \le Ch(T)$ for all $T' \in \omega_0(T)$ and $|\omega_0(T)| \le C'$ with constants C, C' only depending on σ. $\square$

4.5.3 Localized error estimator

We now continue our discussion from Remark 4.44 and use $I_h^{1,0}$ to obtain a localized error estimator:

Theorem 4.49 (Reliable localized error estimator)**.** *With the assumptions of Lemma 4.43, $\sigma(T) \le \sigma$, C the corresponding constant from Corollary 4.48 define*

$$
\eta_h(u_h) := C \cdot \Big(\sum_{T\in\mathcal{T}_h} \eta_T(u_h)^2 + \sum_{S\in\mathcal{F}_h^i} \eta_S(u_h)^2 \Big)^{1/2} \tag{4.20}
$$

with local error indicators

$$
\begin{aligned}
\eta_T(u_h) &:= h(T) \cdot \|f + \Delta u_h\|_{L^2(T)}, \\
\eta_S(u_h) &:= h(S)^{1/2} \cdot \|[\![\nabla u_h \cdot n]\!]\|_{L^2(S)}.
\end{aligned}
$$

Then we have

$$
\|u - u_h\|_{1,2} \le \eta_h(u_h).
$$

The constant C only depends on d, k, σ and c_p.

Proof. By Corollary 4.42 we only need an appropriate bound on $\|R(u_h)\|_{-1,2}$. To obtain this bound, we choose $v_h := I_h^{1,0}(v)$ in (4.14) and pick for each $S \in \mathcal{F}_h^i$ a $S \subset T_S \in \mathcal{T}_h$. Proceeding as in Remark 4.44, we have

$$
\begin{aligned}
&R(u_h)(v)\\
&\quad\le \sum_{T\in\mathcal{T}_h} \|f+\Delta u_h\|_{L^2(T)} \left\|v - I_h^{1,0}(v)\right\|_{L^2(T)}\\
&\qquad + \sum_{S\in\mathcal{F}_h^i} \|[\![\nabla u_h\cdot n]\!]\|_{L^2(S)} \left\|v - I_h^{1,0}(v)\right\|_{L^2(S)}\\
&\quad\overset{4.45}{\le} \sum_{T\in\mathcal{T}_h} \eta_T(u_h) h(T)^{-1} \left\|v - I_h^{1,0}(v)\right\|_{L^2(T)}\\
&\qquad + C\sum_{S\in\mathcal{F}_h^i} \eta_S(u_h) h(S)^{-1/2}\cdot \Big(h(T_S)^{-1/2}\left\|v - I_h^{1,0}(v)\right\|_{L^2(T_S)} + h(T_S)^{1/2}|v - I_h^{1,0}(v)|_{H^1(T_S)}\Big)\\
&\quad\overset{4.48}{\le} C'\sum_{T\in\mathcal{T}_h} \eta_T(u_h)|v|_{H^1(\omega_0(T))}\\
&\qquad + C''\sum_{S\in\mathcal{F}_h^i} \eta_S(u_H) h(S)^{-1/2}\cdot \underbrace{\left(h(T_S)^{-1/2}h(T_S) + h(T_S)^{1/2}\right)}_{h(T_S)^{1/2}\le Ch(S)^{1/2}} |v|_{H^1(\omega_0(T_S))}\\
&\quad\overset{\text{C.S.}}{\le} C'''\Big(\sum_{T\in\mathcal{T}_h}\eta_T(u_h)^2 + \sum_{S\in\mathcal{F}_h^i}\eta_S(u_h)^2\Big)^{1/2}\cdot\Big(\sum_{T\in\mathcal{T}_h}|v|^2_{H^1(\omega_0(T))} + \sum_{S\in\mathcal{F}_h^i}|v|^2_{H^1(\omega_0(T_S))}\Big)^{1/2}\\
&\quad\le C''''\eta_h(u_h)\cdot|v|_{H^1(\Omega)}
\end{aligned}
$$

Dividing by $\|v\|_{2,1}$ and taking the supremum over v yields the claim. □

Remark 4.50. *While we did not track every factor entering the constant C in the definition of $\eta_h(u_h)$, we remark that all constants except c_p only depend on local geometric properties and can be calculated. Upper bounds for c_p can also easily be found.*

4.5.4 Efficiency of η_h

In Theorem 4.49 we have shown that the localized error estimator $\eta_h(u_h)$ given by (4.20) is a reliable upper bound for the finite-element error. Our next goal will be to show the efficiency of $\eta_h(u_h)$. In fact, we will show local efficiency estimates of the form

$$\eta_{T/S}(u_h) \le C\|\nabla(u-u_h)\|_{L^2(U(T/S))} + \text{'data approximation error'},$$

where $U(T/S)$ is some local neighborhood of T or S respectively. To this end, we will test the residual with appropriate 'bubble' functions which are supported on a single element T or around a single facet S.

Lemma 4.51 (Element bubbles)**.** *For each d-simplex T we can define a function $\beta_T \in H_0^1(T)$*

such that for all $q \in \mathbb{P}^k(T)$ we have

$$\|\beta_T q\|_{L^2(T)} \leq \|q\|_{L^2(T)} \leq C_1 \left\|\beta_T^{1/2} q\right\|_{L^2(T)}, \tag{4.21}$$

$$|\beta_T q|_{H^1(T)} \leq C_2 h(T)^{-1} \|q\|_{L^2(T)}, \tag{4.22}$$

where the constants C_1, C_2 only depend on d, k and $\sigma(T)$.

Proof. On the standard simplex $\hat{T}$ choose an arbitrary non-zero smooth function $\beta_{\hat{T}}$ with $0 \leq \beta_{\hat{T}} \leq 1$ that vanishes on $\partial\hat{T}$. Then clearly $\|\beta_{\hat{T}} q\|_2 \leq \|q\|_2$. The other inequalities hold since all (semi-)norms on $\mathbb{P}^k(\hat{T})$ can be bounded by an arbitrary other norm on $\mathbb{P}^k(\hat{T})$. Defining $\beta_T := \beta_{\hat{T}} \circ F^{-1}$, the inequalities for an arbitrary simplex T follow using Corollary 4.32. □

Theorem 4.52 (Efficiency of element indicators)**.** *With the assumptions of Theorem 4.49 and $l \in \mathbb{N}_0$, we have for each $T \in \mathcal{T}_h$:*

$$\eta_T(u_h) \leq C(|u - u_h|_{H^1(T)} + h(T) \inf_{v_h \in \mathbb{P}^l(T)} \|f - v_h\|_{L^2(T)}).$$

The constant C only depends on d, k, l and σ.

Proof. Fix some $v_h \in \mathbb{P}^l(T)$. Then we have

$$\eta_T(u_h) = h(T) \cdot \|f + \Delta u_h\|_{L^2(T)} \leq h(T) \underbrace{\|v_h + \Delta u_h\|_{L^2(T)}}_{A} + h(T)\|f - v_h\|_{L^2(T)}.$$

To bound A we use Lemma 4.51 with $k := l$ to obtain:

$$\begin{aligned}
A^2 &= \|v_h + \Delta u_h\|^2_{L^2(T)} \\
&\overset{(4.21)}{\leq} C\left\|\beta_T^{1/2}(v_h + \Delta u_h)\right\|^2_{L^2(T)} \\
&= C \int_T (v_h + \Delta u_h)\beta_T(v_h + \Delta u_h)\,\mathrm{d}x \\
&= C \int_T (f + \Delta u_h)\beta_T(v_h + \Delta u_h)\,\mathrm{d}x + C\int_T (v_h - f)\beta_T(v_h + \Delta u_h)\,\mathrm{d}x \\
&\overset{(*)}{=} C \int_T (\nabla u - \nabla u_h) \cdot \nabla(\beta_T(v_h + \Delta u_h))\,\mathrm{d}x + C\int_T (v_h - f)\beta_T(v_h + \Delta u_h)\,\mathrm{d}x \\
&\leq C|u - u_h|_{H^1(T)} \cdot |\beta_T(v_h + \Delta u_h)|_{H^1(T)} + C\|v_h - f\|_{L^2(T)} \cdot \|\beta_T(v_h + \Delta u_h)\|_{L^2(T)} \\
&\overset{(4.21),(4.22)}{\leq} C'(h(T)^{-1}|u - u_h|_{H^1(T)} + \|v_h - f\|_{L^2(T)})A.
\end{aligned}$$

In $(*)$ we have used the definition of u and the fact that the support of β_T is contained in T. □

Remark 4.53 (Data approximation error)**.** *The term $\inf_{v_h \in \mathbb{P}^l(T)} \|f - v_h\|_{L^2(T)}$ measures how* *well the 'data' function f is approximated by $S_h^{l,b}$ in the L^2-norm.*

- *To see that this term cannot be neglected in the efficiency estimate, one can consider a function f which is supported on a single element T and which oscillates with a high frequency (exercise).*

- *l can be chosen arbitrarily. However, the constants from Lemma 4.51 explode for $l \to \infty$. In general, $\inf_{v_h \in \mathbb{P}^l(T)} \|f - v_h\|_{L^2(T)}$ should converge with higher order than $|u - u_h|_{H^1(T)}$. For linear finite-elements, this is already the case for $l = 0$.*

- *For general f, the indicator $\|f + \Delta u_h\|_{L^2(T)}$ has to be approximated using a numerical quadrature, which leads to similar error terms.*

Lemma 4.54 (Facet bubbles)**.** *For a facet $\hat{S}$ of the standard simplex $\hat{T}$ let $P_{\hat{S}}^{\hat{T}} : \hat{P} \to \hat{S}$ be the affine map which orthogonally projects $x \in \hat{T}$ to points on $\hat{S}$. For a facet S of an arbitrary simplex T, we define corresponding affine maps $P_S^T := F \circ P_{F^{-1}(S)}^{\hat{T}} \circ F^{-1} : T \to S$ via the reference map F. Using these affine projections, we can define lifting operators $\Lambda_S^T : \mathbb{P}^k(S) \to \mathbb{P}^k(T)$ via $\Lambda_S^T(q) := q \circ P_S^T$.*

Further, for each inner facet $S \in \mathcal{F}_h^i$, let $\omega_0^*(S) := \{T \in \mathcal{T}_h \mid S \subset T\}$. We then can define a function $\beta_S \in H_0^1(\omega_0^*(S))$ such that for all $q \in \mathbb{P}^k(S)$ and $T \in \omega_0^*(S)$ we have:

$$\|\beta_S q\|_{L^2(S)} \le \|q\|_{L^2(S)} \le C_1 \left\|\beta_S^{1/2} q\right\|_{L^2(S)}, \tag{4.23}$$

$$\left\|\beta_S \Lambda_S^T(q)\right\|_{L^2(T)} \le C_2 h(S)^{1/2} \|q\|_{L^2(S)}, \tag{4.24}$$

$$|\beta_S \Lambda_S^T(q)|_{H^1(T)} \le C_3 h(S)^{-1/2} \|q\|_{L^2(S)}, \tag{4.25}$$

where the constants C_1, C_2, C_3 only depend on d, k and $\sigma(T)$.

Proof. The proof for (4.23) is the same as for Lemma 4.51. For Eqs. (4.24) and (4.25) we additionally use inverse inequalities to bound norms on $\hat{T}$ by norms on $\hat{S}$. □

Theorem 4.55 (Efficiency of facet indicators)**.** *With the assumptions of Theorem 4.49 and $l \in \mathbb{N}_0$, we have for each $S \in \mathcal{F}_h^i$:*

$$\eta_S(u_h) \le C \sum_{T \in \omega_0^*(S)} (|u - u_h|_{H^1(T)} + h(T) \inf_{v_h \in \mathbb{P}^l(T)} \|f - v_h\|_{L^2(T)}).$$

The constant C only depends on d, k, l and σ.

Proof. Using Lemma 4.54 with $k := k - 1$ we obtain:

$$\|[\![\nabla u_h \cdot n]\!]\|_{L^2(S)}^2 \overset{(4.23)}{\le} C \left\|\beta_S^{1/2} [\![\nabla u_h \cdot n]\!]\right\|_{L^2(S)}^2$$

$$
\begin{aligned}
&= C \int_S [\![\nabla u_h \cdot n]\!] \beta_S [\![\nabla u_h \cdot n]\!] \,\mathrm{d}x \\
&\overset{(*)}{=} -C \sum_{T \in \omega_0^*(S)} \int_T \nabla \cdot (\nabla u_h \beta_S \Lambda_S^T([\![\nabla u_h \cdot n]\!])) \,\mathrm{d}x \\
&= -C \sum_{T \in \omega_0^*(S)} \int_T \nabla u_h \cdot \nabla(\beta_S \Lambda_S^T([\![\nabla u_h \cdot n]\!])) \,\mathrm{d}x \\
&\qquad\qquad + \int_T \Delta u_h (\beta_S \Lambda_S^T([\![\nabla u_h \cdot n]\!])) \,\mathrm{d}x \\
&\overset{(**)}{=} -C \sum_{T \in \omega_0^*(S)} \int_T \nabla(u_h - u) \cdot \nabla(\beta_S \Lambda_S^T([\![\nabla u_h \cdot n]\!])) \,\mathrm{d}x \\
&\qquad\qquad + \int_T (f + \Delta u_h)(\beta_S \Lambda_S^T([\![\nabla u_h \cdot n]\!])) \,\mathrm{d}x \\
&\leq C \sum_{T \in \omega_0^*(S)} |u_h - u|_{H^1(T)} \cdot |\beta_S \Lambda_S^T([\![\nabla u_h \cdot n]\!])|_{H^1(T)} \\
&\qquad\qquad + \|f + \Delta u_h\|_{L^2(T)} \cdot \left\| \beta_S \Lambda_S^T([\![\nabla u_h \cdot n]\!]) \right\|_{L^2(T)} \\
&\overset{(4.24),(4.25)}{\leq} C' \sum_{T \in \omega_0^*(S)} |u_h - u|_{H^1(T)} \cdot h^{-1/2}(S) \|[\![\nabla u_h \cdot n]\!]\|_{L^2(S)} \\
&\qquad\qquad + \|f + \Delta u_h\|_{L^2(T)} \cdot h(S)^{1/2} \|[\![\nabla u_h \cdot n]\!]\|_{L^2(S)}
\end{aligned}
$$

In $(*)$ we have used the Gauß theorem, the definition of $[\![\nabla u_h \cdot n]\!]$ and the fact that β_S vanishes on all facets of T except S. In $(**)$ we have used the definition of u and the fact that β_S is supported on $\omega_0^*(S)$.

Dividing by $\|[\![\nabla u_h \cdot n]\!]\|_{L^2(S)}$ we obtain:

$$\eta_S(u_h) \leq C' \sum_{T \in \omega_0^*(S)} |u_h - u|_{H^1(T)} + \|f + \Delta u_h\|_{L^2(T)} \cdot h(S).$$

Using $h(S) \leq h(T)$ and the efficiency bound for $\eta_T(u_h)$ yields the claim. □

4.5.5 Adaptive finite-element methods

We assume that we are given a localized error estimator of the form

$$\|u - u_h\|^2 \leq \eta_h(u_h)^2 := C^2 \cdot \sum_{T \in \mathcal{T}_h} \eta_T(u_h)^2.$$

For the estimator from Section 4.5.3, for instance, we could use the local indicators

$$\hat{\eta}_T(u_h)^2 := \eta_T(u_h)^2 + \frac{1}{2} \sum_{\substack{S \in \mathcal{F}_h^i \\ S \subset T}} \eta_S(u_h)^2.$$

Based on this estimator, we want to develop a locally adaptive scheme which ensures that $\|u - u_h\|$ is bounded by some prescribed error tolerance. One such scheme is realized by the equal-distribution strategy:

Definition 4.56 (Equal-distribution strategy)**.** *Given an admissible macro-triangulation $\mathcal{T}_h$ of a domain $\Omega \subset \mathbb{R}^d$, a target error tolerance $TOL > 0$ and a coarsening parameter $\Theta \in (0, 1)$, we define the following adaptive mesh-refinement algorithm:*

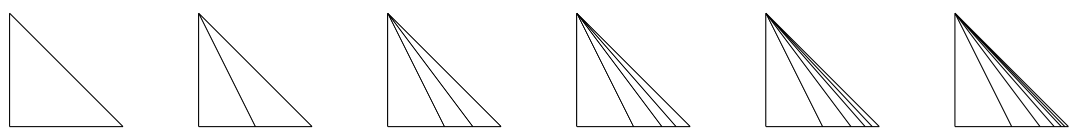
$\textsc{EqualDistribution}(\mathcal{T}_h, TOL, \Theta)$

repeat
    $[u_h, \eta_h(u_h)] := FEM(\mathcal{T}_h)$
    if $\eta_h(u_h) > TOL$
      then
        for $T \in \mathcal{T}_h$
        do *(mark elements for refinement)*
          $M_h(T) := 0$
          if $\eta_T(u_h)^2 > \frac{TOL^2}{|\mathcal{T}_h|}$
            then $M_h(T) := 1$
          if $\eta_T(u_h)^2 < \Theta \cdot \frac{TOL^2}{|\mathcal{T}_h|}$
            then $M_h(T) := -1$
        $\mathcal{T}_h := \textsc{Adapt}(M_h, \mathcal{T}_h)$
  until $\eta_h(u_h) \leq TOL$
return $(\mathcal{T}_h, u_h, \eta_h(u_h))$

Here, $M_h : \mathcal{T}_h \to \{-1, 0, 1\}$ is an indicator function which indicates whether a given element $T \in \mathcal{T}_h$ should be refined, coarsened or remain unchanged.

When convergent, the equal-distribution strategy yields a triangulation $\mathcal{T}_h$ with corresponding finite-element solution u_h such that $\eta_h(u_h) \leq$ TOL. This is achieved by refining or coarsening elements such that all local inicators $\eta_T(u_h)^2$ are of the same order of magnitude $TOL^2/|\mathcal{T}_h|$.

Remark 4.57 (Mesh-adaption strategies)**.** *When refining or coarsening a triangulation, shape regularity ($\sigma(T) \leq \sigma$) has to be ensured to obtain a convergent finite-element scheme. A simple example with $\sigma \to \infty$ is obtained when the same element is repeatedly refined by splitting the element along the same edge:*



To avoid such issues, different refinement strategies are available, such as red-green refinement or marked edge bisection.

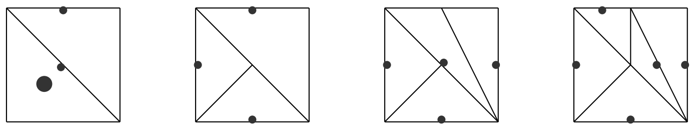
For $d = 2$, the marked edge bisection algortihm works as follows:

1) For the initial triangulation $\mathcal{T}_h$ we mark one facet (edge) of each $T \in \mathcal{T}_h$.

2) When $T \in \mathcal{T}_h$ is marked for refinement, we halve T along the marked edge.

3) For the new elements $T_1 \cup T_2 = T$ we mark those edges which were also an edge of the original element T.

4) In case the resulting triangulation is no longer admissible (due to 'hanging nodes'), we continuously refine neighboring elements (according to their marked edge), until the triangulation is admissible again.



The marking of edges induces a refinement order which guarantees the boundedness of angles. However, to ensure that no refinement cycles occur, the initially marked edges have to be chosen appropriately. Compared to other algorithms, such as red-green refinement, the marked edge bisection algorithm works by only halving existing elements. This ensures that the resulting triangulations are nested, which means that the corresponding finite element spaces are subspaces of one another.

4.6 Quadrature rules

Theorem 4.58 (Strang's first lemma)**.** *Let V be a Hilbert space and $V_h \subseteq V$ a closed subspace of V. Further, let $b \in \mathrm{Bil}(V,V)$, $b_h \in \mathrm{Bil}(V_h,V_h)$ be continuous and coercive bilinear forms on V and V_h respectively, and let $\ell \in V'$, $\ell_h \in V_h'$. Then the solutions $u \in V$, $\tilde{u}_h \in V_h$ of the variational problems*

$$
\begin{aligned}
b(u,v) &= \ell(v) && \forall v \in V\\
b_h(\tilde{u}_h, v_h) &= \ell_h(v_h) && \forall v_h \in V_h
\end{aligned}
$$

are well-defined and we have

$$
\begin{aligned}
&\|u - \tilde{u}_h\|\\
&\quad \le C \left\{ \inf_{v_h \in V_h} \left(\|u - v_h\| + \sup_{0 \neq w_h \in V_h} \frac{|b(v_h,w_h) - b_h(v_h,w_h)|}{\|w_h\|} \right) + \sup_{0 \neq w_h \in V_h} \frac{|f(w_h) - f_h(w_h)|}{\|w_h\|} \right\}.
\end{aligned}
$$

The constant C only depends on $\|b\|$ and the coercivity constant of b_h.

Proof. Exercise. ☐

Definition 4.59 (Numerical quadrature). *Let $X \subset \mathbb{R}^d$ be measurable. The quadrature rule Q on X with weights $\omega_l \in \mathbb{R}, l = 1, \ldots, L$ und quadrature points $q_l \in X, l = 1, \ldots, L$ is the mapping $Q : \{f : X \to \mathbb{R}\} \to \mathbb{R}$ given by*

$$Q(f) := \sum_{l=1}^{L} \omega_l f(q_l).$$

If $P(X) \subset L^1(X)$ is some function space over X, then Q is called exact for $P(X)$ if

$$Q(p) = \int_X p(x)\,\mathrm{d}x \qquad \forall p \in P(X).$$

Example 4.60 (Quadrature rules for triangles). *Let $X := T \subset \mathbb{R}^2$ be a 2-simplex with vertices a_i, $i = 0, \ldots, 2$ and barycenter x_s.*

1) The quadrature rule

$$Q_1(f) := |T| f(x_s)$$

ist exact for $\mathbb{P}^1(T)$.

2) The quadrature rule

$$Q_2(f) := \frac{1}{3}|T| \sum_{0 \le i < j \le 2} f(a_{ij}), \quad a_{ij} := \frac{1}{2}(a_i + a_j),$$

is exact for $\mathbb{P}^2(T)$.

3) The quadrature rule

$$Q_3(f) := \frac{1}{60}|T| \Big(3 \sum_{i=0}^{2} f(a_i) + 8 \sum_{0 \le i < j \le 2} f(a_{ij}) + 27 f(x_s)\Big)$$

is exact for $\mathbb{P}^3(T)$.

Proposition 4.61 (Approximation properties of quadrature rules). *Let $T \subset \mathbb{R}^d$ be a d-simplex and let k be large enough such that we have an embedding $H^{k+1,p}(T) \hookrightarrow C^0(T)$. Further, let Q_0 be a quadrature rule on $\hat{T}$ which is exact for $\mathbb{P}^m(\hat{T})$, $m \ge k$. Then*

$$Q_T(v) := |\det A| Q_0(v \circ F)$$

defines a quadrature rule on T, which is exact on $\mathbb{P}^m(T)$. Moreover, for all $v \in H^{k+1,p}(T)$ we have:

$$\left| \int_T v(x)\,\mathrm{d}x - Q_T(v) \right| \le C h(T)^{d(1-1/p)+k+1} |v|_{H^{k+1,p}(T)},$$

where the constant C only depends on $Q_0, d, k, p,$ and $\sigma(T)$.

Proof. Theorem 4.29 + Corollary 4.32 (exercise). □

Theorem 4.62 (Bramble-Hilbert II). *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain, $p \in [1, \infty]$ and $k \in \mathbb{N}_0$. Further, let $G \in (H^{k+1,p}(T))'$ such that $G(q) = 0$ for all $q \in \mathbb{P}^k(T)$ Then there is a constant C only depending on Ω, p and k such that for all $v \in H^{k+1,p}(\Omega)$ we have*

$$|G(v)| \leq C\|G\||v|_{H^{k+1,p}(\Omega)}.$$

Proof. For arbitrary $q \in \mathbb{P}^k(\Omega)$ we have $G(v) = G(v - q)$. Hence

$$\begin{aligned} |G(v)| &= \inf_{q \in \mathbb{P}^k(\Omega)} |G(v-q)| \\ &\leq \inf_{q \in \mathbb{P}^k(\Omega)} \|G\|_{(H^{k+1,p}(\Omega))'} \|v - q\|_{H^{k+1,p}(\Omega)} \\ &\overset{4.29}{\leq} C\|G\|_{(H^{k+1,p}(\Omega))'} |v|_{H^{k+1,p}(\Omega)} \end{aligned}$$

□

Proposition 4.63 (Approximation of the bilinear form). *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$ with $\sigma(T) \leq \sigma$ and $k \in \mathbb{N}$. Further, let $A \in L^\infty(\Omega)$, such that $A|_T \in H^{k,\infty}(T)$ for all $T \in \mathcal{T}_h$ and such that there is a constant $c_0 > 0$ with*

$$A(x) > c_0 \qquad \forall x \in \Omega.$$

Let $b \in \mathrm{Bil}(H^1(\Omega), H^1(\Omega))$ given by

$$b(u, v) := \int_\Omega A(x)\nabla u(x) \cdot \nabla v(x) \,\mathrm{d}x.$$

With Q_T as in Proposition 4.61, we let $b_h \in \mathrm{Bil}(S_h^k, S_h^k)$ be given by

$$b_h(u_h, v_h) := \sum_{T \in \mathcal{T}_h} Q_T(A\nabla u_h \cdot \nabla v_h).$$

If Q_0 only has positive weights $\omega_l > 0$ and is exact on $\mathbb{P}^{2k-2}(\hat{T})$, then b_h is coercive on $S_{h,0}^k$. Further, if $u \in H^{k+1}(\Omega)$ and $k > \frac{d}{2} - 1$, we have

$$|b(I_h(u), v_h) - b_h(I_h(u), v_h)| \leq Ch^k \|v_h\|_{H^1(\Omega)} \|u\|_{H^{k+1}(\Omega)}.$$

The constant C only depends on d, k, σ and A.

Proof. The proof of the coercivity of b_h is left as an exercise.

Let the functionals G_T, G_0 be given by

$$G_T(v) := \int_T v \,\mathrm{d}x - Q_T(v) \qquad G_0(\hat{v}) := \int_{\hat{T}} \hat{v} \,\mathrm{d}x - Q_0(v).$$

For $\hat{q} \in \mathbb{P}^{k-1}(\hat{T})$ and $\hat{r} \in H^{k,\infty}(\hat{T})$ we then have:

$$\begin{aligned}
|G_0(\hat{r}\hat{q})| &\leq C\|\hat{r}\hat{q}\|_{L^\infty(\hat{T})} \\
&\leq C\|\hat{r}\|_{L^\infty(\hat{T})}\|\hat{q}\|_{L^\infty(\hat{T})} \\
&\leq C'\|\hat{r}\|_{L^\infty(\hat{T})}\|\hat{q}\|_{L^2(\hat{T})} \\
&\leq C'\|\hat{r}\|_{H^{k,\infty}(\hat{T})}\|\hat{q}\|_{L^2(\hat{T})}.
\end{aligned}$$

Note that $G_0(\hat{r}\hat{q})$ is well-defined as $H^{1,\infty}(T)$ embeds into $C^0(T)$ due to Theorem 2.41. Hence, for each fixed $\hat{q} \in \mathbb{P}^{k-1}(\hat{T})$, we have that $G_0(\cdot\hat{q})$ is a continuous linear functional on $H^{k,\infty}(\hat{T})$, which vanishes on $\mathbb{P}^{k-1}(\hat{T})$. Hence, we can apply Theorem 4.62 to obtain:

$$|G_0(\hat{r}\hat{q})| \leq C|\hat{r}|_{H^{k,\infty}(\hat{T})}\|\hat{q}\|_{L^2(\hat{T})}.$$

Let now $\hat{r} = \hat{a}\hat{p}$ where $\hat{a} \in H^{k,\infty}(\hat{T})$ and $\hat{p} \in \mathbb{P}^{k-1}(\hat{T})$. Then:

$$\begin{aligned}
|G_0(\hat{a}\hat{p}\hat{q})| &\leq C|\hat{a}\hat{p}|_{H^{k,\infty}(\hat{T})}\|\hat{q}\|_{L^2(\hat{T})} \\
&\leq C' \max_{s=0}^{k} |\hat{a}|_{H^{k-s,\infty}(\hat{T})}|\hat{p}|_{H^{s,\infty}(\hat{T})}\|\hat{q}\|_{L^2(\hat{T})} \\
&= C' \max_{s=0}^{k-1} |\hat{a}|_{H^{k-s,\infty}(\hat{T})}|\hat{p}|_{H^{s,\infty}(\hat{T})}\|\hat{q}\|_{L^2(\hat{T})} \\
&\leq C'' \max_{s=0}^{k-1} |\hat{a}|_{H^{k-s,\infty}(\hat{T})}|\hat{p}|_{H^{s,2}(\hat{T})}\|\hat{q}\|_{L^2(\hat{T})}
\end{aligned}$$

Here we have exploited that the k-th derivatives of $\hat{p}$ vanish and that $\mathbb{P}^{k-1}(\hat{T})$ is finite-dimensional. With standard scaling arguments it follows that for $p, q \in \mathbb{P}^{k-1}(T)$ and $a \in H^{k,\infty}(T)$ we have

$$\begin{aligned}
|G_T(apq)| &\leq Ch(T)^k \max_{s=0}^{k-1} |a|_{H^{k-s,\infty}(T)}|p|_{H^{s,2}(T)}\|q\|_{L^2(T)} \\
&\leq Ch(T)^k \|a\|_{H^{k,\infty}(T)}\|p\|_{H^{k-1,2}(T)}\|q\|_{L^2(T)}.
\end{aligned}$$

Thus, for the approximation of b we obtain for $u_h, v_h \in S^k_{h,0}$:

$$
\begin{aligned}
|b(u_h, v_h) - b_h(u_h, v_h)| &\leq \sum_{T\in\mathcal{T}_h} \sum_{i=1}^{d} |G_T(A \cdot \partial_i u_h \partial_i v_h)| \\
&\leq C \sum_{T\in\mathcal{T}_h} \sum_{i=1}^{d} h(T)^k \|A\|_{H^{k,\infty}(T)} \|\partial_i u_h\|_{H^{k-1,2}(T)} \|\partial_i v_h\|_{L^2(T)} \\
&\leq C \max_{T\in\mathcal{T}_h} \|A\|_{H^{k,\infty}(T)} \sum_{T\in\mathcal{T}_h} h(T)^k \|\nabla u_h\|_{H^{k-1,2}(T)} \|\nabla v_h\|_{L^2(T)} \\
&\leq C' h^k \Big(\sum_{T\in\mathcal{T}_h} \|u_h\|^2_{H^k(T)} \Big)^{1/2} |v_h|_{H^1(\Omega)}.
\end{aligned}
$$

To complete the proof we choose $u_h := I_h(u)$ and show that:

$$
\Big(\sum_{T\in\mathcal{T}_h} \|I_h(u)\|^2_{H^k(T)} \Big)^{1/2} \leq C \|u\|_{H^{k+1}(\Omega)}.
$$

This, however, follows directly from Theorem 4.33:

$$
\begin{aligned}
\|I_h(u)\|_{H^k(T)} &\leq \|u - I_h(u)\|_{H^k(T)} + \|u\|_{H^k(T)} \\
&\leq C(h(T)|u|_{H^{k+1}(T)} + \|u\|_{H^k(T)} \\
&\leq C' \|u\|_{H^{k+1}(T)}
\end{aligned}
$$

We remark that we have to bound $\|u - I_h(u)\|_{H^k(T)}$ w.r.t $|u|_{H^{k+1}(T)}$ (and not $|u|_{H^k(T)}$ since the assumption $k > d/2 - 1$ only guarantees well-definedness of I_h for u_h in $H^{k+1}(T)$. □

Proposition 4.64 (Approximation of the right-hand side)**.** *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$ with $\sigma(T) \leq \sigma$, and let $k \in \mathbb{N}$ with $k > \frac{d}{2}$. Further, let $f \in L^2(\Omega)$ be given with $f|_T \in H^k(T)$. With Q_T as in Proposition 4.61, we let $\ell_h \in (S^k_{h,0})'$ be given by*

$$
\ell_h(v_h) := \sum_{T\in\mathcal{T}_h} Q_T(f v_h).
$$

If Q_0 is exact on $\mathbb{P}^{2k-2}(\hat{T})$, then for all $v_h \in S^k_{h,0}$:

$$
\left| \int_\Omega f v_h - \ell_h(v_h) \right| \leq C h^k \left(\sum_{T\in\mathcal{T}_h} \|f\|^2_{H^k(T)} \right)^{1/2} \|v_h\|_{H^1(\Omega)}.
$$

Proof. The proof works similarly to the proof of Proposition 4.63. See [Cia02, Theorem 4.1.5] □

Theorem 4.65 (A priori estimate with quadrature)**.** *With the assumptions of 4.63 and 4.64,* *there exists a unique solution* $\tilde{u}_h \in S^k_{h,0}$ *of the variational problem*

$$b_h(\tilde{u}_h, v_h) = \ell_h(v_h), \qquad \forall v_h \in S^k_{h,0}$$

and we have:

$$\|u - \tilde{u}_h\|_{H^1(\Omega)} \leq Ch^k \Big(\|u\|_{H^{k+1}(\Omega)} + \Big(\sum_{T \in \mathcal{T}_h} \|f\|^2_{H^k(T)} \Big)^{1/2} \Big).$$

Proof. The result follows directly from Theorem 4.58, where we choose $v_h := I_h(u)$, using Theorem 4.34, Proposition 4.63 and Proposition 4.64. □

4.7 Solving the linear system

Using appropriate numerical quadrature rules, the finite-element method leads to linear systems

$$\underline{b} \cdot \underline{u}_h = \underline{\ell}, \tag{4.26}$$

which need to be solved in order to determine the coefficients $\underline{u}_h$ of the finite-element approximation u_h w.r.t. the finite element basis (Proposition 4.9). For fine triangulations in $d = 2$ or 3 dimensions, these systems can become very large with 10^6 to 10^8 unknowns. Hence, finding efficient algorithms to solve (4.26) is crucial.

The effort for solving a linear system (4.26) with N unknowns using LU-decomposition (Gauss elimination) is of order $\mathcal{O}(N^3)$. However, the locality of the supports of the nodal finite-element basis functions $\bar{\varphi}_i$ leads to sparse matrices where each row or column of $\underline{b}$ only has C entries, where C only depends on d and the shape regularity σ of the triangulation. Hence, if LU-decomposition is implemented efficiently by exploiting the sparse structure, the computational effort can be significantly reduced. However, for very large systems, the minimal required effort of such methods of order $\mathcal{O}(CN^2)$, where C depends on the structure of the matrix and the 'fill-in' of additional non-zeros during LU-decomposition, can still be too large.

Another approach to solve (4.26) is to use iterative methods, which construct approximations of the solution up to some prescribed error tolerance. Many of these methods only rely on matrix-vector multiplications, which, for a sparse matrix can be computed efficiently with $\mathcal{O}(CN)$ operations. In the following, we discuss the so-called conjugate-gradient (CG) algorithm, the most popular choice to solve a linear system

$$Ax^* = b, \tag{4.27}$$

where A is a symmetric positive matrix.

To this end, consider the energy functional

$$E(x) := \frac{1}{2}x^T Ax - b^T x.$$

As (4.27) can be interpreted as an abstract elliptic problem over the Hilbert space $\mathbb{R}^N$, we get by the same argument as before, that

$$x^* = \operatorname*{argmin}_{x\in\mathbb{R}^N} E(x). \tag{4.28}$$

Thus, an iterative optimization algorithm to solve (4.28) will yield an iterative solver for (4.27).

A straightforward approach to solve (4.28) is given by *gradient descent*. I.e. we iterate using

$$x_{k+1} = x_k + \alpha d_k, \tag{4.29}$$

where the descent direction $d_k \in \mathbb{R}^N$ is given by

$$d_k = -\nabla E(x_k) = -Ax_k + b = R(x_k). \tag{4.30}$$

For a fixed coefficient α, the resulting method is known as *Richardson iteration* (see exercises). However, we can also perform a *line search* to determine an optimal α, i.e.,

$$\alpha = \operatorname*{argmin}_{\alpha\in\mathbb{R}} E(x_k + \alpha d_k) = \frac{d_k^T d_k}{d_k^T A d_k}. \tag{4.31}$$

One can show the following convergence result for the gradient descent algorithm with quadratic goal functional:

Theorem 4.66 (Convergence of gradient descent)**.** *Let $A \in \mathbb{R}^{N\times N}$ be a symmetric positive matrix with associated norm $\|x\|_A := (x^T Ax)^{1/2}$ Let $x_0 \in \mathbb{R}^N$ be an arbitrary start vector and x_k the approximations obtained from the gradient descent algorithm (Eqs. (4.29) to (4.31)). Then we have:*

$$\|x^* - x_k\|_A \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \|x^* - x_0\|_A. \tag{4.32}$$

Here $\kappa := \lambda_{max}(A)/\lambda_{min}(A) = \|A\|_{\mathbb{R}^N\to\mathbb{R}^N}/\|A^{-1}\|_{\mathbb{R}^N\to\mathbb{R}^N}$ denotes the spectral condition number of A.

Proof. See [Saa03]. □

A corresponding result for Richardson iteration is established in the exercises. Note that for large κ we have

$$\frac{\kappa - 1}{\kappa + 1} \approx 1 - \frac{2}{\kappa}.$$

Even for $N = 2$ one can easily construct examples where gradient descent never exactly produces the solution x^* although two linear independent search directions d_k already suffice to exactly represent x^*. This leads to the idea to optimize in (4.31) not only in the current search direction but over the linear span of all search directions considered so far. I.e., assuming w.l.o.g. $x_0 = 0$, we let

$$x_k := \operatorname*{argmin}_{x \in \mathcal{K}_k(A,b)} E(x), \tag{4.33}$$

where

$$\begin{aligned}\operatorname{span}\{d_0, \dots, d_{k-1}\} &= \operatorname{span}\{R(x_0), \dots R(x_{k-1})\} \\ &= \operatorname{span}\{b, Ab, A^2 b, \dots, A^{k-1} b\} =: \mathcal{K}_k(A, b).\end{aligned}$$

$\mathcal{K}_k(A,b)$ is called the *Krylov subspace* generated by A and b. As the Caley-Hamilton theorem implies that $A^{-1} = p(A)$ where $p(X)$ is an $(N-1)$-th order polynomial, we obtain

$$x^* = A^{-1} b = p(A) b \in \mathcal{K}_N(A, b).$$

Hence, the proposed method yields the exact solution x^* after at most N iterations, at least in exact arithmetic.

As we have seen several times before, the minimizer x_k of (4.33) satisfies

$$x^T A x_k = x^T b \qquad \forall x \in \mathcal{K}_k(A, b). \tag{4.34}$$

In other words, x_k is the Galerkin approximation of x^* in $\mathcal{K}_k(A,b)$. In particular, Céa's Lemma yields

$$\|x^* - x_k\|_A = \inf_{x \in \mathcal{K}_k(A,b)} \|x^* - x\|_A.$$

Now, we could assemble the $k \times k$ linear system for (4.34) w.r.t. the basis $b, Ab, \dots, A^{k-1}b$ and solve this system to determine the coefficients of x_k w.r.t. this basis. We can do better, however. To this end, we iteratively build an A-orthogonal basis for $\mathcal{K}_k(A,b)$ using Gram-Schmidt orthogonalization. I.e., with $r_k := R(x_k) := b - Ax_k$, $k = 0, \dots, N-1$, we let

$$p_k := r_k - \sum_{i<k} \frac{p_i^T A r_k}{p_i^T A p_i} p_i \in \mathcal{K}_{k+1}(A, b), \qquad k = 0, \dots, N-1. \tag{4.35}$$

This process is called *Arnoldi iteration*. Writing $x_k := \sum_{i=0}^{k-1} \alpha_i p_i$, we immediately get from (4.34) and the A-orthogonality of the basis that

$$\alpha_i = \frac{p_i^T b}{p_i^T A p_i}.$$

In particular, the coefficients α_k can be computed iteratively along with the orthogonalized search directions p_k. Further, note that $Ap_i \in \mathcal{K}_{i+2}(A,b) \subseteq \mathcal{K}_k(A,b)$ for $i \leq k-2$. Due to Galerkin orthogonality we have that $x^* - x_k$ is A-orthogonal to $\mathcal{K}_k(A,b)$. Hence,

$$p_i^T A r_k = (Ap_i)^T (b - Ax_k) = (Ap_i)^T A (x^* - x_k) = 0, \qquad 0 \leq i \leq k-2.$$

Thus, all summands in (4.35) vanish except for $i = k-1$, and we get:

$$p_k := r_k + \beta_{k-1} p_{k-1}, \qquad \beta_{k-1} := -\frac{p_{k-1}^T A r_k}{p_{k-1}^T A p_{k-1}}.$$

Overall this leads us to the conjugate-gradient algorithm:

Definition 4.67 (Conjugate-gradient algorithm)**.** *Given as symmetric positive $N \times N$-matrix A, $b \in \mathbb{R}^N$ and $\varepsilon > 0$, we define the conjugate-gradient algorithm as follows:*

$CG(A, b, \varepsilon)$

$k := 0,\ x_0 := 0,\ p_0 := r_0 := b$
if $\|r_0\| < \varepsilon$
  then
    return x_0
while *true*
do
  $\alpha_k := \frac{p_k^T b}{p_k^T A p_k}$
  $x_{k+1} := x_k + \alpha_k p_k$
  $r_{k+1} := r_k - \alpha_k A p_k$
  if $\|r_{k+1}\| < \varepsilon$
    then
      return x_{k+1}
  $\beta_k := -\frac{p_k^T A r_{k+1}}{p_k^T A p_k}$
  $p_{k+1} := r_{k+1} + \beta_k p_k$
  $k := k + 1$

Remarks 4.68. *1) The name* conjugate gradient *refers to the fact that the gradients are replaced by A-orthogonalized (= conjugate) gradients as search directions.*

2) In practice, the coefficients α_k and β_k are usually computed using the equivalent expressions,

$$\alpha_k := \frac{r_k^T r_k}{p_k^T A p_k}, \qquad \beta_k := \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k},$$

which improves the efficiency and numerical stability of the algorithm.

Theorem 4.69 (Convergence of the conjugate-gradient algorithm)**.** *Let $A \in \mathbb{R}^{n \times n}$ be a symmetric positive matrix with associated norm $\|x\|_A := (x^T A x)^{1/2}$, $b \in \mathbb{R}^N$, and let $x_k \in \mathbb{R}^N$ be the corresponding iterates of the conjugate-gradient algorithm. Then we have:*

$$\|x^* - x_k\|_A \leq \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|x^* - x_0\|_A. \tag{4.36}$$

Proof. See [Saa03]. □

Theorem 4.69 shows that the CG-algorithms converges significantly faster than the gradient-descent algorithm. Still, the convergence speed crucially depends on the condition number of the system matrix A. Thus, our next aim is to estimate the condition numbers of the system matrices $\underline{\mathrm{b}}$ appearing in the finite-element method.

Proposition 4.70 (Norm of basis isomorphism)**.** *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega \subset \mathbb{R}^d$ with $\sigma(T) \leq \sigma$ and $k \in \mathbb{N}$. For the associated Lagrange finite-element space S_h^k with nodal basis $\bar{\varphi}_i$, $i = 1, \ldots, N$ corresponding to the Lagrange nodes $\bar{g}_i \in G_k(\mathcal{T}_h)$ we define the basis isomorphism*

$$\Phi_{\mathbb{R}^N}^{S_h^k} : \mathbb{R}^N \to S_h^k \subset H^1(\Omega)$$
$$\underline{u}_h \mapsto \Phi_{\mathbb{R}^N}^{S_h^k}(\underline{u}_h) := u_h := \sum_{i=1}^N \underline{u}_{h,i}\bar{\varphi}_i.$$

with inverse $\Phi_{S_h^k}^{\mathbb{R}^N} := \left(\Phi_{\mathbb{R}^N}^{S_h^k}\right)^{-1}$. *We have:*

$$\left\|\Phi_{\mathbb{R}^N}^{S_h^k}\right\| \leq Ch^{-1+d/2} \qquad \text{and} \qquad \left\|\Phi_{S_h^k}^{\mathbb{R}^N}\right\| \leq Ch^{-d/2}.$$

The constant C only depends on d, k and σ. The same estimates hold for corresponding isomorphisms $\Phi_{\mathbb{R}^{N'}}^{S_{h,0}^k}$ and $\Phi_{S_{h,0}^k}^{\mathbb{R}^{N'}}$

Proof. For each $T \in \mathcal{T}_h$ and $1 \leq i \leq |G_k(T)|$ we let $\text{DOF-Map}(T, i) := j$ such that $G_k(T) \ni g_i = \bar{g}_j \in G_k(\mathcal{T}_h)$. For the first inequality we then have:

$$\begin{aligned}
\left|\Phi_{\mathbb{R}^N}^{S_h^k}(\underline{\mathrm{u}}_h)\right|_{H^1(\Omega)}^2 &= |u_h|_{H^1(\Omega)}^2 \\
&= \sum_{T\in\mathcal{T}_h} |u_h|_{H^1(T)}^2 \\
&= \sum_{T\in\mathcal{T}_h} \left|\sum_{i=1}^{|G_k(T)|} \underline{\mathrm{u}}_{h,\text{DOF-Map}(T,i)}\varphi_i^T\right|_{H^1(T)}^2 \\
&\leq C \sum_{T\in\mathcal{T}_h} h(T)^{(-1+d/2)^2} \left|\sum_{i=1}^{|G_k(T)|} \underline{\mathrm{u}}_{h,\text{DOF-Map}(T,i)}\hat{\varphi}_i\right|_{H^1(\hat{T})}^2 \\
&\overset{(*)}{\leq} C' \sum_{T\in\mathcal{T}_h} h(T)^{(-1+d/2)^2} \sum_{i=1}^{|G_k(T)|} \underline{\mathrm{u}}_{h,\text{DOF-Map}(T,i)}^2 \\
&\overset{(**)}{\leq} C'' h^{(-1+d/2)^2} \|\underline{\mathrm{u}}_h\|^2
\end{aligned}$$

In $(*)$ we have used that $\mathbb{P}^k(\hat{T})$ is finite-dimensional. In $(**)$ we have exploited that the maximum number of elements in which a given Lagrange node $\bar{g}_j$ is located is bounded by a function of d and σ. With the same argument we further obtain

$$\|u_h\|^2_{L^2(\Omega)} \leq h^{(d/2)^2}\|\underline{\mathrm{u}}_h\|^2.$$

This proves the first bound.

For the second bound we proceed similarly:

$$\begin{aligned}
\left\|\Phi^{S^k_h}_{\mathbb{R}^N}(\underline{\mathrm{u}}_h)\right\|^2_{H^1(\Omega)} &\geq \|u_h\|^2_{L^2(\Omega)} \\
&= \sum_{T\in\mathcal{T}_h}\left\|\sum_{i=1}^{|G_k(T)|}\underline{\mathrm{u}}_{h,\text{DOF-Map}(T,i)}\varphi_i^T\right\|^2_{L^2(T)} \\
&= \sum_{T\in\mathcal{T}_h}|\det A_T|\left\|\sum_{i=1}^{|G_k(T)|}\underline{\mathrm{u}}_{h,\text{DOF-Map}(T,i)}\hat{\varphi}_i\right\|^2_{L^2(\hat{T})} \\
&\geq C\sum_{T\in\mathcal{T}_h}|\det A_T|\sum_{i=1}^{|G_k(T)|}\underline{\mathrm{u}}^2_{h,\text{DOF-Map}(T,i)} \\
&\geq C'h^d\|\underline{\mathrm{u}}_h\|^2.
\end{aligned}$$

□

Corollary 4.71 (Condition number of the stiffness matrix)**.** *Let $\mathcal{T}_h$ be an admissible triangulation of $\Omega\subset\mathbb{R}^d$ with $\sigma(T)\leq\sigma$ and $k\in\mathbb{N}$. Let $b\in\mathrm{Bil}(S^k_{h,0},S^k_{h,0})$ be a symmetric, coercive bilinear form. For the corresponding stiffness matrix $\underline{b}\in\mathbb{R}^{N\times N}$ w.r.t. the nodal finite-element basis $\bar{\varphi}_i$, $i=1,\dots N$ given by*

$$\underline{b}_{ji} := b(\bar{\varphi}_i,\bar{\varphi}_j)$$

we have the following bound on the condition number:

$$\kappa(\underline{b}) \leq C\frac{\|b\|}{c_b}h^{-2}.$$

The constant C only depends on $d, k,$ and σ.

Proof. Since $\underline{\mathrm{b}}$ is positive definite, we have $\kappa(\underline{\mathrm{b}}) = \frac{\lambda_{\max}(\underline{\mathrm{b}})}{\lambda_{\min}(\underline{\mathrm{b}})}$. We have

$$\begin{aligned}\lambda_{\max}(\underline{\mathrm{b}}) &= \sup_{0\neq \underline{\mathrm{u}}_h \in \mathbb{R}^N} \frac{(\underline{\mathrm{u}}_h)^T \cdot \underline{\mathrm{b}} \cdot \underline{\mathrm{u}}_h}{\|\underline{\mathrm{u}}_h\|^2} \\ &= \sup_{0\neq \underline{\mathrm{u}}_h \in \mathbb{R}^N} \frac{b(u_h, u_h)}{\|\underline{\mathrm{u}}_h\|^2} \\ &\leq \sup_{0\neq \underline{\mathrm{u}}_h \in \mathbb{R}^N} \frac{\|b\|\|u_h\|^2}{\|\underline{\mathrm{u}}_h\|^2} \\ &\leq C\|b\|h^{-2+d}\end{aligned}$$

Similarly we have:

$$\begin{aligned}\lambda_{\min}(\underline{\mathrm{b}}) &= \inf_{0\neq \underline{\mathrm{u}}_h \in \mathbb{R}^N} \frac{(\underline{\mathrm{u}}_h)^T \cdot \underline{\mathrm{b}} \cdot \underline{\mathrm{u}}_h}{\|\underline{\mathrm{u}}_h\|^2} \\ &\geq \inf_{0\neq \underline{\mathrm{u}}_h \in \mathbb{R}^N} \frac{c_b\|u_h\|^2}{\|\underline{\mathrm{u}}_h\|^2} \\ &\geq Cc_b h^d\end{aligned}$$

□

Corollary 4.72. *For small h, the convergence rate of the CG algorithm for the solution of $\underline{b} \cdot \underline{u}_h = \underline{\ell}$ is*

$$\frac{\sqrt{\kappa(\underline{b})} - 1}{\sqrt{\kappa(\underline{b})} + 1} \approx 1 - \frac{2}{\sqrt{\kappa(\underline{b})}} \leq 1 - C\frac{1}{\sqrt{h^{-2}}} = 1 - Ch.$$

For small h, a convergence rate of $1 - Ch$ is insufficient. To overcome this issues, preconditioners $M^{-1} \in \mathbb{R}^{N\times N}$ are introduced, and the iterative solver is applied to the linear system

$$M^{-1}Ax^* = M^{-1}b$$

is solved instead of the original system. The preconditioner M^{-1} should have two properties:

1) M^{-1} is an approximation of A^{-1}, s.t. $\kappa(M^{-1}A) \ll \kappa(A)$.

2) $M^{-1} \cdot x$ can be computed with small effort. In particular, M^{-1} does not need to be known as a matrix, only its action on vectors.

Popular examples for preconditioners are: incomplete LU-factorization, domain decomposition or multigrid methods.

When applying the CG algorithm, the problem arises that $M^{-1}A$ is no longer symmetric positive definite. However, one can easily shows that $M^{-1}A$ is a positive self-adjoint operator on the Hilbert space $(\mathbb{R}^N, M)$, and the entire algorithm can be derived w.r.t. this space. This leads to the preconditioned CG algorithm.

5 Finite element methods for parabolic PDEs

In this chapter, we will consider the finite-element discretization of the heat equation

$$\partial_t u(x,t) - \Delta u(x,t) = f(x,t) \qquad \forall (x,t) \in \Omega \times (0,T) =: \Omega_T,$$

as a prototypical example for a parabolic PDE. Most of what we say equally applies to other parabolic PDEs where the Laplacian is replaced by an arbitrary coercive differential operator.

We recall the notion of a classical solution of the Cauchy-Dirichlet problem for the heat equation. I.e., for initial data $u_0 \in C^2(\Omega) \cap C^0(\bar{\Omega})$ and source term $f \in C^0(\Omega_T)$ we have:

$$\begin{aligned}
u \in C^1([0,T]; C^2(\Omega) \cap C^0(\bar{\Omega})) & & & \text{(regularity)}\\
\partial_t u(x,t) - \Delta u(x,t) = f(x,t) & \quad \forall (x,t) \in \Omega_T & & \text{(PDE)}\\
u(x,t) = 0 & \quad \forall (x,t) \in \partial\Omega \times [0,T] & & \text{(boundary values)}\\
u(x,0) = u_0(x) & \quad \forall x \in \Omega & & \text{(initial values)}
\end{aligned}$$

Central to our analysis will be a priori estimates for the energy of the solution u. We first study such an estimate for classical solutions:

Proposition 5.1 (A priori energy estimate)**.** *If $f \equiv 0$, then the classical solution of the Cauchy-Dirichlet problem for the heat equation with homogeneous Dirichlet boundary values satisfies for all $t \in [0,T]$:*

$$\frac{1}{2}\|u(\cdot,t)\|^2_{L^2(\Omega)} + \int_0^t \|\nabla u(\cdot,s)\|^2_{L^2(\Omega)}\, \mathrm{d}s = \frac{1}{2}\|u_0(\cdot)\|^2_{L^2(\Omega)}. \tag{5.1}$$

Proof. Multiplying the heat equation with the test function $\varphi := u$ and integrating w.r.t. time and space yields:

$$\begin{aligned}
0 &= \int_0^t \int_\Omega \Big(\partial_t u - \Delta u\Big) u \ \mathrm{d}x \, \mathrm{d}t \\
&= \int_0^t \int_\Omega \Big(\partial_t \frac{1}{2} u^2 + |\nabla u|^2\Big) \, \mathrm{d}x \, \mathrm{d}t \\
&= \int_0^t \frac{d}{\mathrm{d}t} \frac{1}{2} \int_\Omega |u(x,t)|^2 \, \mathrm{d}x \, \mathrm{d}t + \int_0^t \|\nabla u(\cdot,s)\|^2_{L^2(\Omega)} \, \mathrm{d}s \\
&= \frac{1}{2} \|u(\cdot,t)\|^2_{L^2(\Omega)} - \frac{1}{2} \|u_0(\cdot)\|^2_{L^2(\Omega)} + \int_0^t \|\nabla u(\cdot,s)\|^2_{L^2(\Omega)} \, \mathrm{d}s.
\end{aligned}$$

□

To derive an appropriate weak formulation for the heat equation, we need to choose an appropriate function space for the solution u. In view of elliptic theory, one obvious choice might be $u \in H^1(\Omega_T)$. However, rewriting the weak formulation

$$\begin{aligned}
\int_{\Omega_T} \partial_t u \varphi + \nabla_x u \cdot \nabla_x \varphi \, \mathrm{d}x \, \mathrm{d}t &= \int_{\Omega_T} f \varphi \, \mathrm{d}x \, \mathrm{d}t \\
\Rightarrow \int_{\Omega_T} \partial_t u \varphi \, \mathrm{d}x \, \mathrm{d}t &= - \int_{\Omega_T} \nabla_x u \cdot \nabla_x \varphi \, \mathrm{d}x \, \mathrm{d}t + \int_{\Omega_T} f \varphi \, \mathrm{d}x \, \mathrm{d}t,
\end{aligned}$$

we see that, in general, the right-hand side, as a functional of φ only has $H^{-1}(\Omega_T)$ regularity, whereas the left-hand side and $\partial_t u \in L^2(\Omega_T)$ would imply that the functional is represented by an $L^2(\Omega_T)$ function. In other words, we expect u to have different regularity w.r.t. time and space. To this end, we will interpret u as a function $u : [0,T] \to H^1(\Omega)$, which for each point in time yields a function of space. In the following section will will study spaces of integrable functions of this type, so-called *Bochner spaces*.

5.1 Bochner spaces

In this section we give a brief introduction to the theory of Bochner spaces. We can only state the most important results. For further details, we refer to [Alt16], [Eva98, p. 5.9.2] or the thorough treatment in [Hyt+16].

Definition 5.2. *We call a function $v : [0,T] \to V$ measurable (w.r.t. the Lebesgue-measure completion of the Borel σ-algebra on $[0,T]$ and the Borel σ-algebra on V) iff $v^{-1}(M)$ is measurable for every measurable $M \subseteq V$.*

We call v strongly measurable iff v is the pointwise limit of a sequence of simple functions almost everywhere (a simple function is a linear combination of characteristic functions of measurable sets).

We call v weakly measurable iff $f \circ v : [0,T] \to \mathbb{R}$ is measurable for each $f \in V'$.

Proposition 5.3. *If v is strongly measurable, then v is measurable. If v is measurable, then v is weakly measurable.*

Proof. Exercise. □

Theorem 5.4 (Pettis)**.** *If V is separable, then all notions of measurability are equivalent.*

Definition and Proposition 5.5 (Bochner spaces)**.** *Let V be a Banach space. For $p \in [1, \infty]$ and $T > 0$, let*

$$L^p(0, T; V) := \{v : [0, T] \to V \text{ strongly measurable} \mid \|v\|_{L^p(0,T;V)} < \infty\},$$

where

$$\|v\|_{L^p(0,T;V)} := \begin{cases} \Big(\int_0^T \|v(t)\|_V^p \,\mathrm{d}t \Big)^{1/p}, & \text{for } 1 \le p < \infty \\ \operatorname{ess\,sup}_{t \in [0,T]} \|v(t)\|_V, & \text{for } p = \infty. \end{cases}$$

Note that $\|v(t)\|$ is measurable since v is measurable and $\|\cdot\|$ is continuous. For each $v \in L^1(0, T; V)$, the Bochner integral

$$\int_0^T v(x) \,\mathrm{d}t \in V$$

can be well-defined via limits of integrals of simple functions converging to v from below, and most of the usual properties of the Lebesgue integral can be proven for the Bochner integral. For each p, $L^p(0, T; V)$ is a Banach space. If V' has the Radon-Nikodym property and $p < \infty$, then the dual space of $L^p(0, T; V)$ is $L^q(0, T; V')$, $1/p + 1/q = 1$, with dual the pairing given by

$$\langle f, v\rangle = \int_0^T \langle f(t), v(t)\rangle \,\mathrm{d}t \qquad \forall f \in L^q(0, T; V'),\ v \in L^p(0, T; V),$$

where $\langle f(t), v(t)\rangle := f(t)[v(t)]$. In particular, $L^p(0, T; V)$ is reflexive for $1 < p < \infty$ if V is reflexive. If V is reflexive, then V' automatically has the Radon-Nikodym property. $L^2(0, T; V)$ is a Hilbert space, if V is a Hilbert space.

Definition and Proposition 5.6 (Weak time derivative)**.** *Let $v \in L^1(0, T; V)$. We call $w \in L^1(0, T; V)$ the weak time-derivative of v if*

$$\int_0^T \varphi'(t)\, v(t) \,\mathrm{d}t = -\int_0^T \varphi(t)\, w(t) \,\mathrm{d}t \qquad \forall \varphi \in C_0^\infty((0, T)), \tag{5.2}$$

Condition (5.2) is equivalent to requiring that $t \to \langle f, v(t)\rangle$ is weakly differentiable w.r.t. t as an $\mathbb{R}$-valued function. As usual, we denote the weak time derivative by $v' = w$ or $\partial_t v = w$.

Proof. We have

$$\begin{aligned}
&\int_0^T \varphi'(t)\, v(t) + \varphi(t)\, w(t)\, \mathrm{d}t = 0 \quad \forall \varphi \in C_0^\infty((0,T)) \\
\Leftrightarrow\ &\underbrace{< f, \int_0^T \varphi'(t)\, v(t) + \varphi(t)\, w(t)\, \mathrm{d}t >}_{=\int_0^T \varphi'(t)\,\langle f, v(t)\rangle + \varphi(t)\,\langle f, w(t)\rangle\,\mathrm{d}t} = 0 \quad \forall f \in V'\ \forall \varphi \in C_0^\infty((0,T))
\end{aligned}$$

□

Consider the dense embedding $H_0^1(\Omega) \subset L^2(\Omega)$. As each continuous linear functional on $L^2(\Omega)$ restricts to a continuous functional on $H_0^1(\Omega)$. Since the embedding is dense, this restriction map is injective. Hence, $(L^2(\Omega))' \subset (H_0^1(\Omega))' = H^{-1}(\Omega)$. Invoking the Riesz isomorphism for $L^2(\Omega)$, we get

$$H_0^1(\Omega) \subset L^2(\Omega) = (L^2(\Omega))' \subset H^{-1}(\Omega).$$

Via these embeddings, each function $v \in H_0^1(\Omega)$ is interpreted as a functional in $H^{-1}(\Omega)$ via the pairing

$$\langle v, w\rangle := (v, w)_{L^2(\Omega)} \qquad \forall w \in H_0^1(\Omega).$$

We emphasize, that this embedding $H_0^1(\Omega) \hookrightarrow H^{-1}(\Omega)$ is *not* the Riesz isomorphism between these spaces. In the literature, the triple $(H_0^1(\Omega), L^2(\Omega), H^{-1}(\Omega))$ is called a *Gelfand triple.*

Theorem 5.7. *Let $v \in L^2(0,T; H_0^1(\Omega))$ such that, as a function in $L^2(0,T;H^{-1}(\Omega))$, v is weakly differentiable with $v' \in L^2(0,T;H^{-1}(\Omega))$. Then we have:*

1) There exists a $\tilde v \in C(0,T;L^2(\Omega))$ s.t. $v = \tilde v$ a.e.

2) The mapping $t \mapsto \|v(t)\|^2_{L^2(\Omega)}$ is absolutely continuous (i.e., a function in $H^{1,1}([0,T])$). We have:

$$\frac{\mathrm{d}}{\mathrm{d}t}\|v(t)\|^2_{L^2(\Omega)} = 2\langle v'(t), v(t)\rangle \qquad a.e.$$

3) There is a constant C only depending on T s.t.

$$\|v\|_{C(0,T;L^2(\Omega))} \le C(\|v\|_{L^2(0,T;H_0^1(\Omega))} + \|v'\|_{L^2(0,T;H^{-1}(\Omega))}).$$

Proof. See [Eva98, §5.9.2, Theorem 3]. □

5.2 Weak formulation of the heat equation

Definition 5.8 (Weak solution). *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain, $u_0 \in L^2(\Omega)$, and $f \in L^2(0,T;H^{-1}(\Omega))$. A function $u \in L^2(0,T;H_0^1(\Omega))$ with $u' \in L^2(0,T;H^{-1}(\Omega))$ is called weak solution of the Cauchy-Dirichlet problem for the heat equation with homogeneous Dirichlet boundary values iff:*

$$\begin{aligned} \langle u'(t), \varphi\rangle + (\nabla u(t), \nabla\varphi)_{L^2(\Omega)} &= \langle f(t), \varphi\rangle \qquad \forall \varphi \in H_0^1(\Omega), t \in (0,T) \text{ a.e.} \\ u(\cdot, 0) &= u_0. \end{aligned}$$

Here, $u(\cdot,0) = u_0$ is to be understood in the sense of Theorem 5.7. I.e., $u(\cdot,0)$ is the value of the continuous representative of u at $t = 0$.

To show the existence of weak solutions of the heat equation, we apply Rothe's method and perform a time discretization of the heat equation:

Definition 5.9 (Time-discrete heat equation). *With the notation and assumption of Definition 5.8, additionally assume $f \equiv 0$ and $u_0 \in H_0^1(\Omega)$. For an equidistant partition $t_n := n\Delta t$, $n = 0,\dots,N$, $\Delta t := T/N$ of $[0,T]$ with intervals $I_n := [t^n, t^{n+1})$, $n = 0,\dots,N-1$, we define the time-discrete approximations $u^n \in H_0^1(\Omega)$ of $u(t^n)$ via*

$$\begin{aligned} u^0 &= u_0, \\ \Big(\frac{u^{n+1} - u^n}{\Delta t}, \varphi\Big)_{L^2(\Omega)} + (\nabla u^{n+1}, \nabla\varphi)_{L^2(\Omega)} &= 0 \qquad \forall \varphi \in H_0^1(\Omega). \end{aligned} \tag{5.3}$$

Further, we define the piecewise constant function $\bar{u}^N : [0,T] \to H_0^1(\Omega)$ via

$$\bar{u}^N(t)|_{I_n} := u^{n+1}, \qquad n = 0, \dots N-1,$$

and the continuous, piecewise linear function $\tilde{u}^N : [0,T] \to L^2(\Omega)$ via

$$\tilde{u}^N(t)|_{I_n} := \frac{t^{n+1} - t}{\Delta t} u^n + \frac{t - t^n}{\Delta t} u^{n+1}, \qquad t \in I_n, \; n = 0, \dots N-1.$$

We want to show that $\bar{u}^N$, $\tilde{u}^N$, or at least some subsequences, converge to the weak solution. To this end, we want to use the following compactness result:

Theorem 5.10 (Weakly sequentially compact unit ball). *Let V be a reflexive Banach space, and let $(v_k)_{k\in\mathbb{N}}$ be a bounded sequence in V. Then there is a sub-sequence (v_{k_l}) of (v_k) which weakly converges to a limit $v \in V$, i.e.,*

$$\lim_{l\to\infty} f(v_{k_l}) \to f(v) \qquad \forall f \in V'.$$

Proof. [Alt16, Theorem 8.10]. □

In particular, the result holds for the Hilbert spaces $L^2(0,T;H_0^1(\Omega))$ and $L^2(0,T;H^{-1}(\Omega))$.

Lemma 5.11. *The sequences of approximations $\bar{u}^N$ and $\tilde{u}^N$, $N \in \mathbb{N}$, are bounded in $L^2(0,T;H_0^1(\Omega))$. The weak time derivative $\partial_t \tilde{u}^N$ exists in $L^2(0,T;H^{-1}(\Omega))$, and we have:*

$$\partial_t \tilde{u}^N(\cdot,t) = \Delta \bar{u}^N(\cdot,t) \qquad t \in (0,T) \text{ a.e.} \tag{5.4}$$

Here, $\Delta : H_0^1(\Omega) \to H^{-1}(\Omega)$ denotes the weak Laplacian given by $\Delta(u)[v] := \int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x$. In particular, the sequence $(\partial_t \tilde{u}^N)$ is bounded in $L^2(0,T;H^{-1}(\Omega))$.

Proof. Testing (5.3) with $\varphi = u^{n+1}$, we obtain:

$$\begin{aligned} 0 &= \frac{1}{\Delta t}(\left\|u^{n+1}\right\|_{L^2(\Omega)}^2 - (u^n, u^{n+1})_{L^2(\Omega)}) + \left\|\nabla u^{n+1}\right\|_{L^2(\Omega)}^2 \\ &\geq \frac{1}{\Delta t}(\left\|u^{n+1}\right\|_{L^2(\Omega)}^2 - \frac{1}{2}(\left\|u^{n+1}\right\|_{L^2(\Omega)}^2 + \|u^n\|_{L^2(\Omega)}^2) + \left\|\nabla u^{n+1}\right\|_{L^2(\Omega)}^2, \end{aligned}$$

where we have used the Cauchy-Schwarz and Young inequalities. Multiplication with Δt and summation gives:

$$\begin{aligned} 0 &\geq \sum_{n=0}^{N-1} \Delta t \left\|\nabla u^{n+1}\right\|_{L^2(\Omega)}^2 + \sum_{n=0}^{N-1} \frac{1}{2}(\left\|u^{n+1}\right\|_{L^2(\Omega)}^2 - \|u^n\|_{L^2(\Omega)}^2) \\ &= \sum_{n=0}^{N-1} \Delta t \left\|\nabla u^{n+1}\right\|_{L^2(\Omega)}^2 + \frac{1}{2}\left\|u^N\right\|_{L^2(\Omega)}^2 - \frac{1}{2}\|u_0\|_{L^2(\Omega)}^2. \end{aligned}$$

Thus, as a discrete analogue to the estimate (5.1), we obtain:

$$\frac{1}{2}\left\|\bar{u}^N(t^N)\right\|_{L^2(\Omega)}^2 + \int_0^T \left\|\nabla \bar{u}^N\right\|_{L^2(\Omega)}^2 \mathrm{d}t \leq \frac{1}{2}\|u_0\|_{L^2(\Omega)}^2.$$

In particular $\left\|\bar{u}^N\right\|_{L^2(0,T;H_0^1(\Omega))} \leq C \cdot \|u_0\|_{L^2(\Omega)}$. The boundedness of $\tilde{u}^N$ immediately follows from the boundedness of $\bar{u}^N$. The weak differentiability and (5.4) directly follow from the definition of $\tilde{u}^N$. Finally, the boundedness of $\partial_t \tilde{u}^N$ in $L^2(0,T;H^{-1}(\Omega))$ follows from (5.4), $\bar{u}^N \in L^2(0,T;H_0^1(\Omega))$ and the continuity of Δ. □

Thus, we can apply Theorem 5.10 to obtain weakly converging subsequences of $(\bar{u}^N)$, $(\tilde{u}^N)$, $(\partial_t \tilde{u}^N)$ converging to functions $\bar{u}, \tilde{u} \in L^2(0,T;H_0^1(\Omega))$ and $g \in L^2(0,T;H^{-1}(\Omega))$. In the following, we simplify notation by assuming that $\bar{u}^N$, $\tilde{u}^N$ and $\partial_t \tilde{u}^N$ themselves already converge.

For each $\varphi \in C_0^\infty((0,T)\times\Omega) \subset L^2(0,T;H_0^1(\Omega)) = \left(L^2(0,T;H^{-1}(\Omega))\right)'$ we have:

$$\langle \varphi, g \rangle \longleftarrow \langle \varphi, \partial_t \tilde{u}^N \rangle = -\langle \partial_t \varphi, \tilde{u}^N \rangle \longrightarrow -\langle \partial_t \varphi, \tilde{u} \rangle.$$

This implies $g = \partial_t \tilde{u}$ (e.g., we can choose $\varphi(x,t) = \hat{\varphi}(t) \cdot \psi(x)$, $\hat{\varphi} \in C_0^\infty((0,T))$, $\psi \in C_0^\infty(\Omega)$, and note that $C_0^\infty(\Omega)$ is dense in $H_0^1(\Omega)$). Next we establish that, in fact, also $\bar{u}^N$ and $\tilde{u}^N$ have the same limit.

Lemma 5.12. *We have*

$$\bar{u} = \tilde{u}.$$

Proof. Exercise. □

Theorem 5.13 (Existence of weak solution)**.** *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain, $u_0 \in L^2(\Omega)$, $f \equiv 0$ and $T > 0$. Then there exists a weak solution of the Cauchy-Dirichlet problem for the heat equation in the sense of Definition 5.8.*

Proof. We first show that u satisfies the weak form of the heat equation. To this end, let $\varphi \in C_0^\infty((0,T) \times \Omega)$. Then from (5.4) we get:

$$\langle \partial_t u, \varphi \rangle \leftarrow \langle \partial_t \tilde{u}^N, \varphi \rangle = \langle \Delta \bar{u}^N, \varphi \rangle = \langle \bar{u}^N, \Delta\varphi \rangle \to \langle u, \Delta\varphi \rangle = \langle \Delta u, \varphi \rangle.$$

Using the fundamental lemma of calculus of variations, it follows that

$$\langle \partial_t u(t), \varphi \rangle = \langle \Delta u(t), \varphi \rangle \qquad \forall \varphi \in C_0^\infty(\Omega)\ \forall t \in (0,T), a.e.$$

As $C_0^\infty(\Omega)$ is dense in $H_0^1(\Omega)$, the claim follows.

It remains to show that $u(0) = u_0$. To this end, first assume $u_0 \in H_0^1(\Omega)$. Then for $\varphi \in C_0^\infty([0,T) \times \Omega)$ we have:

$$\int_0^T \langle \partial_t \tilde{u}^N(\cdot,t), \varphi(\cdot,t) \rangle \,\mathrm{d}t + \int_0^T \langle \tilde{u}^N(\cdot,t), \partial_t\varphi(\cdot,t) \rangle \,\mathrm{d}t = -\langle u_0^N, \varphi(\cdot,0) \rangle.$$

As both $\tilde{u}^N$ and $\partial_t \tilde{u}^N$ converge weakly, we can pass to the limit in each summand and obtain:

$$\int_0^T \langle \partial_t u(\cdot,t), \varphi(\cdot,t) \rangle \,\mathrm{d}t + \int_0^T \langle u(\cdot,t), \partial_t\varphi(\cdot,t) \rangle \,\mathrm{d}t = -\langle u_0, \varphi(\cdot,0) \rangle.$$

Choosing $\varphi(x,t) = \hat{\varphi}(t) \cdot \psi(x)$, $\hat{\varphi} \in C_0^\infty([0,T)$, $\psi \in C_0^\infty(\Omega)$, we get:

$$\int_0^T \langle \partial_t u(\cdot,t), \psi \rangle \hat{\varphi}(t) \,\mathrm{d}t + \int_0^T \langle u(\cdot,t), \psi \rangle \partial_t \hat{\varphi}(t) \,\mathrm{d}t = -\langle u_0, \psi \rangle \hat{\varphi}(0).$$

Noting that $\langle u(\cdot,t)\psi \rangle \in H^{1,1}((0,T))$ with weak derivative $\langle \partial_t u(\cdot,t)\psi \rangle \in L^1((0,T))$, we can perform partial integration to obtain $\langle u(\cdot,t), \psi \rangle \hat{\varphi}(0) = \langle u_0, \psi \rangle \hat{\varphi}(0)$. Hence, $u(\cdot,0) = u_0$.

If $u_0 \in L^2(\Omega) \setminus H_0^1(\Omega)$, one can use a density argument and approximate u_0 by a sequence $u_0^N \in H_0^1(\Omega)$, which we use as initial condition for $\bar{u}^N$, $\tilde{u}^N$. □

Remark 5.14. *The presented proof can be generalized to problems with $f \neq 0$, inhomogeneous Dirichlet conditions or other types of elliptic differential operators.*

Proposition 5.15 (A priori energy estimate for weak solutions)**.** *Let u be a weak solution of the heat equation in the sense of Definition 5.8 and assume $f \in L^2(0,T;L^2(\Omega))$. Then for $t \in [0,T]$ we have:*

$$\|u(\cdot,t)\|^2_{L^2(\Omega)} + \int_0^t \|\nabla u(\cdot,s)\|^2_{L^2(\Omega)}\,\mathrm{d}s \le c_p^2 \int_0^t \|f(\cdot,s)\|^2_{L^2(\Omega)}\,\mathrm{d}s + \|u_0\|^2_{L^2(\Omega)}. \tag{5.5}$$

Proof. Using Theorem 5.7, we obtain for $s \in [0,t]$, a.e.:

$$\begin{aligned}
\frac{1}{2}\frac{d}{dt}\|u(s)\|^2 + \|\nabla u(s)\|^2_{L^2(\Omega)} &= \langle \partial_t u(s), u(s)\rangle + \|\nabla u(s)\|^2_{L^2(\Omega)} \\
&= (f(s), u(s))_{L^2(\Omega)} \\
&\le \|f(s)\|_{L^2(\Omega)} \cdot \|u(s)\|_{L^2(\Omega)} \\
&\le \|f(s)\|_{L^2(\Omega)} c_p \|\nabla u(s)\|_{L^2(\Omega)} \\
&\le \frac{c_p^2}{2}\|f(s)\|^2_{L^2(\Omega)} + \frac{1}{2}\|\nabla u(s)\|^2_{L^2(\Omega)}
\end{aligned}$$

Integration w.r.t. s yields the claim. □

Corollary 5.16. *The weak solution u of the heat equation in the sense of Definition 5.8 is uniquely defined.*

Proof. Assume u_1 and u_2 were weak solutions. Then $u_1 - u_2$ is a weak solution of the homogeneous heat equation with $f \equiv u_0 \equiv 0$. Hence, (5.5) yields $\|u_1 - u_2\|_{L^2(0,T;H_0^1(\Omega))} = 0$. □

5.3 Semi-discrete finite element method

As in the elliptic case, we now replace the space $H_0^1(\Omega)$ in Definition 5.8 by an appropriate finite-dimensional approximation space $V_h \subset H_0^1(\Omega)$:

Definition 5.17 (Semi-discrete heat equation)**.** *Let u be the weak solution of the heat equation in the sense of Definition 5.8. For a finite-dimensional subspace $V_h \subset H_0^1(\Omega)$ and some interpolation operator $I_h : H_0^1(\Omega) \to V_h$, the V_h valued semi-discrete approximation u_h of u is given by the solution of the initial-value problem:*

$$\begin{aligned}
(\partial_t u_h(\cdot,t), \varphi_h)_{L^2(\Omega)} + (\nabla u_h(\cdot,t), \nabla\varphi_h)_{L^2(\Omega)} &= \langle f(\cdot,t), \varphi_h\rangle \qquad \varphi_h \in V_h,\ t \in (0,T), \\
u_h(\cdot,0) &= I_h(u_0).
\end{aligned}$$

Remark 5.18 (ODE for coefficient functions)**.** *Choosing a basis $\varphi_1, \dots, \varphi_N$ of V_h and writing*

$$\underline{u}_h(x,t) = \sum_{i=1}^N \underline{u}_{h,i}(t)\varphi_i(x),$$

we obtain

$$\sum_{i=1}^N \frac{d}{dt}\underline{u}_{h,i}(t)(\varphi_i,\varphi_j)_{L^2(\Omega)} + \sum_{i=1}^N \underline{u}_{h,i}(t)(\nabla\varphi_i,\nabla\varphi_j)_{L^2(\Omega)} = \langle f(\cdot,t),\varphi_j\rangle, \qquad j=1,\dots,N.$$

Defining the mass and stiffness matrices $M, S \in \mathbb{R}^{N\times N}$ via

$$M_{ji} := (\varphi_i,\varphi_j)_{L^2(\Omega)} \qquad S_{ji} := (\nabla\varphi_i,\nabla\varphi_j)_{L^2(\Omega)},$$

and the function $\underline{f}\colon [0,T] \to \mathbb{R}^N$ via

$$\underline{f}_i(t) := \langle f(\cdot,t),\varphi_i\rangle,$$

we obtain the ODE system

$$M\partial_t\underline{u}_h(t) + S\underline{u}_h(t) = \underline{f}(t).$$

This system can be solved using standard ODE solvers.

Now we want to show an a priori error estimate for u_h in the $L^\infty(0,T;L^2(\Omega))$ norm. Instead of using (5.5), we will use another a priori bound, which will lead to a sharper estimate w.r.t. this norm:

Theorem 5.19 (A priori error estimate)**.** *Let u be a weak solution in the sense of Definition 5.8. Let $\mathcal{T}_h$ be an admissible triangulation of Ω with $\sigma(T) \le \sigma$, $k \in \mathbb{N}$ and $S^k_{h,0}$ the corresponding Lagrange finite-element space. Let $u_h(t)$ be the semi-discrete approximation of u in $S^k_{h,0}$ with initial value $u_h(0)$. Further assume that the regularity assumptions of Corollary 4.39 are fulfilled. Then, under the additional regularity assumption $u(t), \partial_t u(t) \in H^{k+1}(\Omega)$, we have:*

$$\begin{aligned}\|u(t) - u_h(t)\|_{L^2(\Omega)} &\\ \le \|u_0 - u_h(0)\|_{L^2(\Omega)} &\\ + Ch^{k+1}\Big(\|u(t)\|_{H^{k+1}(\Omega)} + \|u_0\|_{H^{k+1}(\Omega)} + \big(\int_0^t \|\partial_t u(s)\|^2_{H^{k+1}}\,\mathrm{d}s\big)^{1/2}\Big).&\end{aligned}$$

Proof. Subtracting the weak ODE for u_h from the weak PDE for u, we obtain (using $\partial_t u \in L^2(\Omega)$) the following error equation:

$$(\partial_t(u-u_h)(t),\varphi_h)_{L^2(\Omega)} + (\nabla(u-u_h)(t),\nabla\varphi_h)_{L^2(\Omega)} = 0 \qquad \forall\varphi_h \in S^k_{h,0}, t \in (0,T). \tag{5.6}$$

Define the Galerkin projection $P_h u(t) \in S^k_{h,0}$ of $u(t)$ via

$$(\nabla P_h u(t),\nabla\varphi_h)_{L^2(\Omega)} = (\nabla u(t),\nabla\varphi_h)_{L^2(\Omega)} \qquad \forall\varphi_h \in S^k_{h,0}, t\in(0,T),$$

In other words, P_h is the orthogonal projection onto $S^k_{h,0}$ w.r.t. the inner product $(\nabla\cdot,\nabla\cdot)_{L^2(\Omega)}$ inducing the Sobolev semi-norm $|\cdot|_{H^1(\Omega)}$. Splitting the error as $u-u_h=(u-P_hu)+(P_hu-u_h)$ and using the orthogonality of the projection P_h, we obtain from (5.6) an equation for the error $P_hu-u_h\in S^k_{h,0}$:

$$(\partial_t(P_hu-u_h)(t),\varphi_h)_{L^2(\Omega)}+(\nabla(P_hu-u_h)(t),\nabla\varphi_h)_{L^2(\Omega)}=(\partial_t(P_hu-u)(t),\varphi_h)_{L^2(\Omega)}.$$

Choosing $\varphi_h:=(P_hu-u_h)(t)$, we obtain:

$$(\partial_t(P_hu-u_h)(t),(P_hu-u_h)(t))_{L^2(\Omega)}+\|\nabla(P_hu-u_h)(t)\|^2_{L^2(\Omega)}=(\partial_t(P_hu-u)(t),(P_hu-u_h)(t))_{L^2(\Omega)}.$$

Analogously to the proof of (5.5), it follows that:

$$\begin{aligned}\frac{1}{2}\|(P_hu-u_h)(t)\|^2_{L^2(\Omega)}&+\int_0^t\|\nabla(P_hu-u_h)(t)\|^2_{L^2(\Omega)}\,\mathrm{d}s\\ &\le\frac{c_p^2}{2}\int_0^t\|\partial_t(P_hu-u)(s)\|^2_{L^2(\Omega)}\,\mathrm{d}s+\frac{1}{2}\|(P_hu-u_h)(0)\|^2_{L^2(\Omega)}.\end{aligned}$$

In particular, we have:

$$\|(P_hu-u_h)(t)\|_{L^2(\Omega)}\le c_p\Big(\int_0^t\|\partial_t(P_hu-u)(s)\|^2_{L^2(\Omega)}\,\mathrm{d}s\Big)^{1/2}+\|(P_hu-u_h)(0)\|_{L^2(\Omega)}.$$

Using the triangle inequality, we finally obtain:

$$\begin{aligned}\|(u-u_h)(t)\|_{L^2(\Omega)}&\le\|(u-P_hu)(t)\|_{L^2(\Omega)}+\|(P_hu-u_h)(t)\|_{L^2(\Omega)}\\ &\le\|(u-P_hu)(t)\|_{L^2(\Omega)}+\|(P_hu-u_h)(0)\|_{L^2(\Omega)}\\ &\qquad+c_p\Big(\int_0^t\|\partial_t(P_hu-u)(s)\|^2_{L^2(\Omega)}\,\mathrm{d}s\Big).\end{aligned}$$

Note that we have $\partial_tP_hu=P_h\partial_tu$ as differentiation commutes with continuous linear operators (P_h in our case). Thus, to obtain an a priori error bound, we finally need to bound the projection errors $(u-P_hu)(t)$ and $\partial_t(u-P_hu)(t)=(\partial_tu-P_h\partial_tu)(t)$ in the L^2-norm. As P_h is an orthogonal projection, P_hu is a best-approximation of u in the $|\cdot|_{H^1(\Omega)}$-norm. Thus, using $\|(u-P_hu)(t)\|_{L^2(\Omega)}\le c_p|(u-P_hu)(t)|_{H^1(\Omega)}\le|(u-I_hu)(t)|_{H^1(\Omega)}$ we could use the global interpolation error estimate Theorem 4.34. However, as before, this would give us a suboptimal rate. Instead, note that $u(t)$ is the solution of the weak Poisson problem

$$(\nabla u(t),\nabla\varphi)_{L^2(\Omega)}=(-\Delta u(t),\varphi)_{L^2(\Omega)}$$

where $-\Delta u(t)\in L^2(\Omega)$ since $u(t)\in H^2(\Omega)$. By definition, $P_hu(t)$ is the Galerkin approximation of the solution of this problem in $S^k_{h,0}$. Hence, we can apply the a priori bound from Corollary 4.39 to obtain:

$$\begin{aligned}\|(u-P_hu)(t)\|_{L^2(\Omega)}&\le Ch^{k+1}|u(t)|_{H^{k+1}(\Omega)},\\ \|(\partial_tu-P_h(\partial_tu))(t)\|_{L^2(\Omega)}&\le Ch^{k+1}|\partial_tu(t)|_{H^{k+1}(\Omega)}.\end{aligned}$$

Together with the estimate

$$\begin{aligned}\|(P_h u - u_h)(0)\|_{L^2(\Omega)} &\le \|(P_h u - u)(0)\|_{L^2(\Omega)} + \|(u-u_h)(0)\|_{L^2(\Omega)}\\ &\le Ch^{k+1}|u(0)|_{H^{k+1}(\Omega)} + \|(u-u_h)(0)\|_{L^2(\Omega)},\end{aligned}$$

the claim follows. □

Proposition 5.20 (A priori estimates for weak solutions 2)**.** *Let u be a weak solution of the heat equation in the sense of Definition 5.8 and assume $\partial_t u \in L^2(0,T;L^2(\Omega))$, $f \in L^2(0,T;L^2(\Omega))$. Then for $t \in [0,T]$ we have:*

$$\|u(\cdot,t)\|_{L^2(\Omega)} \le \int_0^t \|f(\cdot,s)\|_{L^2(\Omega)}\,\mathrm{d}s + \|u_0\|_{L^2(\Omega)}, \tag{5.7}$$

$$\|u(\cdot,t)\|_{L^2(\Omega)} \le \int_0^t e^{c_p^{-2}(s-t)}\|f(\cdot,s)\|_{L^2(\Omega)}ds + e^{-c_p^{-2}t}\|u_0\|_{L^2(\Omega)}. \tag{5.8}$$

If, in addition, $\partial_t u \in L^2(0,T;H_0^1(\Omega))$, we have:

$$\int_0^t \|\partial_t u(\cdot,s)\|^2_{L^2(\Omega)} + \|\nabla u(\cdot,t)\|^2_{L^2(\Omega)} \le \int_0^t \|f(\cdot,s)\|^2_{L^2(\Omega)}ds + \|\nabla u_0\|^2_{L^2(\Omega)}. \tag{5.9}$$

Proof. The proofs of (5.7) and (5.9) are exercises. To show (5.8), we again test the weak formulation with $\varphi := u(s)$ to obtain:

$$(\partial_t u(s), u(s))_{L^2(\Omega)} + \|\nabla u(s)\|^2_{L^2(\Omega)} = (f(s), u(s))_{L^2(\Omega)}.$$

Usign the Poincaré inquality $\|u(s)\|_{L^2(\Omega)} \le c_p\|\nabla u(s)\|_{L^2(\Omega)}$ on the left-hand side and the Cauchy-Schwarz inquality on the right-hand side we obtain:

$$\frac{1}{2}\frac{d}{dt}\|u(s)\|^2_{L^2(\Omega)} + c_p^{-2}\|u(s)\|^2_{L^2(\Omega)} \le \|f(s)\|_{L^2(\Omega)}\|u(s)\|_{L^2(\Omega)},$$

As $\partial_t u \in L^2(0,T;L^2(\Omega))$, it follows that $\|u\|_{L^2(\Omega)} \in H^1((0,T))$. In particular, the product rule $1/2 \cdot \frac{d}{dt}\|u(t)\|^2_{L^2(\Omega)} = \|u(t)\|_{L^2(\Omega)}\frac{d}{dt}\|u(t)\|_{L^2(\Omega)}$ applies, and we obtain:

$$\begin{aligned}&\frac{d}{dt}\|u(s)\|_{L^2(\Omega)} + c_p^{-2}\|u(s)\|_{L^2(\Omega)} \le \|f(s)\|_{L^2(\Omega)}\\ \Leftrightarrow\quad &\underbrace{e^{c_p^{-2}s}\frac{d}{dt}\|u(s)\|_{L^2(\Omega)} + c_p^{-2}e^{c_p^{-2}s}\|u(s)\|_{L^2(\Omega)}}_{=\frac{d}{dt}e^{c_p^{-2}s}\|u(s)\|_{L^2(\Omega)}} \le e^{c_p^{-2}s}\|f(s)\|_{L^2(\Omega)}\end{aligned}$$

Integration from 0 to t yields the claim. □

Corollary 5.21 (A priori error estimates 2)**.** *Under the assumptions of Theorem 5.19 we have* *the error bounds:*

$$\begin{aligned}
&\|u(t)-u_h(t)\|_{L^2(\Omega)} \\
&\quad \le e^{-c_p^{-2}t}\|u_0-u_h(0)\|_{L^2(\Omega)} \\
&\qquad + Ch^{k+1}\Big(\|u(t)\|_{H^{k+1}(\Omega)} + e^{-c_p^{-2}t}\|u_0\|_{H^{k+1}(\Omega)} + \int_0^t e^{c_p^{-2}(s-t)}\|\partial_t u(s)\|_{H^{k+1}(\Omega)}\,\mathrm{d}s\Big)
\end{aligned}$$

and

$$\begin{aligned}
&\|\nabla(u-u_h)(t)\|_{L^2(\Omega)} \\
&\quad \le C\|\nabla(u-u_h)(0)\|_{L^2(\Omega)} \\
&\qquad + Ch^k\left(\|u_0\|_{H^{k+1}(\Omega)} + \|u(t)\|_{H^{k+1}(\Omega)} + \left(\int_0^t \|\partial_t u(s)\|^2_{H^k(\Omega)} ds\right)^{1/2}\right).
\end{aligned}$$

5.4 Fully-discrete finite-element method

Definition 5.22 (Implicit-Euler discretization)**.** *With the notation and assumption of Definition 5.8, let $t_m := m\Delta t$, $m = 0,\ldots,M$, $\Delta t := T/M$ be an equidistant partition of $[0,T]$, and let $u_{h,0} \in S^k_{h,0}$. We then define the implicit-Euler approximations $u_h^m \in S^k_{h,0}$ of $u(t^n)$ via*

$$\begin{aligned}
u_h^0 &= u_{h,0} \\
\Big(\frac{u_h^{m+1}-u_h^m}{\Delta t}, \varphi_h\Big)_{L^2(\Omega)} + (\nabla u_h^{m+1}, \nabla\varphi_h)_{L^2(\Omega)} &= \langle f(\cdot, t^{m+1}), \varphi_h\rangle \qquad \forall \varphi_h \in S^k_{h,0}.
\end{aligned}$$

Remark 5.23 (Equation for coefficient vectors)**.** *Let $M, S \in \mathbb{R}^{N\times N}$ be the mass and stiffness matrices of the finite-element discretization and $\underline{f}^m \in \mathbb{R}^N$ with $f_i^m := \langle f(\cdot,t^m), \varphi_i\rangle)$ the right-hand side vectors. Then the coefficient vectors $\underline{u}_h^m$ of the time-discrete approximations u_h^m satisfiy:*

$$(M + \Delta t S)\underline{u}_h^{m+1} = M\underline{u}^m + \Delta t \underline{f}^{n+1}.$$

Thus, in each time step, a linear system with with the positive definite system matrix $A = M+\Delta t S$ has to be solved. In particular, this shows, that the discrete approximation u_h^m is well defined.

We will now imitate the proof of Theorem 5.19 to obtain a corresponding result for the fully-discrete approximation:

Theorem 5.24 (A priori error estimate)**.** *With the assumptions of Theorem 5.19, let $u_h^m \in S^k_{h,0}$, $m = 0,\ldots M$ be the implicit-Euler approximation from Definition 5.22. Then we have the error* *bound:*

$$
\begin{aligned}
&\|u(t^m) - u_h^m\|_{L^2(\Omega)} \\
&\quad \le \left\|u_0 - u_h^0\right\|_{L^2(\Omega)} + \Delta t \int_0^{t^m} \|\partial_{tt} u(s)\|_{L^2(\Omega)} \,\mathrm{d}s \\
&\qquad + Ch^{k+1}\Big(\|u_0\|_{H^{k+1}(\Omega)} + \int_0^{t^m} \|\partial_t u(s)\|_{H^{k+1}(\Omega)} \,\mathrm{d}s\Big).
\end{aligned}
$$

Proof. As in the semidiscrete case, we split the approximation error into the error of Galerkin-projection onto $S_{h,0}^k$ and the error within $S_{h,0}^k$:

$$u(t^m) - u_h^m = (u(t^m) - P_h u(t^m)) + (P_h u(t^m) - u_h^m).$$

As before, the first summand can be estimated using the L^2-error bound for the elliptic case:

$$\|u(t^m) - P_h u(t^m)\|_{L^2(\Omega)} \le Ch^{k+1} \|u(t^m)\|_{H^{k+1}(\Omega)}.$$

Since

$$u(t^m) = u(0) + \int_0^{t^m} \partial_t u(s) \,\mathrm{d}s,$$

this can be further estimated by

$$\|u(t^m) - P_h u(t^m)\|_{L^2(\Omega)} \le Ch^{k+1}(\|u_0\|_{H^{k+1}(\Omega)} + \int_0^{t^m} \|\partial_t u(s)\|_{H^{k+1}(\Omega)} \,\mathrm{d}s).$$

For the second summand $(P_h u(t^m) - u_h^m)$, we again derive an error identity: For u_h^m we have

$$\Big(\frac{u_h^m - u_h^{m-1}}{\Delta t}, \varphi_h\Big)_{L^2(\Omega)} + (\nabla u_h^m, \nabla \varphi_h)_{L^2(\Omega)} = \langle f(\cdot, t^m), \varphi_h \rangle,$$

whereas for $u(t^m)$, we have:

$$\Big(\partial_t u(t^m), \varphi_h\Big)_{L^2(\Omega)} + \underbrace{(\nabla u(t^m), \nabla \varphi_h)_{L^2(\Omega)}}_{=(\nabla P_h u(t^m), \nabla \varphi_h)_{L^2(\Omega)}} = \langle f(\cdot, t^m), \varphi_h \rangle.$$

Thus, for the error $e_h^m := (P_h u(t^m) - u_h^m)$ we obtain:

$$\Big(\partial_t u(t^m), \varphi_h\Big)_{L^2(\Omega)} - \Big(\frac{u_h^m - u_h^{m-1}}{\Delta t}, \varphi_h\Big)_{L^2(\Omega)} + (\nabla e_h^m, \nabla \varphi_h)_{L^2(\Omega)} = 0.$$

Rearranging, we obtain the error identity:

$$\begin{aligned}
&\Big(\frac{e_h^m - e_h^{m-1}}{\Delta t}, \varphi_h\Big)_{L^2(\Omega)} + (\nabla e_h^m, \nabla\varphi_h)_{L^2(\Omega)} \\
&\quad = \Big(\frac{P_h u(t^m) - P_h u(t^{m-1})}{\Delta t}, \varphi_h\Big)_{L^2(\Omega)} - \Big(\partial_t u(t^m), \varphi_h\Big)_{L^2(\Omega)} \\
&\quad = \Big(P_h\Big(\frac{u(t^m) - u(t^{m-1})}{\Delta t}\Big) - \partial_t u(t^m), \varphi_h\Big)_{L^2(\Omega)} \\
&\quad = \Big((P_h - Id)\Big(\frac{u(t^m) - u(t^{m-1})}{\Delta t}\Big), \varphi_h\Big)_{L^2(\Omega)} \\
&\qquad + \Big(\Big(\frac{u(t^m) - u(t^{m-1})}{\Delta t}\Big) - \partial_t u(t^m), \varphi_h\Big)_{L^2(\Omega)}.
\end{aligned}$$

Choosing as test function $\varphi_h := e_h^m$, it follows:

$$\begin{aligned}
&\Big(\frac{e_h^m - e_h^{m-1}}{\Delta t}, e_h^m\Big)_{L^2(\Omega)} + \|\nabla e_h^m\|^2_{L^2(\Omega)} \\
&\quad \le \Big\{ \underbrace{\Big\|(P_h - Id)\Big(\frac{u(t^m) - u(t^{m-1})}{\Delta t}\Big)\Big\|_{L^2(\Omega)}}_{=:\varepsilon_{1,m}} + \underbrace{\Big\|\Big(\frac{u(t^m) - u(t^{m-1})}{\Delta t}\Big) - \partial_t u(t^m)\Big\|_{L^2(\Omega)}}_{=:\varepsilon_{2,m}} \Big\} \cdot \|e_h^m\|_{L^2(\Omega)}
\end{aligned}$$

With $\varepsilon_m := \varepsilon_{1,m} + \varepsilon_{2,m}$ we obtain after multiplication with Δt:

$$\|e_h^m\|^2_{L^2(\Omega)} - (e_h^{m-1}, e_h^m)_{L^2(\Omega)} + \Delta t\|\nabla e_h^m\|^2_{L^2(\Omega)} \le \Delta t \varepsilon_m \|e_h^m\|_{L^2(\Omega)},$$

from which we obtain

$$\|e_h^m\|_{L^2(\Omega)} \le \left\|e_h^{m-1}\right\|_{L^2(\Omega)} + \Delta t \varepsilon_m.$$

Summing over m yields:

$$\|e_h^m\| \le \left\|e_h^0\right\| + \Delta t \sum_{j=1}^m \varepsilon_j.$$

As in the semi-discrete case, we bound the error at $t = 0$ by:

$$\begin{aligned}
\left\|e_h^0\right\|_{L^2(\Omega)} &\le \left\|u_h^0 - u_0\right\|_{L^2(\Omega)} + \|u_0 - P_h u_0\|_{L^2(\Omega)} \\
&\le \left\|u_h^0 - u_0\right\|_{L^2(\Omega)} + Ch^{k+1}\|u_0\|_{H^{k+1}(\Omega)}.
\end{aligned}$$

It remains to bound the terms $\varepsilon_{1,m}$ and $\varepsilon_{2,m}$. We begin with $\varepsilon_{1,m}$:

$$\begin{aligned}
\varepsilon_{1,m} &= \left\|(P_h - Id)\frac{1}{\Delta t}\int_{t^{m-1}}^{t^m} \partial_t u(s)\,\mathrm{d}s\right\|_{L^2(\Omega)} \\
&\le \frac{1}{\Delta t}\int_{t^{m-1}}^{t^m} \|(P_h - Id)\partial_t u(s)\|_{L^2(\Omega)}\,\mathrm{d}s \\
&\le Ch^{k+1}\frac{1}{\Delta t}\int_{t^{m-1}}^{t^m} \|\partial_t u(s)\|_{H^{k+1}(\Omega)}\,\mathrm{d}s.
\end{aligned}$$

Summing over m and multiplying with Δt yields:

$$\Delta t \sum_{j=1}^{m} \varepsilon_{1,j} \leq C h^{k+1} \int_0^{t^m} \|\partial_t u(s)\|_{H^{k+1}(\Omega)} \,\mathrm{d}s.$$

Finally, we consider $\varepsilon_{2,m}$. We have:

$$\begin{aligned}\Big(\frac{u(t^m) - u(t^{m-1})}{\Delta t}\Big) - \partial_t u(t^m) &= \frac{1}{\Delta t}\int_{t^{m-1}}^{t^m} \partial_t u(s) ds - \partial_t u(t^m) \\ &= \frac{1}{\Delta t}\int_{t^{m-1}}^{t^m} \Big(\partial_t u(s) - \partial_t u(t^m)\Big)\,\mathrm{d}s \\ &= \frac{1}{\Delta t}\int_{t^{m-1}}^{t^m} \int_s^{t^m} \partial_{tt} u(r)\,\mathrm{d}r\,\mathrm{d}s.\end{aligned}$$

Summing over m and multiplying with Δt yields:

$$\Delta t \sum_{j=1}^{m} \varepsilon_{2,j} \leq \sum_{j=1}^{m} \int_{t^{j-1}}^{t^j} \int_s^{t^j} \|\partial_{tt} u(r)\|_{L^2(\Omega)} \,\mathrm{d}r\,\mathrm{d}s \leq \Delta t \int_0^{t^m} \|\partial_{tt} u(s)\|_{L^2(\Omega)} \,\mathrm{d}s.$$

Collecting all bounds yields the claim. □

Definition 5.25 (θ-method)**.** *With the assmuptions of Definition 5.22. and $\theta \in [0,1]$, the θ-method approximations $u_h^m \in S_{h,0}^k$ of $u(t^n)$ are given by:*

$$\begin{aligned} u_h^0 &= u_{h,0}, \\ \Big(\frac{u_h^{m+1} - u_h^m}{\Delta t}, \varphi_h\Big)_{L^2(\Omega)} &+ \theta(\nabla u_h^{m+1}, \nabla\varphi_h)_{L^2(\Omega)} + (1-\theta)(\nabla u_h^m, \nabla\varphi_h)_{L^2(\Omega)} \\ &= \langle \theta f(\cdot, t^{m+1}) + (1-\theta) f(\cdot, t^m), \varphi_h\rangle \qquad \forall \varphi_h \in S_{h,0}^k.\end{aligned}$$

In particular, for $\theta = 1$ we recover the implict Euler method. The explict Euler method is obtained for $\theta = 0$. For $\theta = 1/2$ we obtain the Crank-Nicolson method.

Theorem 5.26 (Stability of the θ-method)**.** *Let $u_h^m \in S_{h,0}^k$, $m = 1, \dots, M$ be the approximations of the θ-method from Definition 5.25. Assume that $f \in L^\infty(0,T; L^2(\Omega))$. If $\theta \in [0, \frac{1}{2})$, further assume that*

$$(1-\theta)\Delta t \leq \frac{h_{\min}^2}{c^2},$$

where $h_{\min} := \min_{T\in\mathcal{T}_h} h(T)$ and c is the optimal constant in the inverse inequality

$$\|\nabla v_h\|_{L^2(\Omega)} \leq \frac{c}{h_{\min}} \|v_h\|_{L^2(\Omega)} \quad \forall v_h \in S_{h,0}^k. \tag{5.10}$$

Then the θ-method is stable in the sense that the following bound holds:

$$\|u_h^m\|_{L^2(\Omega)} \leq \left\|u_h^0\right\|_{L^2(\Omega)} + C \max_{t\in[0,T]} \|f(\cdot,t)\|_{L^2(\Omega)}.$$

Here, the constant C is given by $C = \sqrt{\frac{c_p^2 T}{2}}$ if $\theta \geq \frac{1}{2}$, and by $C = \sqrt{\frac{c_p^2 T}{1+\theta}}$ if $\theta < \frac{1}{2}$.

Proof for $\theta \geq \frac{1}{2}$. Choose the test function $\varphi_h = \theta u_h^{m+1} + (1-\theta) u_h^m$ to obtain

$$\underbrace{\Big(\frac{u_h^{m+1} - u_h^m}{\Delta t}, \theta u_h^{m+1} + (1-\theta) u_h^m\Big)}_{T_1} + \underbrace{\Big\|\nabla(\theta u_h^{m+1} + (1-\theta) u_h^m)\Big\|^2}_{T_2}$$
$$= \underbrace{\big(\theta f(\cdot, t^{m+1}) + (1-\theta) f(\cdot, t^m), \theta u_h^{m+1} + (1-\theta) u_h^m\big)}_{T_3}.$$

We have:

$$\begin{aligned}
T_1 &= \frac{1}{\Delta t}\Big(\theta\Big\|u_h^{m+1} - u_h^m\Big\|^2 + (u_h^{m+1} - u_h^m, u_h^m)\Big) \\
&= \frac{1}{\Delta t}\Big(\theta\Big\|u_h^{m+1} - u_h^m\Big\|^2 - \frac{1}{2}\Big\|u_h^{m+1} - u_h^m\Big\|^2 + \frac{1}{2}\Big\|u_h^{m+1}\Big\|^2 - \frac{1}{2}\|u_h^m\|^2\Big) \\
&= \frac{(\theta - \frac{1}{2})}{\Delta t}\Big\|u_h^{m+1} - u_h^m\Big\|^2 + \frac{1}{2\Delta t}\Big(\Big\|u_h^{m+1}\Big\|^2 - \|u_h^m\|^2\Big).
\end{aligned}$$

For T_3, we get:

$$\begin{aligned}
T_3 &\leq \Big\|\theta f(\cdot, t^{m+1}) + (1-\theta) f(\cdot, t^m)\Big\| \cdot \Big\|\theta u_h^{m+1} + (1-\theta) u_h^m\Big\| \\
&\leq \Big\|\theta f(\cdot, t^{m+1}) + (1-\theta) f(\cdot, t^m)\Big\| \cdot c_p\Big\|\nabla(\theta u_h^{m+1} + (1-\theta) u_h^m)\Big\| \\
&\leq \frac{c_p^2}{4}\Big\|\theta f(\cdot, t^{m+1}) + (1-\theta) f(\cdot, t^m)\Big\|^2 + \underbrace{\Big\|\nabla(\theta u_h^{m+1} + (1-\theta) u_h^m)\Big\|^2}_{=T_2}.
\end{aligned}$$

Multiplication with $2\Delta t$ yields:

$$\Big\|u_h^{m+1}\Big\|^2 - \|u_h^m\|^2 + \underbrace{2(\theta - \frac{1}{2})\Big\|u_h^{m+1} - u_h^m\Big\|^2}_{\geq 0} \leq \frac{c_p^2 \Delta t}{2}\Big\|\theta f(\cdot, t^{m+1}) + (1-\theta) f(\cdot, t^m)\Big\|^2. \quad (5.11)$$

Note that the coefficient $\theta - \frac{1}{2}$ is only non-negative since $\theta \geq \frac{1}{2}$. Thus, summation over m yields

$$\begin{aligned}
\|u_h^m\|^2 &\leq \Big\|u_h^0\Big\|^2 + \frac{c_p^2}{2}\sum_{j=0}^{m-1} \Delta t\Big\|\theta f(\cdot, t^{m+1}) + (1-\theta) f(\cdot, t^m)\Big\|^2 \\
&\leq \Big\|u_h^0\Big\|^2 + \frac{c_p^2 T}{2}\max_{t\in[0,T]}\|f(\cdot, t)\|^2.
\end{aligned}$$

For $\theta < \frac{1}{2}$, we have to take a different approach, as we have a negative term on the left-hand side of (5.11). This time, we choose as test function $\varphi = u_h^{m+1}$ and obtain after multiplication with Δt:

$$(u_h^{m+1} - u_h^m, u_h^{m+1}) + \Delta t\theta\left\|\nabla u_h^{m+1}\right\|^2 + \Delta t(1-\theta)(\nabla u_h^m, \nabla u_h^{m+1}) \\ = \Delta t(\theta f(\cdot, t^{m+1}) + (1-\theta)f(\cdot, t^m), u_h^{m+1}).$$

Using the identities $(a-b)a = \frac{1}{2}(a^2 + (a-b)^2 - b^2)$ and $ab = -\frac{1}{2}((a-b)^2 - a^2 - b^2)$ to rewrite the inner products on the left-hand side in terms of norms, we get:

$$\frac{1}{2}\Big(\left\|u_h^{m+1}\right\|^2 + \left\|u_h^{m+1} - u_h^m\right\|^2 - \|u_h^m\|^2\Big) + \Delta t\theta\left\|\nabla u_h^{m+1}\right\|^2 \\ - \frac{\Delta t(1-\theta)}{2}\Big(\left\|\nabla(u_h^{m+1} - u_h^m)\right\|^2 - \left\|\nabla u_h^{m+1}\right\|^2 - \|\nabla u_h^m\|^2\Big) \\ \le \Delta t\left\|\theta f(\cdot, t^{m+1}) + (1-\theta)f(\cdot, t^m)\right\| \cdot \left\|u_h^{m+1}\right\|.$$

Multiplication with 2 and using the Poincaré inequalities and Young's inequality with some factor $\varepsilon > 0$, we obtain:

$$(\left\|u_h^{m+1}\right\|^2 - \|u_h^m\|^2) + \left\|u_h^{m+1} - u_h^m\right\|^2 + \Delta t(1+\theta)\left\|\nabla u_h^{m+1}\right\|^2 \\ + \Delta t(1-\theta)\|\nabla u_h^m\|^2 - \Delta t(1-\theta)\left\|\nabla(u_h^{m+1} - u_h^m)\right\|^2 \\ \le \frac{\Delta t c_p^2}{2\varepsilon}\left\|\theta f(\cdot, t^{m+1}) + (1-\theta)f(\cdot, t^m)\right\|^2 + 2\Delta t\varepsilon\left\|\nabla u_h^{m+1}\right\|^2.$$

Finally applying the inverse inequality (5.10), we get:

$$(\left\|u_h^{m+1}\right\|^2 - \|u_h^m\|^2) + \left\|u_h^{m+1} - u_h^m\right\|^2 + \Delta t(1+\theta-2\varepsilon)\left\|\nabla u_h^{m+1}\right\|^2 \\ + \Delta t(1-\theta)\|\nabla u_h^m\|^2 - \frac{\Delta t(1-\theta)c^2}{h_{\min}^2}\left\|(u_h^{m+1} - u_h^m)\right\|^2 \\ \le \frac{\Delta t c_p^2}{2\varepsilon}\left\|\theta f(\cdot, t^{m+1}) + (1-\theta)f(\cdot, t^m)\right\|^2.$$

Choosing $\varepsilon = \frac{1}{2}(1+\theta)$ as large as possible while not creating a negative term on the left-hand side, we end at:

$$(\left\|u_h^{m+1}\right\|^2 - \|u_h^m\|^2) + \Delta t(1-\theta)\|\nabla u_h^m\|^2 + \Big(1 - \frac{\Delta t(1-\theta)c^2}{h_{\min}^2}\Big)\left\|(u_h^{m+1} - u_h^m)\right\|^2 \\ \le \frac{\Delta t c_p^2}{1+\theta}\left\|\theta f(\cdot, t^{m+1}) + (1-\theta)f(\cdot, t^m)\right\|^2.$$

With the assumptions on θ and Δt, the second and third summands on the left-hand side are positive. Hence, summing over m yields the claim:

$$\begin{aligned}\|u_h^m\|^2 &\le \left\|u_h^0\right\|^2 + \frac{c_p^2}{1+\theta}\sum_{j=0}^{m-1}\Delta t\left\|\theta f(\cdot,t^{m+1}) + (1-\theta)f(\cdot,t^m)\right\|^2\\ &\le \left\|u_h^0\right\|^2 + \frac{c_p^2 T}{1+\theta}\max_{t\in[0,T]}\|f(\cdot,t)\|^2.\end{aligned}$$

□

Theorem 5.27 (A priori error estimate for the θ-method)**.** *With the assumptions of Theorem 5.19, let $u_h^m \in S_{h,0}^k$, $m = 0,\dots M$ be the θ-method approximation from Definition 5.25. In case $\theta \in [0,\frac{1}{2})$, additionally assume that the time-step restriction*

$$(1-\theta)\Delta t \le \frac{h_{\min}^2}{c^2}.$$

from 5.26 is satisfied. Then, for a sufficiently regular weak solution u we have the error bound:

$$\begin{aligned}&\|u_h^m - u(t^m)\|_{L^2(\Omega)}\\ &\quad\le \left\|u_h^0 - u_0\right\|_{L^2(\Omega)} + \frac{|2\theta-1|}{2}\Delta t\max_{j=1}^{m}\left\|\partial_{tt}u(t^j)\right\|_{L^2(\Omega)}\\ &\qquad + \frac{2(1-\theta)+\theta}{2}\Delta t^2\max_{j=0}^{m-1}\frac{1}{\Delta t}\int_{t^j}^{t^{j+1}}\left\|\partial_t^3 u(s)\right\|_{L^2(\Omega)}\mathrm{d}s\\ &\qquad + Ch^{k+1}\Big(\|u_0\|_{H^{k+1}(\Omega)} + \max_{j=0}^{m-1}\frac{1}{\Delta t}\int_{t^j}^{t^{j+1}}\|\partial_t u(s)\|_{H^{k+1}(\Omega)}ds\Big).\end{aligned}$$

Proof. As before, we split the error as $u(t^m)-u_h^m = (u(t^m)-P_hu(t^m))+(P_hu(t^m)-u_h^m)$. For the projection error $u(t^m)-P_hu(t^m)$ we proceed as before. For the second error $e_h^m := u_h^m - P_hu(t^m)$, we obtain from the definition of the θ-method:

$$\frac{1}{\Delta t}(e_h^{m+1}-e_h^m,\varphi_h) + (\nabla(\theta e_h^{m+1} + (1-\theta)e_h^m),\nabla\varphi_h) = (\varepsilon_h^m,\varphi_h),\ \forall\varphi_h\in S_{h,0}^k,$$

where ε_h^m is given by

$$\begin{aligned}(\varepsilon_h^m,\varphi_h) = (\theta f(\cdot,t^{m+1}) &+ (1-\theta)f(\cdot,t^m),\varphi_h)\\ &- \frac{1}{\Delta t}\Big(P_h(u(t^{m+1})-u(t^m)),\varphi_h\Big)\\ &- (\nabla(\theta u(t^{m+1}) + (1-\theta)u(t^m)),\nabla\varphi_h)\qquad \forall\varphi_h\in S_{h,0}^k.\end{aligned}$$

Using that u is a weak solution of the heat equation, we obtain:

$$\begin{aligned}(\theta f(\cdot,t^{m+1}) + (1-\theta)f(\cdot,t^m),\varphi_h) - (\nabla(\theta u(t^{m+1}) + (1-\theta)u(t^m)),\nabla\varphi_h)&\\ = (\theta\partial_t u(t^{m+1}) + (1-\theta)\partial_t u(t^{m+1},\varphi_h).&\end{aligned}$$

Hence,

$$(\varepsilon_h^m, \varphi_h) = (\theta\partial_t u(t^{m+1}) + (1-\theta)\partial_t u(t^{m+1}, \varphi_h) - \frac{1}{\Delta t} P_h(u(t^{m+1}) - u(t^m)), \varphi_h).$$

Since this identity holds for all $\varphi_h \in S_{h,0}^k$ and $\varepsilon_h^m \in S_{h,0}^k$, it follows that

$$\varepsilon_h^m = \theta\partial_t u(t^{m+1}) + (1-\theta)\partial_t u(t^m) - \frac{1}{\Delta t} P_h(u(t^{m+1}) - u(t^m)).$$

Hence, the error e_h^m is the solution of θ-method for the problem with right-hand side ε_h^m instead of $\theta f(\cdot, t^{m+1}) + (1-\theta) f(\cdot, t^m)$.

Following the proof of Theorem 5.26, we obtain:

$$\|e_h^m\| \leq \left\|e_h^0\right\| + C \max_{j=0}^{m} \|\varepsilon_h^m\|.$$

As before, we bound $\|e_h^0\|_{L^2(\Omega)}$ by:

$$\left\|e_h^0\right\| \leq \left\|u_h^0 - u_0\right\| + Ch^{k+1}\|u_0\|_{H^{k+1}(\Omega)}.$$

It remains to bound $\|\varepsilon_h^m\|$. Splitting ε_h^m as

$$\varepsilon_h^m = \underbrace{\theta\partial_t u(t^{m+1}) + (1-\theta)\partial_t u(t^m) - \frac{1}{\Delta t}(u(t^{m+1}) - u(t^m))}_{\varepsilon_{h,1}^m} + \underbrace{\frac{1}{\Delta t}(Id - P_h)(u(t^{m+1}) - u(t^m))}_{\varepsilon_{h,2}^m},$$

we note that $\varepsilon_{h,2}^m$ can be bounded as before by:

$$\frac{1}{\Delta t}\int_{t^m}^{t^{m+1}} \|(Id - P_h)\partial_t u(s)\| \,\mathrm{d}s \leq ch^{k+1}\frac{1}{\Delta t}\int_{t^{m-1}}^{t^m} \|\partial_t u(s)\|_{H^{k+1}(\Omega)} \,\mathrm{d}s.$$

To bound $\varepsilon_{h,1}^m$, we consider Taylor expansions of u around t^m and t^{m+1}:

$$\begin{aligned}
u(t^{m+1}) &= u(t^m) + \Delta t\partial_t u(t^m) + \frac{1}{2}\Delta t^2\partial_t^2 u(t^m) + \frac{1}{2}\int_{t^m}^{t^{m+1}} (t^{m+1} - s)^2\partial_t^3 u(s) \,\mathrm{d}s,\\
u(t^m) &= u(t^{m+1}) - \Delta t\partial_t u(t^{m+1}) + \frac{1}{2}\Delta t^2\partial_t^2 u(t^{m+1}) - \frac{1}{2}\int_{t^m}^{t^{m+1}} (t^m - s)^2\partial_t^3 u(s) \,\mathrm{d}s,
\end{aligned}$$

which we rewrite as:

$$\begin{aligned}
\frac{1}{\Delta t}(u(t^{m+1}) - u(t^m)) &= \partial_t u(t^m) + \frac{1}{2}\Delta t\partial_t^2 u(t^m) + \frac{1}{2\Delta t}\int_{t^m}^{t^{m+1}} (t^{m+1} - s)^2\partial_t^3 u(s) \,\mathrm{d}s,\\
\frac{1}{\Delta t}(u(t^{m+1}) - u(t^m)) &= \partial_t u(t^{m+1}) - \frac{1}{2}\Delta t\partial_t^2 u(t^{m+1}) + \frac{1}{2\Delta t}\int_{t^m}^{t^{m+1}} (t^m - s)^2\partial_t^3 u(s) \,\mathrm{d}s.
\end{aligned}$$

Multiplying the first equation by $(1-\theta)$, the second by θ and adding the results, we obtain:

$$\varepsilon_{h,1}^m = \frac{\Delta t}{2}(\theta\partial_t^2(t^{m+1}) - (1-\theta)\partial_t^2 u(t^m)) - \frac{\theta}{2\Delta t}\int_{t^m}^{t^{m+1}} (t^m - s)^2 \partial_t^3 u(s)\,\mathrm{d}s$$
$$- \frac{1-\theta}{2\Delta t}\int_{t^m}^{t^{m+1}} (t^{m+1} - s)^2 \partial_t^3 u(s)\,\mathrm{d}s.$$

Using

$$\partial_t^2 u(t^m) = \partial_t^2 u(t^{m+1}) - \int_{t^m}^{t^{m+1}} \partial_t^3 u(s)\,\mathrm{d}s,$$

we obtain

$$\varepsilon_{h,1}^m = \frac{\Delta t}{2}(2\theta - 1)\partial_t^2 u(t^{m+1}) + \frac{\Delta t}{2}(1-\theta)\int_{t^m}^{t^{m+1}} \partial_t^3 u(s)\,\mathrm{d}s$$
$$- \frac{\theta}{2\Delta t}\int_{t^m}^{t^{m+1}} (t^m - s)^2 \partial_t^3 u(s)\,\mathrm{d}s - \frac{1-\theta}{2\Delta t}\int_{t^m}^{t^{m+1}} (t^{m+1} - s)^2 \partial_t^3 u(s)\,\mathrm{d}s.$$

Hence,

$$\left\|\varepsilon_{h,1}^m\right\| \le \frac{|2\theta - 1|}{2}\Delta t\left\|\partial_t^2 u(t^{m+1})\right\| + \frac{2(1-\theta)+\theta}{2}\Delta t^2 \frac{1}{\Delta t}\int_{t^m}^{t^{m+1}} \left\|\partial_t^3 u(s)\right\|\mathrm{d}s.$$

$\square$

Remark 5.28. *We see that among all θ-methods, the Crank-Nicolson method ($\theta = 1/2$) is the only scheme which is unconditionally stable and second-order convergent.*

Bibliography

[Alt12] Hans Wilhelm Alt. *Lineare Funktionalanalysis*. Springer Berlin Heidelberg, 2012. DOI: 10.1007/978-3-642-22261-0.

[Alt16] Hans Wilhelm Alt. *Linear Functional Analysis*. Universitext. Springer London, 2016. DOI: 10.1007/978-1-4471-7280-2.

[Bra07a] Dietrich Braess. *Finite Elemente: Theorie, schnelle Löser und Anwendungen in der Elastizitätstheorie*. 4th ed. 2007.

[Bra07b] Dietrich Braess. *Finite Elements: Theory, Fast Solvers, and Applications in Solid Mechanics*. 3rd ed. Cambridge University Press, 2007. DOI: 10.1017/CBO9780511618635.

[BS08] Susanne C. Brenner and L. Ridgway Scott. *The mathematical theory of finite element methods*. Third. Vol. 15. Texts in Applied Mathematics. New York: Springer, 2008, pp. xviii+397.

[Cia02] Philippe G. Ciarlet. *The Finite Element Method for Elliptic Problems*. Classics in Applied Mathematics. Society for Industrial and Applied Mathematics, Jan. 2002. DOI: 10.1137/1.9780898719208.

[Clé75] Ph. Clément. “Approximation by Finite Element Functions Using Local Regularization”. In: *Rev. Française Automat. Informat. Recherche Opérationnelle Sér. RAIRO Analyse Numérique* 9.R-2 (1975), pp. 77–84. ISSN: 0399-0516.

[EG04] Alexandre Ern and Jean-Luc Guermond. *Theory and Practice of Finite Elements*. Ed. by S. S. Antman, J. E. Marsden, and L. Sirovich. Vol. 159. Applied Mathematical Sciences. Springer New York, 2004. DOI: 10.1007/978-1-4757-4355-5.

[EG17] Alexandre Ern and Jean-Luc Guermond. “Finite Element Quasi-Interpolation and Best Approximation”. In: *ESAIM: Mathematical Modelling and Numerical Analysis* 51.4 (July 2017), pp. 1367–1385. ISSN: 0764-583X, 1290-3841. DOI: 10.1051/m2an/2016066.

[EG21a] Alexandre Ern and Jean-Luc Guermond. *Finite Elements I: Approximation and Interpolation*. Vol. 72. Texts in Applied Mathematics. Springer International Publishing, 2021. DOI: 10.1007/978-3-030-56341-7.

[EG21b] Alexandre Ern and Jean-Luc Guermond. *Finite Elements II: Galerkin Approximation, Elliptic and Mixed PDEs*. Vol. 73. Texts in Applied Mathematics. Springer International Publishing, 2021. DOI: 10.1007/978-3-030-56923-5.

[EG21c] Alexandre Ern and Jean-Luc Guermond. *Finite Elements III: First-Order and Time-Dependent PDEs*. Vol. 74. Texts in Applied Mathematics. Springer International Publishing, 2021. DOI: 10.1007/978-3-030-57348-5.

[EG21d] Alexandre Ern and Jean-Luc Guermond. *Finite Elements IV: Exercises and Solutions*. 2021. URL: https://hal.archives-ouvertes.fr/hal-03226052.

[Eva98] Lawrence C. Evans. *Partial differential equations*. Vol. 19. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 1998.

[Gri11] Pierre Grisvard. *Elliptic Problems in Nonsmooth Domains*. Classics in Applied Mathematics. Society for Industrial and Applied Mathematics, 2011. DOI: 10.1137/1.9781611972030.

[Hyt+16] Tuomas Hytönen et al. *Analysis in Banach Spaces*. Cham: Springer International Publishing, 2016. DOI: 10.1007/978-3-319-48520-1.

[Saa03] Yousef Saad. *Iterative Methods for Sparse Linear Systems*. Second. Philadelphia, PA: Society for Industrial and Applied Mathematics, 2003, pp. xviii+528. ISBN: 0-89871-534-2. DOI: 10.1137/1.9780898718003.